

PHYSICS GRADE 10: CURRENT ELECTRICITY

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.2 (12 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Physics
Strand	Strand 3.0: Electricity and Magnetism
Sub-Strand	Sub-Strand 3.2: Current Electricity
Total Duration	12 lessons × 40 minutes = 480 minutes total
Sub-Strand Content	– Electric circuits: components and circuit symbols (cell, battery, switch, bulb, resistor, ammeter, voltmeter) – Electric current, potential difference and electromotive force (e.m.f.) – Measurement of current and potential difference using ammeters and voltmeters – Ohm's law and resistance: $V = IR$ – Factors affecting the resistance of a conductor (length, cross-sectional area, material, temperature) – Resistors in series and in parallel: effective resistance – Cells in series and in parallel – Electrical energy and power: $P = IV$, $E = IVt$ – Domestic wiring and electrical safety (fuses, earthing, circuit breakers) – Applications of series and parallel circuits in day-to-day life (lighting circuits, Christmas lights, solar home systems)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - draw circuit diagrams using standard circuit symbols, - explain the relationship between current, potential difference and resistance in a circuit, - investigate factors affecting the resistance of a conductor, - determine the effective resistance of resistors connected in series and in parallel, - apply Ohm's law and the relations $P = IV$ and $E = IVt$ to solve problems on current electricity, - appreciate the application of series and parallel circuits in day-to-day life and the need for electrical safety. Key concepts: current is the flow of charge and is the same at every point of a series circuit; potential difference is shared among series components and is the same across parallel branches; resistance opposes current; total resistance increases in series and decreases in parallel; power depends on current and voltage.
Core Competencies	– Critical thinking and problem solving: the learner analyses why bulbs in series are dimmer than bulbs in parallel and why one failed tube does not switch off the others, then solves circuit problems. – Communication and collaboration: the learner works in groups to build circuits, share measurements and justify claims using evidence. – Creativity and imagination: the learner designs a safe household lighting circuit for a classroom or home.

	– Digital literacy: the learner uses circuit simulations to test predictions about series and parallel circuits. – Learning to learn: the learner revises their circuit model after each investigation and reflects on how their understanding has changed.
Core Values	– Responsibility: the learner handles electrical equipment safely and cares for shared apparatus. – Integrity: the learner records and reports measurements honestly, even when they differ from predictions. – Respect: the learner values the ideas of group members when discussing explanations.
Science & Engineering Practices	– Asking questions and defining problems: students generate questions about the failed tube and dim bulbs for the Driving Question Board. – Developing and using models: students draw and revise circuit models showing current flow and energy transfer. – Planning and carrying out investigations: students measure current and voltage in series and parallel circuits. – Analyzing and interpreting data: students tabulate ammeter and voltmeter readings and identify patterns. – Using mathematics and computational thinking: students apply $V = IR$, $P = IV$ and the resistance formulae. – Constructing explanations: students use evidence to explain the phenomenon. – Engaging in argument from evidence: students defend explanations in class discussions.
Pertinent & Contemporary Issues (PCIs)	– Safety and security education: the learner observes electrical safety (fuses, earthing, avoiding wet hands and damaged wires) when handling circuits and household appliances. – Education for sustainable development: the learner explores energy saving by using efficient lighting such as LEDs and solar power in homes and schools. – Financial literacy: the learner calculates the cost of electricity use (kWh) for household appliances.
Career Connections	– Electrician or electrical technician: wiring homes, schools and businesses in towns like Nairobi, Kisumu and Mombasa. – Electrical engineer at Kenya Power or KenGen: designing and maintaining power distribution networks. – Solar technician: installing solar home systems in rural areas of the Rift Valley and Turkana. – Electronics technician: repairing appliances and phones. – Physics teacher: explaining circuits and electrical safety.
Focus for Lessons	This unit is anchored in a familiar puzzle: why the other fluorescent tubes stay lit when one blows during evening study, and why bulbs in series are dimmer than those in parallel. Students predict, investigate with real circuits and simulations, explain using current, voltage and resistance, and progressively build and revise a class circuit model. The unit moves from qualitative ideas to quantitative reasoning with Ohm's law, power and energy, and links to safe domestic wiring in Kenyan homes.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why do the other tubes stay lit when one blows out, and why do bulbs connected in series glow dimmer than identical bulbs connected in parallel to the same battery? KICD KEY INQUIRY QUESTIONS: 1. How do we determine the relationship between current, potential difference and resistance in an electric circuit? 2. How is current electricity applied in day-to-day life?
Anchoring Phenomenon	You are in class studying at night and one of the fluorescent tubes blows out! Why do the other tubes remain lit? Students also observe that three identical bulbs connected one after another (in series) to a 9 V battery light dimmer than identical bulbs connected in parallel to an identical battery. In a Kenyan school or home, such as a boarding school dormitory in Kericho or a classroom in Nairobi, the lights keep working when one tube fails, yet when one bulb in an old string of Christmas lights fails, the whole string may go dark. The puzzle is that the same type of battery and the same bulbs give very different brightness and

	behaviour depending only on how they are connected. Why does the connection pattern change brightness and what happens when one component fails?
Storyline Thread	Lesson 1: Anchoring phenomenon. Students observe the blown tube and the series vs parallel bulb demo, predict why, and generate questions for the Driving Question Board. Initial models drawn. Lesson 2: Circuit components and symbols. Students build simple circuits, draw circuit diagrams, and identify what is needed for a bulb to light. Lesson 3: Electric current. Students investigate current as flow of charge, measure with an ammeter in series, and discover that current is the same at all points in a series circuit. Lesson 4: Potential difference and e.m.f. Students measure voltage across bulbs and cells with a voltmeter and relate voltage to energy transferred per charge. Lesson 5: Ohm's law. Students vary voltage across a resistor, plot V against I, and determine resistance from the gradient. Lesson 6: Factors affecting resistance. Students investigate length, thickness, material and temperature using wires and explain why long thin extension cables heat up. Lesson 7: Resistors in series. Students measure current and voltage in series circuits, derive total resistance, and explain why the three series bulbs are dimmer. Lesson 8: Resistors in parallel. Students investigate current branches and equal voltage, derive effective resistance, and explain why parallel bulbs stay bright and why one failed tube does not affect others. Lesson 9: Combined circuits and cells. Students solve problems with mixed circuits and compare cells in series and in parallel, linking to solar home systems. Lesson 10: Electrical energy and power. Students use $P = IV$ and $E = IVt$ to compare bulbs and appliances and calculate electricity costs in kWh. Lesson 11: Domestic wiring and safety. Students examine fuses, earthing and circuit breakers and explain why a fuse blows when a socket is overloaded. Lesson 12: Model revision and application. Students finalise their circuit model, resolve the Driving Question Board, and design a safe lighting circuit for a classroom that keeps lights on when one tube fails.
Supporting Phenomena	– In an old string of festival lights, when one bulb is removed the whole string goes out, but in a house lighting circuit one bulb can be removed without affecting the others. – A long thin extension cable makes a kettle heat more slowly than when it is plugged directly into a wall socket, as the cable heats up. – A fuse in a plug blows when too many appliances such as an iron, kettle and heater are plugged into one socket. – Adding more bulbs in parallel to a solar home system battery in a rural home drains the battery faster, while adding bulbs in series makes all of them dim.

ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (80 minutes (2 x 40-minute periods)): The Blown Tube and the Dim Bulbs: Anchoring Phenomenon

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the unit by confronting learners with a puzzling phenomenon: in a Nairobi classroom the other tubes stay lit when one blows, and three identical bulbs on a 9 V pack glow dim in series but bright in parallel. Learners predict, observe, draw initial models and generate questions. No explanation is given in this lesson.
Knowledge	Learners will describe what they observe: three identical bulbs in series glow dim and all go out when one is removed, while three identical bulbs in parallel glow brightly and the others stay lit when one is removed. They will state that the same battery and bulbs give different results depending only on how they are connected. They will recognise that lights in a Kenyan classroom or dormitory behave like the parallel arrangement, and an old Christmas-light string behaves like the series arrangement.
Skills	Making and recording predictions (NGSS SEP: Asking Questions and Defining Problems); careful qualitative observation and comparison of brightness; developing and using an initial model by drawing a circuit sketch with arrows or labels (NGSS SEP: Developing and Using Models); generating testable investigable questions; collaborating and communicating ideas in groups.
Attitudes	Curiosity about everyday electrical devices; willingness to risk an incomplete idea; respect for the ideas of others (Respect); responsibility when handling batteries and bulbs (Responsibility); appreciation of how Physics explains day-to-day life in Kenyan homes and schools.
Key Inquiry Question	How is current electricity applied in day-to-day life? How do we determine the relationship between current, potential difference and resistance in an electric circuit?
Purpose in Storyline	This is the anchoring lesson. It gives learners a shared experience of the phenomenon and opens the Driving Question Board (DQB) with their own questions. Their initial models become the baseline that is revised in later lessons. Lessons 2 to 11 build the ideas of closed circuits, current, p.d., Ohm's law, resistance, and series and parallel rules. Lesson 12 returns to the DQB to resolve it. Nothing is explained today, and learners will explain why only after they have gathered evidence.
Safety Notes	Use only the 9 V pack (six 1.5 V dry cells) with 9 V, 2.7 W bulbs, and never connect learners to mains 240 V. Mr. Otieno demonstrates the mains classroom tubes by observation only. Do not touch the tubes, sockets or switches with wet hands, and do not let learners test sockets. Bulbs may become warm after a minute, so switch off between trials and do not short-circuit the battery pack with a bare wire. Keep the Christmas-light string unplugged and use it as a visual discussion object or on a low-voltage pack only.

B. LESSON OVERVIEW

Learners begin with a story: during evening prep in a Nairobi school, one of four fluorescent tubes blows out, yet the other three remain lit. Mr. Otieno then shows two bench circuits using a 9 V pack and three identical bulbs. In series the bulbs glow dim, and removing one puts all out. In parallel the bulbs glow brightly, and removing one leaves the others lit. Learners also discuss an old Christmas-light string that goes completely dark when one bulb fails. They record predictions before seeing the demo, observe carefully, and compare predictions with observations without being given the answer. They draw initial models of what is happening inside the wires and bulbs, share them in a gallery walk, and turn their puzzles into questions for the Driving Question Board. The lesson ends with a first summary table (what we observed, what we think, what we need to find out).

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Simplifying resistor networks <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Series and Parallel Circuits (Flexbook) Source: CK-12  Search ARES for similar readings	Learners read or hear the Nairobi scenario: four fluorescent tubes in a classroom, one blows, and three stay lit. Individually they write in their notebooks what they think happens inside the wiring so that the other tubes stay lit. Mr. Otieno then shows a 9 V pack, three identical bulbs and wires. He describes the two arrangements: bulbs one after another in a single loop (series) and bulbs on separate branches (parallel). Before anything is connected, learners predict (1) how bright the series bulbs will be compared with the parallel bulbs and (2) what will happen to the others if one bulb is removed in each case. Kipchirchir may predict that the battery gets used up faster in one of the circuits, and learners are invited to include ideas like this in their predictions. Learners record predictions with a short reason (I think... because...).	Say: "Imagine you are studying at night in Nairobi and one tube blows out. What do you expect to happen to the other lights, and why?" Give silent WAIT TIME of 10-15 seconds, then ask learners to write for 2 minutes. Then ask for talk in pairs for 2 minutes. Cold-call 4 learners with: "Tell us your prediction in one sentence and the reason you have for it." Record all ideas on the board without saying right or wrong, and use the phrase: "Thank you, that's an idea we will test." Show the equipment and say: "The same battery and the same type of bulbs are used in both circuits. Will the brightness be the same? Draw what you expect." Give another WAIT TIME of 10-15 seconds. Ask for a show of hands: same, series brighter, or parallel brighter, and count each. Note the numbers on the board. Say: "In a boarding school dormitory in Kericho, would the lights go out when one tube fails? Why do you think so?" Do not correct any misconceptions, such as 'current gets used up by the first bulb' or 'bulbs in parallel share the light'.	Strategy: Predict-and-Justify (Prediction Table with Reasons). Writing a prediction and a reason before seeing evidence makes learners' existing ideas visible to themselves and to the teacher. This creates a mental commitment, so later evidence either confirms or challenges their thinking. That generates cognitive conflict and a real need to know, which drives the unit.	Check that at least 80% of learners have written a prediction with a reason in their notebooks (circulate and tick on a class list). Look for the indicators: states a prediction about brightness, states a prediction about what happens when one component is removed, and gives any reason at all. Note common ideas (for example, 'the first bulb uses up the current', 'the battery pushes harder in parallel', 'all lights are on one wire') on a sticky-note list for later lessons. Learners who cannot give a reason are asked: "What have you seen at home that makes you think so?"
Observe Phase  VIDEO: Simplifying resistor networks <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML:	Learners watch Mr. Otieno's demonstrations in groups of 5-6 around the bench, and then (if enough equipment is available) repeat the circuits themselves. Demo A: three identical bulbs (9 V, 2.7 W) connected one after another to the 9 V pack; learners note the brightness (dim) and then unscrew one bulb, noting that all the bulbs go out. Demo B: three identical bulbs on separate branches connected to the same 9 V pack; learners note that they	Say: "Watch carefully. Your job now is to be a scientist: describe exactly what you see, without explaining it yet." Connect Demo A and give WAIT TIME of 10-15 seconds in silence while learners observe. Ask: "What do you notice about the brightness of each bulb?" Cold-call 3 learners. Then say: "I will now unscrew the middle bulb. What do you predict will happen? Write it down." Wait 10 seconds, then unscrew it. Ask 2 cold-calls: "What did you see?"	Strategy: Structured Observation with a Comparison Table (Notice-and-Wonder). Learners separate what they see from what they infer by recording only observations first. Comparing two circuits that differ in one variable (the arrangement) helps them focus on the important feature. Linking the bench circuits to the classroom and the Christmas lights shows that the phenomenon is a real-life pattern rather than a one-off trick.	Check the observation tables for specific, accurate statements: series bulbs dim and all go out when one is removed; parallel bulbs bright and the others stay lit. Ask 3 groups to share their tables and check that observations are separated from explanations (look for words like 'dimmer', 'brighter', 'went out', 'stayed on'). Check that learners link the series circuit to the Christmas lights and the parallel circuit to the classroom tubes. Tick the groups that mark

Resistance in Parallel Circuits Vs. Series Circuits (PLIX) Source: CK-12 Search ARES for similar readings	glow brightly. One bulb is unscrewed and the other two remain lit at the same brightness as before. Learners record their observations in a table with the headings: circuit, brightness of each bulb, and what happens when one bulb is removed. They then look at a photograph or the actual old string of Christmas lights (unplugged) and discuss why the entire string goes dark when one bulb fails. Finally they look at the classroom lighting itself: with the tubes on, the teacher points out that each tube has its own switch-controlled path from the same supply, and learners note that when one is switched off, the others remain lit. Learners compare their observations with their predictions and mark each prediction as 'matched', 'partly matched' or 'surprised me'. Wanjiru's puzzle, 'Why are the series bulbs so dim?', is posted to the class.	Was it what you predicted?" Repeat for Demo B. Keep the same bulbs and the same pack, and say aloud: "Notice that nothing has changed except how the wires are arranged." Do not name or use the words 'current', 'voltage' or 'resistance' as explanations, and if learners use them, say: "Hold that thought, we will put it on our board as something to find out." Circulate with the question: "What evidence tells you that?" Ask groups to compare the two circuits for 3 minutes using sentence frames: 'In the series circuit... but in the parallel circuit...'. Show the Christmas-light string and ask: "Which of our two circuits does this string behave like? Which does the Nairobi classroom behave like?" Take 3 cold-calls. Switch off the bulbs between demos so the filaments and the pack stay cool.		their predictions as matched or not matched honestly. Learners who write explanations in the observation column are asked: "What did you actually see that makes you say that?"
Explain Phase  VIDEO: Simplifying resistor networks <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Series and parallel resistors Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners produce initial explanations only, not final ones. In pairs they discuss: "Why might the connection pattern change the brightness and what happens when a component fails?" Each learner draws an Initial Model on a page: a diagram of the 9 V pack, the three bulbs in series and in parallel, and the Nairobi classroom tubes. Using arrows, words and colours they show what they think is moving or being shared inside the wires and the bulbs and why the bulbs behave differently. They add a short caption: 'I think the bulbs in series are dim because...'. Learners also write one thing they are unsure of. Models are placed on desks for a quiet gallery walk in	Say: "You are not expected to have the right answer today. Draw what you currently think is happening. Use arrows and labels to show what is moving or what is being shared." Give 8 minutes of drawing time, then WAIT TIME of 10-15 seconds before moving on to ask: "What do you think is happening inside the wires?" Cold-call 3 learners to describe their model and ask each: "What in your model explains why the bulbs are dim in series?" Do not evaluate, and respond with: "What would we see if your idea were true?" Start the gallery walk for 8 minutes and say: "Read at least three models. Write one question or one 'I agree because...' on a sticky note."	Strategy: Initial Model Drawing and Gallery Walk (Developing and Using Models). Drawing forces learners to turn vague ideas into specific, visible claims about what is happening in the circuit. The gallery walk exposes them to alternative explanations, creating productive disagreement. The models become a baseline against which the growth of understanding in later lessons can be measured.	Check that each model shows both the series and the parallel arrangement with the 9 V pack and three bulbs, and that it includes at least one arrow or label indicating what the learner thinks is moving or being shared. Look for any reasoning that links arrangement to brightness or to failure of one bulb. Record the dominant ideas on a class tally (for example, 'used up', 'shared', 'more push'). Check that each learner has added a gallery-walk comment and one revision or question to their own model. Learners with a blank or copied model are asked to explain their thinking aloud to a partner first and then draw.

	which learners read at least three other models and leave a sticky note with one question or one 'I agree because...'. Learners then return to their own model and add one idea or question prompted by the gallery walk. Kipchirchir's idea that the battery gets used up faster in one of the circuits is displayed as an idea to be tested later.	Circulate and photograph or note 3-4 contrasting models (for example, 'current used up', 'current shared equally', 'battery pushes more in parallel'). Select 2 contrasting models and ask their authors to explain them in 1 minute each. Then ask the class: "What do these ideas have in common and where do they differ?" Take 3 cold-calls. Close with: "We do not yet have enough evidence to decide. What would we need to find out?"		
Driving Question Board (DQB) Creation  VIDEO: Simplifying resistor networks <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Series and Parallel Circuits (Flexbook) Source: CK-12  Search ARES for similar readings	Learners turn their wonderings into questions. In groups of 4 they write 2-3 questions on strips of paper beginning with 'Why...', 'How...' or 'What...', for example: 'Why do the other tubes stay lit when one blows out?' or 'Why do series bulbs glow dimmer than parallel bulbs on the same battery?'. Groups post their strips on the DQB under the central driving question: Why do the other tubes stay lit when one blows out, and why do bulbs connected in series glow dimmer than identical bulbs connected in parallel to the same battery? Together the class sorts the questions into categories, such as 'brightness', 'what is moving in the wire', 'battery', 'when one fails' and 'safety at home and school'. Learners add their names or initials to the strips. Questions from Wanjiru (series bulbs dim), Kipchirchir (battery lasting) and Achieng (how a solar system at home keeps lights on) may be included as examples that the class has to consider.	Say: "Scientists begin with questions. Write questions you would need answered to explain the blown tube and the dim bulbs. Make them questions we can investigate." Give WAIT TIME of 10-15 seconds for thinking, then 5 minutes for group writing. Post the driving question on the board's centre. Invite each group to bring 1 strip and read it aloud, and cold-call 5 groups. Ask: "Which category does this belong with?" and cluster the strips with learners' help. Ask: "What do we notice about the groups of questions? Which one do you think we should investigate first?" Cold-call 3 learners. Tell learners: "We will return to this board in every lesson, and we will move questions to 'answered' when we have evidence." Point out the connection to KICD Key Inquiry Questions: "How is current electricity applied in day-to-day life?" and "How do we determine the relationship between current, potential difference and resistance in an electric circuit?" Ask learners which of their questions fit under each. Do not answer any of the questions now, and say: "Good	Strategy: Driving Question Board (Question Clustering). Turning confusion into learner-owned questions gives learners a purpose for each later lesson. Sorting questions into categories reveals the structure of the problem (for example, brightness, what is moving, and failure), so learners can see how the pieces connect and plan what to investigate.	Check that every group has posted at least 2 questions and that at least half are investigable (starting with How or Why and linked to the phenomenon, not simply yes/no). Look for questions covering both parts of the driving question (the blown tube and the dim series bulbs). Ask 2 learners to reword a vague question such as 'Why is electricity?' into a more specific one. Note which categories have few questions and prompt with: "What about the battery? What about safety at home?"

		question. Let's keep it on the board until we have evidence." Ensure the board stays visible for the coming lessons.		
Model Building Phase  VIDEO: Simplifying resistor networks <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Resistance in Parallel Circuits Vs. Series Circuits (PLIX) Source: CK-12  Search ARES for similar readings	Learners review their initial models and, in a class discussion, build a shared first class model on chart paper showing the series and parallel circuits and the Nairobi classroom. They agree on what is clearly observed (for example, 'series bulbs are dim and all go out if one is removed; parallel bulbs are bright and the rest stay lit') and mark with question marks the parts they cannot yet explain (for example, 'what is moving?' 'why dim?'). Each learner then completes a Summary Table in their notebook with three columns: what we observed, what we think (initial ideas), and what we need to find out. They finish with an exit ticket: one sentence that begins 'I wonder whether...' and one thing they would test next with the pack and bulbs.	Say: "Let's build our class model. What can we all agree we observed? What are we unsure about?" Draw the circuits on the chart and ask learners to come forward to add labels. Give WAIT TIME of 10-15 seconds after each question. Cold-call 4 learners: "What should we put in the 'we know' part and what goes in the 'we wonder' part?" Use a coloured marker to mark question marks where the class is unsure. Ask: "Which of these question marks could we investigate with the pack and bulbs we have?" Take 3 cold-calls. Announce the next lesson: "Next time we will look closely at what a bulb needs in order to light, and we'll learn how to draw circuits with symbols so we can talk about them clearly." Do not give any explanation of brightness or of why the other tubes stay lit. Collect exit tickets and keep the class model on the wall.	Strategy: Consensus Model with Summary Table. Combining individual models into one class model, and separating 'what we know' from 'what we wonder', helps learners see the limits of their current understanding. This keeps the model open to revision as new evidence arrives in later lessons and shows that scientific models grow through evidence.	Check the Summary Table for accurate entries in the 'observed' column (brightness differences, behaviour when one component is removed) and honest entries in the 'need to find out' column. Review exit tickets for investigable wonderings such as 'I wonder whether the bulbs are dimmer because the battery is shared by three bulbs'. Success indicators: at least 80% of learners identify the key observable difference between the series and parallel circuits, and at least 70% pose a question that could be tested. Note learners who still confuse observation with explanation for support in the next lesson.

D. TEACHER REFLECTION

After the lesson, reflect: Did learners separate what they saw from what they think? Which initial ideas were most common (for example, 'current is used up by the first bulb', 'the battery pushes harder in parallel', 'the battery runs out faster in one circuit')? Did I resist the urge to explain? Which questions on the DQB are most likely to guide the next lessons, and which categories need more prompting? Were all learners, including quieter ones, given enough WAIT TIME and cold-call turns? Note equipment issues (loose bulbs, weak cells) so the demo can be repeated reliably, and keep photographs of the initial models for comparison in Lesson 12.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Describe what you saw: how bright the three identical bulbs were in series and in parallel on the same 9 V battery pack, what happened when one bulb was removed in each circuit, and what happened when one of the four tubes in the Nairobi classroom blew out and when a bulb in an old Christmas-light string failed.
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What did I learn?	Write your first ideas about why the connection pattern changes brightness and why the other tubes stay lit. Say what you are unsure about, and note that these are initial ideas which you will test with evidence.
How does this explain the phenomenon?	Write one 'I wonder...' question for the Driving Question Board and say what you would test next with the 9 V pack and bulbs. Do not give a final explanation yet, because the full explanation will be built in later lessons.

LESSON 2 (80 minutes (2 x 40-minute periods)): Circuit Components and Symbols: What Does a Bulb Need to Light?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use the 9 V pack and one bulb to find out what a bulb needs in order to light, and learn to represent circuits with standard symbols. This gives them the language to describe the series and parallel arrangements from Lesson 1 and to explain why a broken loop puts out a bulb.
Knowledge	A bulb lights only when it is part of a complete, closed loop that includes a source. Any break in the loop (a loose wire, an open switch, a blown filament) makes an open circuit and the bulb goes out. Standard symbols represent the cell, battery, bulb, switch, wire, fuse, ammeter (in series) and voltmeter (in parallel). A circuit diagram shows how the components are connected. Series means one loop. Parallel means branches.
Skills	Building a simple circuit from a 9 V pack, wires and a bulb; troubleshooting an open circuit; drawing and reading circuit diagrams with correct symbols; translating between a real circuit and its diagram; planning and carrying out a simple investigation; constructing explanations from evidence (NGSS SEP: Developing and Using Models, Planning and Carrying Out Investigations).
Attitudes	Curiosity about how everyday lighting works; precision and neatness when drawing symbols; cooperation in group work; responsibility in handling batteries and wires; open-mindedness when a prediction proves wrong. Values: Responsibility and Respect. Core competencies: Communication and collaboration, Critical thinking and problem solving, Learning to learn.
Key Inquiry Question	How do we determine the relationship between current, potential difference and resistance in an electric circuit? (Starting point: what is needed for current to flow at all?) How is current electricity applied in day-to-day life?
Purpose in Storyline	In Lesson 1 learners saw three series bulbs glow dim, three parallel bulbs glow bright, and the Nairobi tubes stay lit when one blew out. Nobody explained these observations. Lesson 2 gives learners the basic requirement for a lit bulb (a complete closed loop with a source) and the symbols to draw the demo circuits. Learners can then say precisely what 'in series' (one loop) and 'in parallel' (branches) mean, and they see that one loop means one break stops everything. Current is measured in Lesson 3 and p.d. in Lesson 4, so no numbers are explained here.
Safety Notes	Use only the 9 V battery pack (six 1.5 V dry cells) and 9 V bulbs. Never connect a wire straight across the battery terminals with no bulb, because this short circuit heats the wire and drains the cells quickly. Do not connect any circuit to mains (240 V). Switch off or disconnect when changing the circuit. Keep water away from batteries. Dispose of used dry cells in a collection box and not by burning.

B. LESSON OVERVIEW

Mr. Otieno begins by reminding the class of Lesson 1: a Nairobi classroom with four tubes where one blew and the other three stayed lit, and an old string of Christmas lights that goes dark when one bulb fails. Learners predict what a single bulb needs to light, then test it with the 9 V pack, one 9 V bulb and wires. Groups try different arrangements (one wire, two wires touching the base and side, a gap in the loop, a switch open and closed) and record which ones light. They draw circuit diagrams using standard symbols, then analyse a diagram of the series and parallel demo from Lesson 1, tracing the loop(s) with a finger. They apply the idea to the Christmas-light string and the classroom tubes. They write what a bulb needs (a closed loop with a source) and what they still wonder about (why series bulbs are dim, why parallel bulbs stay lit). They add new questions to the Driving Question Board and revise their initial models using symbols. No explanation of brightness is given in this lesson. The numbers (0.10 A, 0.30 A, 3.0 V, 9.0 V) appear in later lessons.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: 01 Symbols Used In An Electric Circuit Source: Seavuria  Search ARES for similar videos  HTML: Circuit Construction Kit: DC Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners recall Lesson 1 (three series bulbs dim, three parallel bulbs bright, the Nairobi tubes and the Christmas-light string). Mr. Otieno holds up one 9 V bulb and the 9 V pack and asks, 'What is the minimum you need to make this one bulb light?' Individually, learners sketch on paper how they would connect the pack, wires and bulb. Wanjiru predicts that 'one wire from the battery to the bulb is enough'. Kipchirchir predicts that you need two wires. Learners then pair up, compare sketches and mark where they disagree. Each pair writes one prediction: 'The bulb will light if ____, and will not light if ____.'	Say: 'Remember our Nairobi classroom, where one tube blew and the others stayed lit. Today we start with the simplest question: what does ONE bulb need to light?' Show the bulb, pack and wires. Say: 'Draw how you would connect them. Do not talk yet, you have 2 minutes.' After drawing, WAIT TIME 10-15 seconds before calling on anyone. Cold-call 4 learners from different parts of the room to describe their sketch; ask each one, 'Where does the electricity go in your sketch?' Do not say which sketch is right. Write 2 or 3 different sketches on the board and label them A, B, C. Ask: 'Hands up for A, B or C.' Record the counts. Say: 'We will test these today with real equipment.'	Predict-Observe-Explain (POE) with Individual Sketching then Pair Comparison. Learners commit to a prediction in a sketch before testing, which makes their current ideas visible to the teacher and to themselves. When the test results come in, they can compare them with what they expected, which makes conceptual change more likely. Sketching first stops the class copying one confident learner.	Look for: (1) how many learners draw a single wire to the bulb (a common misconception that current is 'used up' in one direction from battery to bulb); (2) how many draw a loop with two connections to the battery; (3) whether any sketch includes a switch. Record the A/B/C counts on the board. Note names of learners whose sketches show one wire so you can check their understanding during the investigation. Success indicator: learners write a prediction using 'if' and 'will not light if'.
Observe Phase  VIDEO: Circuit Diagrams - Overview Source: CK-12  Search ARES for similar videos  HTML: Circuit Construction Kit: DC - Virtual Lab Source: PhET Interactive Simulations (English)  Search ARES for similar readings	In groups of 4, learners get one 9 V pack, one 9 V bulb in a holder, 3 insulated wires with crocodile clips, and one switch. They carry out Investigation 1: 'What arrangements make the bulb light?' They test and record (sketch plus tick/cross) at least 5 arrangements: (a) one wire from the + terminal to the bulb only; (b) one wire from the + terminal to the bulb and another from the bulb to the - terminal; (c) the same as (b) with one clip disconnected; (d) a switch added to (b), tested open and closed; (e) wires connected to the bulb base only vs base and side contact. Learners discuss in groups and tick the arrangements that light. Then they carry out a short Investigation 2: they open the loop at different places (a wire,	Distribute equipment and say: 'Connect the battery pack last. Keep your hands on the clips, not on bare metal.' Reinforce safety: 'Never join the + and - terminals with a wire alone. Always include the bulb.' Circulate and ask each group, 'Show me where the loop is complete.' Use the question: 'Which arrangement made the bulb light, and what do all of the lit arrangements have in common?' WAIT TIME 10-15 seconds after asking. Cold-call 3 groups to report one arrangement that lit and one that did not. Press for evidence: 'What did you see? How do you know the loop was complete?' Say: 'Trace the path from the + terminal through the bulb back to the - terminal with your finger. Is there any gap?' For groups still	Structured Investigation with Data Table and Gallery of Class Results. Each group tests the same set of arrangements, so the class table contains repeated evidence across groups and reveals the pattern: the bulb lit only in arrangements with an unbroken loop through the source. The troubleshooting task (open the loop at different places) requires learners to treat the loop as a system and find where the break is, which supports the claim that a break anywhere has the same effect. Using the Christmas-light string connects the lab evidence to the phenomenon.	Observe: (1) Do groups connect the bulb with two connections to the pack and trace the loop? (2) Do they record results accurately (a tick for lit, a cross for not lit) with a sketch? (3) Do they identify that opening the switch or removing a wire has the same effect (the bulb goes out)? (4) Can a learner explain, 'The bulb lights only when the path is complete'? Success indicator: every group's table shows lit only for complete loops, and at least 3 groups correctly name where the loop was broken in the troubleshooting test. Note any group still saying one wire is enough and revisit their test with them.

	the bulb, the switch) and observe that the bulb goes out wherever the loop is broken. Each group notes the place of the break and the effect. Learners finally look at the simple Christmas-light string used by Mr. Otieno (unplugged, with bulbs removable at low voltage in the demo version) and predict what removing one bulb does.	holding the one-wire idea, ask, 'What happens when you remove the second wire? What does that tell you?' Record group results in a class table on the board (arrangement, lit or not lit). Then do the break test: 'Break the loop at the switch. What happens? Now break it at the wire. What changes?' Link back: 'Think about the Christmas-light string. All the bulbs are in one loop. What do you think a dead bulb does to the loop?'		
Explain Phase  VIDEO: 01 Symbols Used In An Electric Circuit Source: Seavuria  Search ARES for similar videos  HTML: Circuit Construction Kit: DC Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners use their results to write a class claim: 'A bulb lights only when ____.' Mr. Otieno then introduces the standard symbols for cell, battery, bulb, switch (open and closed), wire, fuse, ammeter (circle with A) and voltmeter (circle with V). Learners redraw their working circuit as a circuit diagram using symbols and a ruler. They swap diagrams with a partner, who builds the circuit from the diagram and checks whether the bulb lights. Next, learners study a symbol diagram of the Lesson 1 demos: three bulbs in one loop (series) and three bulbs on separate branches across the same 9 V pack (parallel). They trace the loop(s) with different coloured pencils, count the number of loops, and answer: 'In which diagram is there only one path? In which are there branches? If one bulb were removed from each, where would the loop be broken?' Learners write a short explanation for the Christmas-light string and the Nairobi tubes using the words 'closed loop', 'open circuit', 'series' and 'parallel'. Achieng from Kisumu adds that her family's 12 V	Ask: 'Looking at all the tables, finish this sentence: A bulb lights only when ____.' WAIT TIME 10-15 seconds, then cold-call 3 learners and write the best wording on the board. Introduce symbols one at a time by holding up the real component, then drawing its symbol. Say: 'Scientists in Nairobi and in Nanjing draw a cell the same way, so everyone can read the diagram.' Give a rule: 'Straight lines, drawn with a ruler, with corners and no gaps.' Circulate during redrawing and ask, 'Where does current leave the battery and where does it return?' Show the series and parallel demo diagrams on the board. Ask: 'How many separate paths can charge take in the series diagram? How many in the parallel diagram?' WAIT TIME 10-15 seconds. Cold-call 4 learners. Then ask: 'If I remove one bulb in the series diagram, what happens to the loop? What about in the parallel diagram?' Allow learners to propose ideas but do NOT explain brightness: 'We will find out why the series bulbs are dim in later lessons. Today we only describe the loop.' Ask: 'Use today's idea to describe	Claim-Evidence-Reasoning (CER) with Symbol Translation. Learners state a claim (a bulb lights only in a complete closed loop), support it with evidence from their own tables, and give reasoning about the loop. Translating between the real circuit and a symbol diagram, and then reading a diagram of a circuit they have not drawn, builds the modelling skill of using a shared representation. Tracing paths with coloured pencils makes the difference between one loop (series) and branches (parallel) visible, so learners can make a structural, not numerical, explanation of why one failure breaks the series string but not necessarily the parallel one.	Check: (1) the claim sentence includes both a source and a complete closed loop; (2) symbols are correct (cell with long and short lines, bulb as a circle with a cross, open switch with a gap); (3) the partner-built circuit lights from the diagram; (4) learners correctly identify one path in series and several branches in parallel; (5) learners use 'open circuit' and 'closed circuit' correctly. Success indicator: at least 80% of diagrams are correct enough to be built by a partner. Look out for common errors: drawing the bulb with no return path, or drawing a voltmeter in the loop in series (note it and return to it in Lessons 3 and 4).

	solar system also needs a complete loop to power the lights at home.	the Christmas-light string and the tubes in the Nairobi classroom.' Correct only vocabulary, such as using 'open circuit' for a broken loop.		
Driving Question Board (DQB) Creation  VIDEO: Circuit Diagrams - Overview Source: CK-12  Search ARES for similar videos  HTML: Circuit Construction Kit: DC - Virtual Lab Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners return to the Driving Question Board started in Lesson 1: 'Why do the other tubes stay lit when one blows out, and why do bulbs connected in series glow dimmer than identical bulbs connected in parallel to the same battery?' In groups, they write on sticky notes (1) what they learned today that helps answer the question, and (2) new questions they now have. Examples may include: 'What flows through the loop?', 'Why does the bulb in series get dimmer if the loop is still complete?', 'Does the battery push the same in every part of the loop?', 'Why does the whole Christmas string go off but not the classroom tubes?' Learners place notes on the board under headings such as 'We now know' and 'We still wonder', and cluster questions that are similar. Wanjiru places her question about why the series bulbs are dim under 'We still wonder', because a complete loop does not yet explain brightness.	Say: 'Go back to our big question about the tubes and the dim bulbs. What did today's work tell us, and what did it NOT tell us?' Give 3 minutes silent writing. Then say: 'Place one note on the board and read it aloud.' Cold-call 5 learners to read and place notes. Ask: 'Is there another note that connects to yours?' and group them. WAIT TIME 10-15 seconds when asking: 'Which of our questions can we now partly answer?' Highlight questions that will be answered in later lessons: 'What flows in the loop? That is our next lesson, on current.' Do not answer questions about brightness. Say: 'Great question, keep it on the board; the evidence will come.' Add a marker to any question that is answered by today's evidence (for example, 'Does a bulb need a complete loop?').	Driving Question Board with Know/Wonder Sorting. Making questions public and sorting them into 'We now know' and 'We still wonder' shows how today's evidence advances the phenomenon, and it sets the purpose for the following lessons (current, p.d., Ohm's law). Learners see that they have contributed to the direction of the unit, which builds ownership and motivation.	Look for: (1) at least one note per group that links today's finding (complete loop needed) to the phenomenon; (2) new questions that probe more deeply (for example, about what flows or why brightness differs) rather than repeating Lesson 1 questions; (3) sorted notes in the correct column. Success indicator: every learner writes at least one question or finding, and at least 3 new questions on the board go beyond 'does it need a loop?'. Record learners whose questions show confusion between a complete loop and brightness.
Model Building Phase  VIDEO: Introduction to circuits and Ohm's law <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES	Learners take out the initial models they drew in Lesson 1 to explain why the other tubes stay lit and why series bulbs are dim. Individually, they revise these models using what they learned today. They redraw the model in two panels using standard symbols: (1) three bulbs in series on the 9 V pack, and (2) three bulbs in parallel on the 9 V pack. They mark the loop(s) with arrows (conventional current from + to -,	Say: 'Take out your Lesson 1 model. Use a different coloured pen to add what we learned today.' Provide success criteria on the board: standard symbols, ruler lines, a complete loop shown for every lit bulb, an X for a break, and a label for what is not yet explained. Circulate and ask: 'Show me the path for the bulb on the far right. If I remove the middle bulb, which bulbs still have a loop?' WAIT TIME 10-15 seconds.	Iterative Model Revision with Symbolic Representation. Learners keep their earlier model and add to it in a new colour, which shows how their thinking has changed. Using standard circuit symbols turns their model into a scientific representation that others can read and test. Marking a break with an X helps them link model structure to what is observed in the phenomenon. Identifying what remains	Check models for: (1) correct symbols and complete loops; (2) series drawn with one loop and parallel with separate branches from the battery; (3) correct X placement showing that removing a series bulb stops all bulbs, whereas removing a parallel bulb leaves the other loops complete; (4) the classroom tubes drawn as parallel branches; (5) a statement of what is still unexplained. Success indicator: at least 75% of

for similar videos  HTML: Combined Series-Parallel Circuits (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	noting that this is a convention) and show with an X what happens when one bulb is removed, indicating where the loop is broken and which bulbs still have a complete loop. They add a third small sketch of the Nairobi classroom with 4 tubes drawn as parallel branches on 240 V mains. Using a different colour, they add a label: 'Still to explain: why is the brightness different?' Learners share models in pairs and add one improvement based on their partner's feedback.	Cold-call 3 learners to explain their model. Challenge: 'Your model shows bulbs in parallel; where does each branch connect to the battery?' Remind learners that numbers are not needed yet: 'We will add current and p.d. values in later lessons.' Collect models as an exit ticket and say: 'Next lesson we investigate what is actually flowing round the loop and how we measure it.'	unexplained makes the model's limits explicit and prepares them for Lesson 3.	models show correct loops for both circuits. Misconceptions to note for follow-up: current drawn leaving only one terminal or being 'used up' by the first bulb; parallel bulbs drawn with a single shared path; no battery in the diagram.
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D. TEACHER REFLECTION

After the lesson, reflect: (1) How many groups arrived at 'complete closed loop with a source' from their own evidence before I stated it? (2) Which symbols caused the most confusion (for example the cell versus battery, or an open switch)? (3) Did learners keep to describing the loop and avoid claiming a reason for dim bulbs? How did I respond when they asked about brightness? (4) Did my cold-calling reach different learners, including quieter ones like those who rarely speak in group work? (5) Did the Kenyan contexts (the Nairobi classroom, the Kericho dormitory, the Christmas-light string, Achieng's solar system) help learners see the relevance? (6) What will I change in the next lesson on current so that learners can link 'something flows in the loop' to the measurements of 0.10 A and 0.30 A they have not yet explained? Record learners who still hold the one-wire idea and give them extra practice with building from diagrams.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the bulb lit only when the 9 V pack, wires and bulb formed a complete closed loop. When we opened the switch, removed a wire or left a gap, the bulb went out. In the diagrams of the demo, the three series bulbs share one loop, whereas the parallel bulbs are on separate branches across the same 9 V pack.
What did I learn?	We learned that a circuit needs a source and an unbroken path to work, and that standard symbols let us draw circuit diagrams that anyone can read and build. 'In series' means one loop and 'in parallel' means branches. A break in a single loop gives an open circuit, so everything in that loop stops.
How does this explain the phenomenon?	The Christmas-light string goes dark when one bulb fails because all the bulbs are in one series loop, so one break opens the loop. In the Nairobi classroom, each tube seems to have its own branch, so the loop for each of the other tubes is still complete when one blows. We still need evidence to explain why series bulbs are dim and parallel bulbs are bright.

LESSON 3 (40 minutes (double lesson of 80 minutes recommended if the practical is done by every group)): Electric Current: Charge Flow and the Ammeter in Series

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students investigate current as the flow of charge, use $I = Q/t$, connect an ammeter in series, and discover that the current is the same at every point in a series loop. They also record, but do not yet explain, the parallel currents (0.30 A in each branch and 0.90 A from the battery).
Knowledge	Electric current is the rate of flow of charge, $I = Q/t$, in amperes (1 A = 1 C/s). An ammeter is connected in SERIES. In a series loop the current is the same at every point (0.10 A). Conventional current flows from + to -, while electrons move the opposite way in wires. Measured only: parallel branch current 0.30 A each, battery current 0.90 A.
Skills	Predicting, assembling a circuit from a diagram (building on the L2 closed circuits and symbols), reading an ammeter, recording data in a table, calculating charge using $Q = It$, constructing an explanation from evidence (NGSS SEP: Planning and Carrying Out Investigations; Analyzing and Interpreting Data; Using Mathematics).
Attitudes	Responsibility and safety when handling batteries and meters; respect for the ideas of group members; curiosity about why the Nairobi tubes and the Christmas-light string behave differently.
Key Inquiry Question	How do we determine the relationship between current, potential difference and resistance in an electric circuit? (Current is the first quantity we learn to measure.) How is current electricity applied in day-to-day life?
Purpose in Storyline	L1 raised the puzzle of dim series bulbs, bright parallel bulbs and the blown tube. L2 taught that a bulb lights only in a closed loop with a source, and gave us the symbols. L3 supplies the first measurable quantity: current. Students find 0.10 A everywhere in the series loop, and see that the parallel battery supplies 0.90 A. L4 will add potential difference. Why the numbers differ is left open for L7 and L8.
Safety Notes	Use only the 9 V pack of six 1.5 V dry cells, never mains. Connect the ammeter in series only; never place it directly across the battery. Check the circuit with Mr. Otieno before closing the switch. Disconnect when not measuring, because bulbs and wires warm up. Keep liquids away from the equipment. Do not touch any mains socket in the classroom.

B. LESSON OVERVIEW

Mr. Otieno opens by recalling the Nairobi evening prep class where one tube blew and three stayed lit. Students predict what an ammeter will read in the series circuit, at the battery, between bulbs and after the last bulb. They then build the 9 V pack plus three identical 9 V, 2.7 W, 30 Ω bulb series loop and measure the current at four points: 0.10 A at each. They measure the parallel circuit: 0.30 A in each branch and 0.90 A at the battery, recorded but not yet explained. In the Explain phase they define current as $I = Q/t$ (0.10 A moves 0.10 C every second, so 6 C in a minute), clarify conventional current versus electrons, and correct the misconception that bulbs use up current. The DQB and the initial model are then revised with what is now known about current.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Electric current Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Electric Current and Circuits (Flexbook) Source: CK-12  Search ARES for similar readings	Students recall the Nairobi prep class (4 tubes, one blown, three lit) and the dim series bulbs from L1. Mr. Otieno holds up the 9 V pack, three bulbs and an ammeter. Individually, in their notebooks, students draw the series loop with L2 symbols and mark where they would place the ammeter. They write predictions for the reading at four points (before bulb 1, between bulbs 1 and 2, between bulbs 2 and 3, after bulb 3) and justify them. Wanjiru's idea is given: 'Bulbs use up some current, so less comes out at the end.' Students vote on whether they agree with her and why.	Say: 'Last lesson we learnt that a bulb lights only in a complete loop. Today we measure what is moving round that loop. Draw the series circuit and mark four points where you could place an ammeter.' Give 2 minutes of silent drawing. Say: 'Wanjiru says the bulbs use up current so less returns to the battery. Show me with fingers: 1 = agree, 2 = disagree, 3 = not sure.' Count the votes on the board. Cold-call 3 students from different parts of the room: 'Which point do you think has the biggest current, and what is your reason?' WAIT TIME 10-15 seconds before calling on anyone. Do not say who is right. Say: 'Keep your predictions; the ammeter will be the judge.'	Prediction with justification (Predict-Observe-Explain). Committing to a written prediction and a reason exposes the 'current is used up' preconception, so the later measurement of 0.10 A everywhere produces productive conflict. Students must connect the drawn loop (from L2) to a measurable quantity.	Check notebooks for: a closed loop drawn with correct symbols (cell or battery, three lamps, wires); the ammeter shown IN the loop, not across the bulb; a stated reason for the prediction. Tally how many predict 'decreasing current' (the misconception). Note students who draw the ammeter in parallel; address them at the observe stage.
Observe Phase  VIDEO: Conventional current direction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Electric Current (Flexbook) Source: CK-12  Search ARES for similar readings	Groups of 4 collect the 9 V pack (six 1.5 V cells), three identical 9 V, 2.7 W bulbs, an ammeter, a switch and wires. Part A, series: they connect the loop, place the ammeter at four points in turn, close the switch briefly and record the readings in a table (expected 0.10 A at each; the bulbs glow dim). They then unscrew one bulb and note that the ammeter drops to 0 A and all bulbs go out. Part B, parallel: Mr. Otieno or one group demonstrates the three bulbs in parallel with the ammeter measuring each branch and the battery lead. Students record 0.30 A in each branch and 0.90 A at the battery, and note the bulbs are brighter. They record these numbers only, without explanation. Kipchirchir comments that the battery must be working harder in parallel; this is noted on the board as an idea to test later. Roles: builder, ammeter reader, recorder, safety checker.	Say: 'Before you close the switch, call me to check that the ammeter is in the loop, as one of the wires, and not across a bulb.' Circulate to every group within 5 minutes. Ask each group: 'What does your reader say at point 1? At point 3? Is that what you predicted?' If readings differ slightly, say: 'Check that the ammeter is on the same scale and the connections are tight.' Ask: 'What happened when you unscrewed one bulb? Which L2 idea does this match?' (a broken loop). For Part B say: 'Record only the numbers. We will explain them in later lessons.' WAIT TIME 10-15 seconds after asking 'What do you notice about the four readings?' Cold-call 4 groups to report one reading each and write them on the class table on the board.	Data table and pattern-spotting (Analyzing and Interpreting Data). Collecting repeated measurements at several points lets students see a pattern (the same value) rather than being told. Class pooling of results shows consistency and reduces measurement error. Keeping parallel data as 'recorded, not explained' maintains the open questions for the DQB.	Observe: ammeter connected in series by every group; correct reading of the scale to 0.01 A; table completed with units; the group explains that unscrewing one bulb gives 0 A. Look for students saying 'the same everywhere' without prompting. Exit check for the parallel data: can the recorder state the three numbers (0.30 A, 0.30 A, 0.30 A at branches; 0.90 A at the battery) correctly?

Explain Phase  VIDEO: Electric current Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Electric Current Practice Source: CK-12  Search ARES for similar readings	Students compare the class readings with their predictions and explain in pairs why the current is the same at all points (charge has nowhere to go, it keeps circulating round the single loop, so the same amount passes each point every second). Mr. Otieno introduces $I = Q/t$: 0.10 A means 0.10 C of charge passes each point every second. Students calculate $Q = It$ for $t = 60 \text{ s}$ ($0.10 \times 60 = 6 \text{ C}$) and for the parallel battery current ($0.90 \times 60 = 54 \text{ C}$) and compare. They sketch a 'bucket-brigade' picture of the loop. They clarify conventional current (+ to -) versus electron movement (- to +) on the diagram. They revise Wanjiru's claim: bulbs do not use up current; they transfer energy while charge continues round. Finally they write a 2-3 sentence claim-evidence-reasoning statement.	Say: 'Turn to your partner. Using the table, explain why the reading at point 4 is not smaller than at point 1.' WAIT TIME 10-15 seconds before calling on pairs. Cold-call 3 pairs. Then say: 'Current is the rate of flow of charge: $I = Q$ divided by t. If I is 0.10 A, how many coulombs pass in 1 second? In 60 seconds?' Give 1 minute to calculate, cold-call 2 students to the board. Ask: 'In a Kisumu village, think of matatus passing a point on a road. If every minute the same number pass the first and last stage, what does that say about the road having no turn-offs?' Link: 'A series loop has no turn-offs, so the same current passes everywhere.' Draw conventional current arrows and electron arrows in opposite directions; say: 'We use conventional current in all our diagrams.' Ask: 'Wanjiru, do you still agree that bulbs use up current? What energy does the bulb change instead?' Do not fully explain why the bulbs are dim; say: 'We will need potential difference and resistance for that.'	Claim-Evidence-Reasoning with analogy. Students make a claim ('current is the same at all points in a series loop'), cite the 0.10 A data, and reason with the idea of conserved charge flow. The matatu and bucket-brigade analogies make an invisible flow concrete, and $I = Q/t$ gives it a mathematical form (Using Mathematics and Computational Thinking).	Listen for: 'the same amount of charge passes every point each second'; correct calculation $Q = 6 \text{ C}$; a diagram with conventional current arrows from + to -; rejection of 'current is used up'. Check the written CER statement for the claim, the data (0.10 A at four points) and a reason. Common errors: confusing coulombs with amperes, or putting the ammeter in parallel; note these and re-teach in the next lesson opening.
Driving Question Board (DQB) Creation  VIDEO: Conventional current direction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Electric Current and Circuits (Flexbook)	Students revisit the class DQB with the driving question about the blown tube and dim series bulbs. In groups, they write on sticky notes what they have now learnt about current (for example 'series current is 0.10 A everywhere') and place them beside the relevant questions. They add new questions that arose, such as 'Why is the battery current 0.90 A in parallel?', 'Why does each parallel branch have 0.30 A?', 'What makes the series current so small?' and Kipchirchir's 'Does the battery get used up faster in	Say: 'Look at our DQB. Which questions did today's evidence help with? Write the evidence on a note and stick it next to that question.' Give 3 minutes. Cold-call 4 groups to read one note aloud and say where they placed it. Say: 'What new question do you have after seeing 0.90 A at the battery?' WAIT TIME 10-15 seconds. Sort the new questions into 'about current' (answered today), 'about energy and p.d.' (L4) and 'about resistance' (later). Say: 'Notice that we measured the parallel currents but we cannot yet	Driving Question Board as a public record of growing understanding. Linking each note to the evidence makes the logic visible, helps students see which parts of the driving question remain unanswered and motivates the next lessons.	Check that notes cite specific evidence with numbers and units; that at least one new question per group connects to the phenomenon (Christmas lights, blown tube or dim bulbs); that no note claims that the parallel results are already explained. Record which students contribute questions.

Source: CK-12 Search ARES for similar readings	parallel!?. They star questions that the next lesson on potential difference might answer.	say why. That is our next job.'		
Model Building Phase  VIDEO: Conductors and insulators <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Electric Current (Flexbook) Source: CK-12 Search ARES for similar readings	Students return to their initial models from L1 (and the L2 version). Using a fresh colour pen, each draws the series loop and the parallel circuit with L2 symbols and adds: the ammeter placed in series; arrows for conventional current; the value 0.10 A at several points in the series loop; 0.30 A in each parallel branch and 0.90 A leaving the battery. They add a labelled box 'still unknown: why the bulbs are dim and why one broken branch does not stop the others'. In pairs, they use the model to predict what the ammeter would show if a bulb were removed from the series circuit (0 A) and then explain it. They also write one sentence linking the model to the old Christmas-light string.	Say: 'Take out your model. Use a new colour and add only what today's evidence supports.' Circulate and ask: 'Where on your model is the evidence for 0.10 A? What does the arrow show?' Say: 'Add a box for what you cannot yet explain; being honest about gaps is good science.' Cold-call 3 students to share one revision. WAIT TIME 10-15 seconds. Ask: 'Using your model, what will the ammeter read in the series string if one bulb is dead?' Then ask them to predict how the current would change in the Nairobi classroom if one tube fails (to be tested later; do not explain). Collect models for review.	Cumulative model revision (Developing and Using Models). Adding to the model in a new colour shows how the idea of current changes the model, while the 'unknown' box maintains the link to the phenomenon and prepares students for potential difference in L4 and resistance in L5-L8.	Check models for: the ammeter drawn in series; current direction arrows from + to -; 0.10 A at four series points; 0.30 A per branch and 0.90 A at the battery; an unknown box; a correct prediction of 0 A when a series bulb is removed. Use a 3-level rubric (complete, partial, needs support) and note who needs help with series versus parallel ammeter placement.

D. TEACHER REFLECTION

After the lesson ask: Did every group place the ammeter in series and obtain about 0.10 A at all four points? Which students still think current is used up, and what evidence changed (or did not change) their view? Did anyone confuse current with potential difference or charge? Were the parallel readings (0.30 A per branch and 0.90 A total) kept as data only, without premature explanation? Did Kipchirchir's idea about the battery get recorded on the DQB for later testing? Which Kenyan analogies (matatus, Christmas lights, Nairobi tubes) worked best, and what should change in the next lesson on potential difference? If meters were scarce, consider a station rotation; if cells were weak, use fresh ones so the readings stay close to 0.10 A.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We measured the current in the series loop of three identical 9 V, 2.7 W bulbs on a 9 V battery pack: 0.10 A at every point, and 0 A (all bulbs out) when one bulb was removed. In the parallel circuit we measured 0.30 A in each branch and 0.90 A at the battery, and the bulbs were brighter.
What did I learn?	Current is the flow of charge, $I = Q/t$, in amperes (1 A = 1 C/s). An ammeter is connected in series. The current is the same at every point in a series loop, so bulbs do not use up current. Conventional current flows from + to -, while electrons move the opposite way. In 60 s a current of 0.10 A moves 6 C.
How does this explain the	So far, a series loop has a single path so the same 0.10 A passes every point, and a broken bulb opens the loop, which is why an

phenomenon?	old Christmas-light string goes dark. We cannot yet explain why the series bulbs are dim, why the parallel battery gives 0.90 A, or why the other Nairobi tubes stay lit. These need potential difference and resistance, which we study next.
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LESSON 4 (40 minutes): Potential Difference and e.m.f.: Energy Per Charge in the Dim and Bright Bulbs

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate why the same 9 V battery gives 3.0 V across each of three identical series bulbs but 9.0 V across each parallel bulb. They use a voltmeter to measure potential difference and relate it to energy transferred per unit charge. This gives the energy-per-charge evidence needed to explain the blown tube and the dim bulbs.
Knowledge	Potential difference (p.d.) is the energy transferred per unit charge as charge moves between two points of a circuit ($1 \text{ V} = 1 \text{ J/C}$). e.m.f. is the energy supplied per unit charge by the source (battery or cell). A voltmeter is connected in PARALLEL across the component being measured. Each 1.5 V dry cell supplies 1.5 J to every coulomb; six cells in series give 9 V. In the series demo the p.d. across each bulb is 3.0 V ($3 \times 3.0 = 9.0 \text{ V}$); in the parallel demo it is 9.0 V across each bulb.
Skills	Predicting voltage readings; connecting a voltmeter correctly in parallel; measuring and recording p.d.; reading analogue and digital scales; comparing data across circuits; using claim-evidence-reasoning; developing and revising a model (NGSS SEP: Planning and Carrying Out Investigations; Analyzing and Interpreting Data; Developing and Using Models).
Attitudes	Curiosity about how energy moves in circuits; care and responsibility when handling batteries and meters; respect for the ideas of group members; appreciation of how physics explains the lights in our schools and homes.
Key Inquiry Question	How do we determine the relationship between current, potential difference and resistance in an electric circuit? (This lesson supplies the potential difference part.)
Purpose in Storyline	Lesson 3 showed that current is the flow of charge, measured by an ammeter in series, with 0.10 A at every point in the series loop. Lesson 4 now asks what pushes that charge and how much energy each coulomb carries and gives up. The voltage data (3.0 V per series bulb, 9.0 V per parallel bulb) are the evidence for Lesson 5 (Ohm's law, $V = IR$) and Lessons 7 and 8 (why series bulbs share the p.d. and parallel branches each get the full p.d.). We do not yet explain resistance or why the p.d. shares as it does.
Safety Notes	Use only the low-voltage 9 V battery pack and dry cells. Never connect a voltmeter or any lab wiring to mains (240 V) sockets. Do not short-circuit the battery: no bare wire directly across the terminals. Disconnect the circuit when changing connections. Bulbs and wires may become warm. Keep water away from the equipment.

B. LESSON OVERVIEW

Mr. Otieno returns to the Nairobi evening prep class where one of four fluorescent tubes has blown out while the others stay lit. Learners predict how much voltage each bulb will get from the 9 V battery in series and in parallel. They then measure p.d. with voltmeters across single cells, the whole battery, each series bulb (3.0 V) and each parallel bulb (9.0 V). They make sense of the readings using the idea of energy per charge ($1 \text{ V} = 1 \text{ J/C}$) and distinguish e.m.f. (energy the source supplies) from p.d. (energy transferred in a component). The lesson ends by adding energy-per-charge ideas to the DQB and to their models. Learners do not yet explain why the p.d. divides this way; that is for the resistance lessons.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Voltmeters and Ammeters Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Energy Transfer in Electric Circuits (Flexbook) Source: CK-12  Search ARES for similar readings	Learners sit in groups of four. Mr. Otieno shows the 9 V battery pack (six 1.5 V dry cells) and three identical bulbs (9 V, 2.7 W). He recalls the Nairobi classroom where one tube blew and the others stayed lit, and the Christmas-light string that went dark. Learners draw two circuit diagrams using Lesson 2 symbols (three bulbs in series; three bulbs in parallel) and predict on paper: 'What voltage will a voltmeter read across ONE bulb in each circuit? Why?' Wanjiru's question is posted: 'Why are the series bulbs dim?' Kipchirchir suggests that each bulb gets the same 9 V in both circuits. Learners write their predictions with a reason and sketch where the voltmeter would be placed.	Say: 'Last lesson we saw 0.10 A at every point in the series loop. Today we ask: what is the battery giving each bulb?' Hold up one 1.5 V cell: 'What do you think the 1.5 V printed on this cell means?' WAIT TIME 10-15 seconds. Cold-call 3 learners for ideas about what 'volts' means without correcting them. Say: 'Draw both circuits. Predict the reading across one bulb and write WHY.' Walk around and photograph or note 4 predictions. Ask: 'Kipchirchir thinks every bulb gets 9 V in both circuits. Who agrees, who disagrees, and what is your reason?' Take a quick hand count. Do not give the answer yet.	Eliciting prior ideas with a Predict-Explain-Justify routine: writing a prediction with a reason before measurement makes learners commit to a mental model, so the later data produce real conceptual conflict and reveal ideas such as 'voltage is used up like water' or 'each bulb gets the same as the battery'.	Look for: (1) correct voltmeter placement drawn across the bulb (in parallel) in at least 3 of 4 groups; (2) predictions with a stated reason; (3) whether learners think voltage is the same as current or gets 'used up'. Record the tally of predictions (9 V in both / 3 V in series / other) on the board for revisiting in the Explain Phase.
Observe Phase  VIDEO: Voltage Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Energy Transfer in Electric Circuits Practice Source: CK-12  Search ARES for similar readings	In groups with a voltmeter, a 9 V battery pack, single 1.5 V cells and three identical bulbs, learners do four measurements. (1) They connect the voltmeter across ONE 1.5 V cell, then across three cells in series (4.5 V) and across all six (9.0 V), and record e.m.f. values. (2) They build the series circuit of three bulbs, check the bulbs glow dim, and connect the voltmeter in parallel across each bulb in turn: 3.0 V each. They add the three readings to get 9.0 V and compare with the battery terminal reading. (3) They build the parallel circuit of three bulbs, which glows bright, and measure across each bulb: 9.0 V each. (4) In the series circuit, they carefully remove one bulb, see everything go out, and measure across the gap and across the battery. Learners record all values in a table and sketch each circuit with the	Model once: 'The voltmeter goes ACROSS the component, like a bridge, in parallel. The ammeter went IN the loop, in series. Which is which?' Cold-call 2 learners to say where the voltmeter goes. Circulate and ask each group: 'Where is your voltmeter connected, and what does it read?' Prompt: 'Add your three series readings. What do you notice?' WAIT TIME 10-15 seconds before taking answers. Ask: 'Why do the series bulbs get only 3.0 V when the battery gives 9 V?' Say: 'Hold that question; just record for now.' Remind learners: 'Disconnect before changing wires and never short the battery.' Ensure all groups record the readings 3.0 V and 9.0 V accurately and note the bulb brightness beside each value.	Structured data collection with a comparison table: placing e.m.f., series p.d. and parallel p.d. side by side lets learners spot patterns (the three series p.d. add up to the battery e.m.f.; parallel bulbs each get the full e.m.f.) and links numbers to the brightness seen in the phenomenon.	Observe: voltmeter connected in parallel across the component (not in the loop); readings recorded with units (V) to the correct precision; groups noticing that $3.0 + 3.0 + 3.0 = 9.0$ V. Check the tables for 1.5 V, 4.5 V, 9.0 V, 3.0 V and 9.0 V. Note groups that connect the voltmeter in series (bulb goes out, reading odd) and re-teach on the spot with the symbol diagram.

	voltmeter position marked.			
Explain Phase  VIDEO: Voltage - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Series Circuits: Voltage Drops (PLIX) Source: CK-12  Search ARES for similar readings	Learners meet the key ideas through a short teacher-led explanation using the Kenyan context of carrying maize. Imagine each coulomb of charge as a porter carrying energy from the battery to the bulbs: a 9 V battery gives 9 J to every coulomb (e.m.f. = 9 V, 1 V = 1 J/C). When charge passes through a bulb, it transfers some of that energy as light and heat; the p.d. across the bulb is the energy transferred per coulomb. In series each coulomb passes through three bulbs, 3.0 J given up at each, 9 J altogether. In parallel each coulomb passes through only one bulb, which transfers all 9 J. Groups write a claim-evidence-reasoning (CER) answer: 'Why does a parallel bulb get 9.0 V but a series bulb 3.0 V?' and 'What is the difference between e.m.f. and p.d.?' They then use the formula energy = charge x p.d. to calculate, for example, the energy given up by 2 C passing through a 3.0 V bulb (6 J) and through a 9.0 V bulb (18 J). They revisit their predictions and mark whether they were right.	Say: 'Think of each coulomb as a porter leaving the battery carrying 9 joules. Where does it drop its load?' Draw both circuits on the board with energy bars (9 J split 3 + 3 + 3 in series; 9 J dropped at one bulb in parallel). Ask: 'So what does 1 V mean?' WAIT TIME 10-15 seconds, then cold-call 3 learners to complete: '1 volt is one ____ per ____.' Contrast: 'e.m.f. belongs to the source; p.d. belongs to the component.' Pose: 'If 2 C go through a 3.0 V bulb, how much energy becomes light and heat?' Give 2 minutes of pair work, cold-call 2 pairs. Return to the tally: 'Whose prediction matched the data?' Address the misconception that voltage is used up like current: 'The current stays 0.10 A; it is the energy per charge that is dropped.' Do not yet explain why the share is equal (Lesson 7) or compute resistance.	Analogy plus Claim-Evidence-Reasoning: the 'porter carrying energy' analogy gives a concrete picture of energy per charge, while the CER structure forces learners to tie each claim to measured readings (3.0 V, 9.0 V, 9 V) and to state the reasoning in terms of energy per coulomb. The analogy is then explicitly critiqued so that learners do not confuse charge with energy.	Check written CER statements for: (1) the claim that the parallel bulb has a larger p.d.; (2) evidence quoting 3.0 V and 9.0 V; (3) reasoning using 'energy per coulomb' or '1 V = 1 J/C'. Check calculations: $2\text{ C} \times 3.0\text{ V} = 6\text{ J}$; $2\text{ C} \times 9.0\text{ V} = 18\text{ J}$. Listen for correct wording: e.m.f. = energy supplied by source per coulomb; p.d. = energy transferred in component per coulomb. Flag learners who say 'voltage flows' or 'current is used up' and ask them to point to the 0.10 A from Lesson 3.
Driving Question Board (DQB) Creation  VIDEO: Voltmeters and Ammeters Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Energy Transfer	Learners return to the Driving Question Board begun in Lesson 1. In groups they write on sticky notes (a) one question that this lesson answered, for example 'What is a volt?' or 'Why does a voltmeter go in parallel?', and (b) one new question, for example 'Why do the three series bulbs share the 9 V equally?', 'Why does the parallel bulb have the full 9 V and a bigger current?' or 'Does a thin cable have a different p.d.?'. They place notes under	Say: 'Take two minutes: one question we can now answer, one new question this raises.' WAIT TIME 10-15 seconds before groups write. Cold-call 3 groups to read and place a note, asking: 'Which evidence from today helps us answer this?' Point to the original driving question: 'Why do the other tubes stay lit, and why do the series bulbs glow dimmer? Which part did we move forward today?' Circle the questions about why the p.d. shares as 'to be	Driving Question Board with question sorting: learners track their growing understanding and see how new evidence generates new, more precise questions, keeping the storyline coherent and giving them ownership of the next lessons.	Check that at least one new question per group is physics-focused and links to the phenomenon, and that learners can name the evidence from today that supports an answered question. Look for the shift from vague questions ('What is voltage?') to causal ones ('Why does the p.d. split in series?').

in Electric Circuits (Flexbook) Source: CK-12 Search ARES for similar readings	categories: Energy and p.d., Current, Resistance, Wiring at home. Achieng adds: 'Our 12 V solar system in Kisumu: does it give 12 J to each coulomb?' The class marks the question about p.d. as partly answered.	answered in Lessons 5, 7 and 8'. Tell Achieng: 'We will return to your 12 V system in a later lesson on cells.'		
Model Building Phase  VIDEO: Voltage Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Energy Transfer in Electric Circuits Practice Source: CK-12 Search ARES for similar readings	Learners revise their initial model of the Nairobi classroom and the dim bulbs from Lesson 1. Using the Lesson 2 symbols, they redraw the series and parallel circuits and add (a) arrows or labels for p.d. across each bulb (3.0 V each; 9.0 V each), (b) the battery e.m.f. of 9 V, and (c) a note on energy per coulomb (9 J per C supplied; 3 J per C per series bulb; 9 J per C per parallel bulb). They add a caption on why the series bulb is dim, using only today's evidence, for example 'each coulomb gives only 3 J to each bulb', and mark clearly what they cannot yet explain (why the energy shares equally; why the parallel battery supplies a larger current). They add a short line linking this to the whole class: 'Each tube in the Nairobi classroom is across the full mains supply of 240 V, so a blown tube does not change the others.' They mark this as a hypothesis to be tested.	Say: 'Open your model from Lesson 1. Using a different colour, add what we learned about p.d. and e.m.f.' Give 8 minutes. Circulate and ask: 'Where is the voltmeter in your diagram? Where does the energy go in your series circuit?' WAIT TIME 10-15 seconds after each question. Cold-call 2 groups to share a revised drawing under the document camera or on the board. Prompt: 'What is still a gap in your model?' Record gaps on the board such as 'Why 3.0 V in each?' and 'Why is the parallel current bigger?'. Remind learners: 'Keep the model in your folder; we add to it again after Ohm's law.'	Cumulative model revision with gaps: learners add new evidence in a different colour so they can see how their thinking has grown, and explicitly marking unexplained parts builds a habit of honest scientific modelling and creates a purpose for the next lessons.	Check that revised models: (1) show the voltmeter in parallel across the bulbs; (2) have the right numbers (9 V, 3.0 V, 9.0 V); (3) use 'energy per coulomb' language; (4) list at least one unanswered question. Exit prompt: 'Complete the sentence: 1 volt means ____.' Learners who cannot should review the porter analogy and 1 J/C before Lesson 5.

D. TEACHER REFLECTION

Which learners still treated voltage as something that flows or gets used up, and what evidence (for example the unchanged 0.10 A) helped them revise? Did every group connect the voltmeter in parallel, or did some place it in series? Which part of the energy-per-coulomb explanation (the porter analogy, the 6 J and 18 J calculations, the 3.0 V readings) made the most sense to learners, and which need another representation? Did the DQB questions move from 'what is a volt' to 'why does the p.d. split'? Note what to adjust before Lesson 5, where learners use the measured V and I to find resistance.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We measured 1.5 V across one dry cell, 4.5 V across three cells in series and 9.0 V across the whole battery. In the series circuit, each of the three identical bulbs had 3.0 V across it ($3.0 + 3.0 + 3.0 = 9.0$ V) and glowed dim. In the parallel circuit each bulb had 9.0 V across it and glowed bright. When a series bulb was removed, all the lights went out.
What did I learn?	Potential difference (p.d.) is the energy transferred per unit charge between two points, measured in volts ($1 \text{ V} = 1 \text{ J/C}$) with a voltmeter connected in parallel across the component. e.m.f. is the energy per unit charge supplied by the source, such as the 9 V battery. A series bulb receives only part of the e.m.f. (3.0 V), while a parallel bulb receives all of it (9.0 V).
How does this explain the phenomenon?	The parallel bulbs glow brighter because each coulomb of charge gives up all 9 J in one bulb, while in series the 9 J is divided between three bulbs, 3 J each. This helps explain the dim series bulbs and suggests that each tube in the Nairobi classroom is across the full supply. We still need to explain why the p.d. is shared equally and why the parallel circuit draws more current, which needs resistance and Ohm's law.

LESSON 5 (80 minutes (2 x 40-minute periods)): Ohm's Law: How Much Current Does a Resistor Allow for Each Volt?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students investigate how the current through a fixed $30\ \Omega$ resistor changes as the p.d. across it is varied, plot a V-I graph, and use the gradient to find resistance. This gives them the quantitative rule ($V = IR$) that links the current and p.d. measured in Lessons 3 and 4. They will use it later to explain why three $30\ \Omega$ bulbs in series pass only $0.10\ \text{A}$ while each parallel bulb passes $0.30\ \text{A}$.
Knowledge	$V = IR$ (Ohm's law) holds for an ohmic conductor at constant temperature. For such a conductor the graph of V against I is a straight line through the origin, and the gradient equals the resistance in ohms (Ω). For the $30\ \Omega$ resistor: $1.5\ \text{V}$ gives $0.05\ \text{A}$, $3.0\ \text{V}$ gives $0.10\ \text{A}$, $4.5\ \text{V}$ gives $0.15\ \text{A}$, $6.0\ \text{V}$ gives $0.20\ \text{A}$ and $7.5\ \text{V}$ gives $0.25\ \text{A}$, so $R = V/I = 30\ \Omega$ every time. The same $30\ \Omega$ is used for the demo bulb filament, which is a simplification because real filaments get hotter and their resistance rises.
Skills	Connect a circuit with the ammeter in series and the voltmeter in parallel across the resistor (NGSS SEP 3: Planning and Carrying Out Investigations). Record paired readings in a table, plot a V-I graph with a line of best fit and calculate the gradient (SEP 4: Analyzing and Interpreting Data). Use mathematical reasoning with $V = IR$ (SEP 5). Construct an explanation linking resistance to current (SEP 6).
Attitudes	Responsibility in handling electrical equipment and recording honestly. Respect for group members' roles and ideas. Curiosity and persistence in resolving unexpected readings. Appreciation of how precise measurement in Physics explains everyday devices such as the lights in a Kericho dormitory or a Nairobi classroom.
Key Inquiry Question	How do we determine the relationship between current, potential difference and resistance in an electric circuit?
Purpose in Storyline	Lessons 3 and 4 gave students the tools to measure current (ammeter in series, the same everywhere in a series loop) and p.d. (voltmeter in parallel, $1\ \text{V} = 1\ \text{J/C}$). Lesson 5 combines the two measurements into one rule, $V = IR$, and shows that resistance can be found from data. Lesson 6 will explore what makes resistance larger or smaller. Lessons 7 and 8 will then apply $V = IR$ to series and parallel bulbs, so students can finally explain $90\ \Omega$ versus $10\ \Omega$ and why the series bulbs are dim. This lesson does not yet explain the full phenomenon. It builds the quantitative tool.
Safety Notes	Use only the low-voltage dry-cell supply (maximum $7.5\ \text{V}$ from $1.5\ \text{V}$ cells) and never mains $240\ \text{V}$. Check that the ammeter is in series and the voltmeter is in parallel before switching on. Open the switch between readings, because the resistor warms slightly if current flows for long. Do not touch the resistor during or just after use. Keep water away from the equipment. Report loose or hot connections to the teacher immediately.

B. LESSON OVERVIEW

Mr. Otieno reminds the class of the Nairobi classroom where one tube blows and the other three stay lit, and of the $9\ \text{V}$ demo where three identical bulbs in series glow dim. Wanjiru asked why the series bulbs were dim, and the class has measured $0.10\ \text{A}$ in the series loop and $3.0\ \text{V}$ across each bulb. Today students investigate the rule that connects p.d. and current. They predict how the current in a $30\ \Omega$ resistor changes as the p.d. is raised in steps of $1.5\ \text{V}$ (one extra cell each time), then measure V and I , plot V against I and find R from the gradient. They finish by testing the rule on the $30\ \Omega$ bulb in the demo ($3.0\ \text{V}$ gives $0.10\ \text{A}$) and listing what they still need to explain: why adding more bulbs in series reduces the current, and why parallel branches draw $0.30\ \text{A}$ each. The DQB and the class model are updated with the new relationship.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Introduction to circuits and Ohm's Law Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Ohm's Law Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Students recall the 9 V demo, in which each series bulb had 3.0 V across it and 0.10 A flowed, while each parallel bulb had 9.0 V and 0.30 A. In groups of four they individually sketch a prediction graph on a sheet: if we add one 1.5 V cell at a time across a fixed resistor (1.5 V, 3.0 V, 4.5 V, 6.0 V, 7.5 V), what will a graph of V (y-axis) against I (x-axis) look like? They write one sentence of reasoning, e.g. 'when the push doubles, the current ...'. Kipchirchir suggests the current will rise faster and faster, while Wanjiru thinks it rises evenly. Groups agree on one prediction and draw it on mini whiteboards.	Say: 'Think back to the series bulbs. Each had 3.0 V across it and 0.10 A flowing. If the push from the battery gets bigger, what do you think happens to the current in the same bulb?' Give WAIT TIME of 10-15 seconds in silence while students write. Then say: 'Draw your prediction graph on your own first. No talking yet.' After 3 minutes let groups compare. Cold-call 4 different groups (not volunteers) to hold up whiteboards and explain in one sentence. Say: 'Kipchirchir says the line will curve upward. Wanjiru says straight. Who can add a reason for either idea?' Record the 3 types of predictions (straight, curved up, curved down) on the board without judging. Say: 'We will test these today with real readings.'	Predict-Observe-Explain with competing ideas: making students commit to a graph shape before measuring creates a cognitive conflict they want to resolve. Seeing several predicted shapes on the board shows that the class has not agreed, so the data will be used as the judge. This links to the phenomenon by asking whether the current responds to p.d. in a regular way, which is needed to explain the bulbs.	Check each whiteboard for: (a) the axes labelled V and I with units, (b) a clear graph shape, (c) a reason that mentions a change in p.d. causing a change in current. Note how many groups predict a straight line (expect about half). Listen for the misconception that more volts means resistance changes, or that current is 'used up' by the resistor. Tally the three predictions on the board to compare later.
Observe Phase  VIDEO: Topic - current electricity - Ohm's Law Video Primer Source: ARES Local Curriculum - Digital Literacy Content  Search ARES for similar videos  HTML: Ohm's Law: Looking at a Circuit (PLIX) Source: CK-12  Search ARES for similar readings	Each group of four builds the circuit: 1.5 V dry cells in a holder, a switch, a 30 Ω resistor and an ammeter in series, with a voltmeter connected in parallel across the resistor. Roles: connector, reader of the ammeter, reader of the voltmeter, recorder. They begin with one cell (1.5 V), close the switch, read V and I, open the switch, and repeat with 2, 3, 4 and 5 cells (up to 7.5 V). The expected results are 1.5 V with 0.05 A, 3.0 V with 0.10 A, 4.5 V with 0.15 A, 6.0 V with 0.20 A and 7.5 V with 0.25 A (small differences from real equipment are recorded honestly). They add a column V/I. Then they replace the resistor with the 30 Ω demo bulb, read 3.0 V and 0.10 A,	Demonstrate for 2 minutes how to connect the ammeter in series and the voltmeter in parallel, asking: 'Where does the current have to pass through the ammeter? So how must it be connected?' Cold-call 3 students to answer. Circulate and check every circuit BEFORE the switch is closed: 'Show me the path of the current with your finger.' Say: 'Open the switch between readings. Why?' (to keep the resistor from warming). Remind: 'Record exactly what you see, even if it is 0.24 A instead of 0.25 A. Real scientists report real readings.' After the table is done, ask: 'Look at your V/I column. What do you notice?' WAIT TIME of 10-15 seconds. Prompt: 'Now	Data table and graph analysis (Analyzing and Interpreting Data): students turn raw numbers into a pattern by adding a V/I column and plotting a graph. The constant V/I value and the straight line give the evidence needed to judge the earlier predictions. Using the same bulb at 3.0 V and 0.10 A links the lab to the phenomenon from the demo.	Observable indicators: ammeter in series and voltmeter in parallel on first connection; switch opened between readings; values recorded to 2 decimal places with units; V/I column computed correctly (about 30 Ω each row); points plotted accurately (within half a small square); line of best fit drawn with a ruler, passing through or near the origin. Note groups that join the dots instead of drawing a best-fit line, and address this immediately.

	and compare V/I . They plot V on the y-axis and I on the x-axis on graph paper and draw a line of best fit.	plot. Which axis is which? Place your ruler so the line passes close to as many points as possible.' Ask groups with outlier points to repeat that reading.		
Explain Phase  VIDEO: Introduction to circuits and Ohm's law Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Ohm's Law Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Students calculate the gradient of their $V-I$ graph by choosing two points far apart on the line, e.g. (0.05 A, 1.5 V) and (0.25 A, 7.5 V): $\text{gradient} = (7.5 - 1.5)/(0.25 - 0.05) = 6.0/0.20 = 30 \Omega$. They compare this with the V/I column and with the resistor's label of 30Ω . They state Ohm's law in their own words, then write $V = IR$ and rearrange it to $I = V/R$ and $R = V/I$. They check whether the 30Ω bulb at 3.0 V and 0.10 A fits the same rule ($3.0/0.10 = 30 \Omega$), noting that real filaments heat up so this is a simplification. They then revisit their predictions and decide which was correct. Finally they use the rule to answer a short problem: 'In the series loop the three bulbs were 90Ω in total with a 9 V supply. Using $V = IR$, what current do you expect?' ($I = 9/90 = 0.10$ A, matching the measurement).	Ask: 'What does the gradient of your graph tell us? Which quantity are we getting out of it?' WAIT TIME of 10-15 seconds, then cold-call 3 students. Say: 'Explain to your partner why the line is straight, using the words p.d. and current. Then give me one sentence.' Cold-call 2 pairs. Write on the board: 'For an ohmic conductor at constant temperature, V is directly proportional to I , and $V/I = R$.' Ask: 'Kipchirchir predicted a curve. What does the evidence say about his prediction? What did he get right about the idea?' Make clear that it is fine to revise ideas. Pose: 'If I doubled the resistance to 60Ω and kept 9 V, what would the current be?' WAIT TIME 10-15 seconds, cold-call 2 students. Clarify that Lesson 6 will explore why different wires and filaments have different resistance, and Lessons 7-8 will combine resistors. Do not derive series and parallel rules today. Correct the misconception that 'resistance is used up' or that the resistor stores current.	Claim-Evidence-Reasoning (CER): students make the claim ' $V = IR$ for this resistor', give evidence (table, straight-line graph through the origin, gradient 30Ω) and reason (current is directly proportional to p.d. when R is constant). This builds disciplined explanations and links the measurements from Lessons 3 and 4 to a single mathematical rule that can later be applied to the bulbs.	Look for: a correct gradient calculation with units (30Ω) using two widely spaced points; use of $V = IR$ with correct rearrangements; an explicit statement that the line passes through the origin; a CER written response that cites at least two data points; correct answer to the 60Ω question (0.15 A). Flag students who say 'resistance changes with voltage' for a follow-up question: 'What did V/I stay at?'
Driving Question Board (DQB) Creation  VIDEO: Topic - current electricity - Ohm's Law Video Primer Source: ARES Local Curriculum - Digital Literacy Content	Students return to the class DQB. In groups they review the sticky notes and move any questions that today's work helps answer, such as 'Why do series bulbs glow dimmer?', placing them in a 'partly answered' zone. They write new questions on sticky notes, for example: 'Why does the current fall when we add more bulbs in series?', 'What makes the	Say: 'Look at our board. Which questions can we now partly answer using $V = IR$? Come and move one and tell us why.' Cold-call 3 students to move notes and justify. Say: 'Write two new questions that came from today's lab. Start with Why or How.' WAIT TIME 10-15 seconds before groups write. Select 4 new questions to read aloud and ask:	Driving Question Board with a 'partly answered' zone: sorting questions by what the evidence supports and what is still missing makes students' progress visible and keeps the anchoring phenomenon (the dim series bulbs and the tubes that stay lit) at the centre of the inquiry. It builds ownership, since the next lessons respond directly to student	Check that new questions are specific and linked to evidence (for example, mention of resistance, current or the 30Ω bulb) rather than vague. Count how many groups moved at least one question with a data-based reason. Look for questions that show the common misconception that current is 'used up'. Note these for the next lesson's

 Search ARES for similar videos  HTML: Ohm's Law: Looking at a Circuit (PLIX) Source: CK-12  Search ARES for similar readings	resistance of a wire bigger?', 'Why do parallel bulbs each get 0.30 A?', 'Does the bulb really stay 30 Ω when it gets hot?' and 'Why does the extension cable get warm?'. Each group posts at least two new questions and sorts them by lesson: 'Resistance factors', 'Series and parallel', 'Energy and safety'.	'Which of today's data helps us with this question, and what do we still need?' Mark the question 'What makes resistance bigger or smaller?' as the target for Lesson 6 and the question about the number of bulbs in series and parallel as the target for Lessons 7 and 8.	questions.	discussion.
Model Building Phase  VIDEO: Ideal circuit elements <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Slope from graph <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Students open their individual models of the phenomenon, which show the series and parallel circuits and the Nairobi classroom. They add today's evidence: next to each resistor or bulb they write the rule $V = IR$ and an arrow showing that a bigger p.d. pushes more current through the same resistance. In the series circuit they write 'Total $R = 90 \Omega$? (to be tested in L7), $I = 0.10 \text{ A}$ measured' and in the parallel circuit 'each bulb 9 V, 0.30 A, $R = 30 \Omega$ by $V = IR$'. They check this with the calculation $9/30 = 0.30 \text{ A}$. They mark a 'still to explain' box listing what they cannot yet explain: why the series total is 90Ω, and why the battery current is 0.90 A in parallel. They write a short 2-3 sentence revision: 'What I now understand about why brightness depends on current ...'.	Say: 'Add today's evidence to your model in a different colour pen so you can see your thinking grow.' Circulate and ask: 'Where on your diagram does $V = IR$ apply? Show me with an arrow.' Ask a student: 'Using $V = IR$, check the parallel bulb: 9 V across 30Ω. What current?' WAIT TIME 10-15 seconds, then cold-call 2 students to share (0.30 A). Ask: 'Can you use $V = IR$ to say anything yet about the series current? What do you need to know first?' (the total resistance). Say: 'Keep that question for Lesson 7.' Ask 3 students to share one revision and one remaining gap. Collect models as an exit ticket.	Cumulative model revision with colour-coded updates: adding each lesson's evidence in a new colour shows how the explanation develops and keeps the model tied to the phenomenon. The 'still to explain' box makes gaps explicit, which prepares students for the series and parallel rules in later lessons.	Look for models that (a) include $V = IR$ labelled on at least one component, (b) use correct values (30 Ω, 9 V, 0.30 A, 0.10 A, 3.0 V), (c) use arrows or words showing that more p.d. gives more current for the same resistance, and (d) list at least one honest gap. Common errors to check: ammeter drawn in parallel, resistance described as 'using up' current, or the V-I gradient given without units.

D. TEACHER REFLECTION

Before the lesson, check that each group has working 1.5 V cells, a 30 Ω resistor, a switch, an ammeter, a voltmeter and graph paper, and test one set yourself to know typical readings. Afterwards, consider: Did students connect the ammeter and voltmeter correctly the first time? Which groups could explain why the graph is a straight line through the origin, and which only read off the gradient? How did students react when their data differed slightly from 30 Ω ? Did any students claim that resistance changes with voltage, and how did the V/I column help to correct this? Were the DQB questions about resistance and number of bulbs strong enough to drive Lessons 6-8? Note which students need extra support with graph skills and rearranging $V = IR$ for the next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We measured the p.d. across and the current through a 30 Ω resistor at 1.5 V, 3.0 V, 4.5 V, 6.0 V and 7.5 V and found currents of 0.05 A, 0.10 A, 0.15 A, 0.20 A and 0.25 A. The graph of V against I was a straight line through the origin, and V/I stayed at 30 Ω . The 30 Ω bulb at 3.0 V carried 0.10 A.
What did I learn?	For an ohmic conductor at constant temperature, the current is directly proportional to the p.d. ($V = IR$). Resistance is found from V/I or from the gradient of the V-I graph (30 Ω here). A larger p.d. across the same resistance gives a larger current, and a larger resistance gives a smaller current for the same p.d.
How does this explain the phenomenon?	This gives us a rule to explain the dim series bulbs and the bright parallel bulbs: a parallel bulb has 9 V across 30 Ω , so $I = 9/30 = 0.30$ A. We still need to find out how the three bulbs in series combine to give 90 Ω and 0.10 A, and why a failed tube in the Nairobi classroom does not stop the others. This will come in the lessons on resistance and on series and parallel circuits.

LESSON 6 (80 minutes (2 x 40-minute periods)): Factors Affecting Resistance: Why Long Thin Extension Cables Heat Up

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate how the length, thickness (cross-sectional area), material and temperature of a conductor affect its resistance. They use $R = V/I$ from Lesson 5 as the measure of resistance and explain why a long thin extension cable heats up and why a bulb filament has a much higher resistance than a copper connecting wire.
Knowledge	Resistance increases with the length of a conductor (R is proportional to L), decreases as the cross-sectional area increases (thicker wire means lower R), depends on the material (copper has low resistance, nichrome and the tungsten filament have high resistance), and increases with temperature for metals. Resistance is calculated as $R = V/I$ (Lesson 5). A long thin cable has higher resistance and so transfers more energy as heat for the same current.
Skills	Planning a fair test and controlling variables (SEP 3: Planning and Carrying Out Investigations); measuring current with an ammeter in series and p.d. with a voltmeter in parallel; calculating $R = V/I$; graphing and describing trends in data (SEP 4: Analyzing and Interpreting Data); constructing explanations using evidence (SEP 6); communication and collaboration in groups.
Attitudes	Responsibility in using electrical equipment safely; curiosity about why cables heat up; respect for peers' ideas during Gallery Walk and Talk Moves; appreciation of the importance of choosing the correct cable for appliances in Kenyan homes and schools.
Key Inquiry Question	1. How do we determine the relationship between current, potential difference and resistance in an electric circuit? 2. How is current electricity applied in day-to-day life?
Purpose in Storyline	In Lesson 5 learners found that $R = V/I$ measures resistance (30 Ω resistor, gradient method). In this lesson they ask what makes resistance bigger or smaller. This gives the reason why the 30 Ω bulb filament differs from a copper wire and prepares for Lesson 7 (resistors in series add up) and Lesson 11 (why overloaded extension cables are a fire risk and need fuses). It advances the driving question by showing that resistance is a property of the component itself, which later explains why connection patterns change current and brightness. No new circuit numbers are introduced; the 9 V, 30 Ω bulb data are used only for comparison.
Safety Notes	Use only low-voltage supplies (1.5 V dry cells, a maximum of 4.5 V) for wire tests. Never connect any wire to mains 240 V in class. Thin nichrome or constantan wire can get hot, so keep circuits closed for no more than 10 seconds at a time and do not touch the wire until it cools. Use an ammeter only in series and never directly across a cell. Keep water away from the benches. Mr. Otieno switches off and disconnects cells at the end of each trial. The extension cable shown is unplugged and used only as a demonstration object.

B. LESSON OVERVIEW

The class returns to the Nairobi evening prep room with four tubes on 240 V mains, where one tube blew and three stayed lit, and to the 9 V demo where three identical 30 Ω bulbs in series glow dim. Mr. Otieno holds up a long thin extension cable that has become warm in a school workshop and asks: why do some wires heat up more than others? Learners predict what affects resistance, then test length, thickness, material and temperature using low-voltage circuits and $R = V/I$ from Lesson 5. They look for patterns in data, compare a 30 Ω filament with a short copper wire, and explain why a long thin cable heats up. They add a new row to the DQB and update their models by showing that resistance depends on the conductor itself. The lesson builds the idea that each component has its own resistance, a step toward understanding why bulbs in series and in parallel behave differently.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Video: Resistance in a Wire Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Resistance in a Wire Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners see a long thin extension cable (unplugged) and hear Mr. Otieno describe how one in the school workshop got warm when a heater was used on it. Wanjiru recalls the dim series bulbs and asks, 'Is a wire like a bulb? Does it also have resistance?' Individually, learners write in their notebooks: 'What do you think makes a wire have more or less resistance?' They list at least two factors and draw a quick sketch of a wire that would resist current strongly and one that would resist it weakly. Pairs then share, and the class records ideas such as long, short, thick, thin, copper, hot and cold on the board. Learners also predict: 'Which will heat up more: a long thin cable or a short thick one carrying the same current?'	Hold up the cable and say: 'Remember our Nairobi classroom with four tubes on 240 V. The wires inside the walls carry current all evening. What do you think decides how much a wire resists the current?' Give 2 minutes of silent writing. Say: 'Write first, talk second. I want your own idea.' Then say: 'Turn to your partner and compare.' Cold-call 4 learners (not volunteers) to share one factor each and record every idea on the board without judging. Ask: 'Wanjiru raised a good question. Is a bulb filament different from a copper wire? What makes you think so?' WAIT TIME 10-15 seconds before taking answers. Ask a follow-up of 2 learners: 'What evidence from Lesson 5, where we used $R = V/I$, could help us test your idea?' Do not confirm or reject predictions yet.	Individual Prediction then Think-Pair-Share (eliciting prior ideas). Writing before talking commits each learner to an idea so that they can later compare it with evidence. Listing all the factors on the board gives the class a shared set of testable variables, so that Observe Phase investigations answer questions the learners themselves raised. Linking back to $R = V/I$ shows that resistance can be measured.	Check notebooks as you circulate: Do at least 80% of learners name two or more possible factors? Look for the misconception that 'thicker wire resists more because it is bigger' or 'resistance depends on the current'. Listen for learners who link the cable heating to resistance. Note the 4 cold-called learners' ideas on a class list. Indicator of readiness: learners can say what they would measure (I and V) to find R for each wire.
Observe Phase  VIDEO: Resistivity and conductivity Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Resistance in a Wire Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Groups of 4 carry out four short investigations using a 4.5 V supply (three 1.5 V dry cells in a holder), an ammeter in series, a voltmeter in parallel across the test wire, a switch and crocodile clips. STATION A (Length): nichrome wire of fixed thickness is tested at 25 cm, 50 cm, 75 cm and 100 cm. STATION B (Thickness): equal lengths of nichrome wire of two different thicknesses (thin and thick) are tested, or one strand versus two strands twisted together to double the area. STATION C (Material): equal-length, equal-thickness copper, nichrome and constantan (or iron) wires are compared. STATION D	Before starting, say: 'We must test one factor at a time. Which things must we keep the same for the length test?' Cold-call 3 learners to name controlled variables (same material, same thickness, same temperature, same supply). Demonstrate ammeter in series and voltmeter in parallel across the test wire, and say: 'Close the switch for 10 seconds only, read, then open it. Why do you think I am asking for this?' WAIT TIME 10-15 seconds, then take 2 answers (the wire heats and its resistance changes). Circulate with prompts: 'What happens to the ammeter reading as the wire gets longer? What does that tell you	Structured Investigation with Controlled Variables and Data Tables (SEP 3 and SEP 4). Changing only one variable at a time shows cause and effect, and calculating $R = V/I$ from Lesson 5 turns each trial into a number that can be compared. Plotting R against length shows a straight-line trend for learners to describe. Using stations lets the class gather evidence on four factors, and the copper versus filament comparison links the findings to the bulb in the phenomenon.	Observe whether groups: (1) connect the ammeter in series and the voltmeter in parallel correctly without prompting; (2) change only one variable per station; (3) calculate $R = V/I$ with correct units (Ω); (4) record at least four readings at the length station; (5) state a trend in words, e.g. 'when length doubled, R doubled'. Look for the misconception that a thick wire has a higher R. Use a quick check: ask 3 learners to show their R values for Station A and explain the pattern.

	(Temperature): a coil of thin nichrome or a torch bulb filament is tested briefly at low current and again after the circuit has been on a little longer so that it warms, noticing the ammeter reading drop. For each trial learners record V and I, calculate $R = V/I$, and note any warming felt by cautiously holding a hand near the wire. Roles rotate: circuit builder, ammeter reader, voltmeter reader, recorder. Groups then plot R against length and discuss the trend. Finally, the teacher shows a copper connecting wire next to the 30 Ω bulb filament (treated as 30 Ω in the demo) and learners observe that the copper wire barely resists current while the filament resists strongly.	about R?' and 'Your calculated R for 50 cm is double the 25 cm value; is that a coincidence?' Ask the recorder in each group: 'What pattern do you see in your table?' At the demonstration, hold up copper wire and the bulb: 'Both are conductors. Why does the filament glow while the copper wire stays cool for the same current? Think silently first.' WAIT TIME 10-15 seconds. Remind groups that no mains electricity is allowed and that the wire must be allowed to cool before it is touched.		
Explain Phase  VIDEO: Video: Resistance in a Wire Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Ohm's Law <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners first compile class results in a summary table: factor changed, effect on resistance, evidence from our data. They write a Claim-Evidence-Reasoning (CER) explanation answering: 'Why does a long thin extension cable heat up?' Their claim should state that a long thin cable has high resistance. Evidence uses their data (resistance increases with length, thinner wire gives higher R and a larger rise in the temperature) and the copper versus filament comparison (material matters). Reasoning links resistance to the collisions that charge carriers experience as they move through the wire: more collisions in a longer or thinner wire means more energy transferred as heat. Learners also use the temperature station to explain that a hot metal wire resists current more, so the ammeter reading falls as the wire	Say: 'Use your own table as evidence. Do not tell me what you think the textbook says; tell me what your data show.' Provide the CER sentence frames: 'I claim that... My evidence is... This happens because...' Cold-call 4 groups to read one claim and say what evidence they used. Ask: 'Kipchirchir, if the length is doubled and everything else is kept the same, what happens to R? How do you know?' WAIT TIME 10-15 seconds. Prompt: 'Think of the charge carriers moving along the wire. Why would a longer path or a narrower path make it harder for current to flow?' Take 3 responses and refine the idea of more collisions and a narrower path. Address the common misconception: 'Does a thicker wire resist more?' Ask learners to defend with data. Introduce the idea that resistance depends on both the material and the shape,	Claim-Evidence-Reasoning (CER) with Gallery Walk (SEP 6: Constructing Explanations). CER requires learners to connect a claim about the extension cable to their own measurements, so explanations are grounded in evidence rather than memory. The reasoning step introduces a particle-level mechanism (charge carriers colliding in the wire) that explains both the factor trends and the heating. The Gallery Walk exposes learners to alternative explanations and lets them refine theirs.	Use a CER rubric: (1) the claim states that a long thin cable has high resistance and heats up; (2) the evidence cites at least two measured values or trends; (3) the reasoning links resistance to energy transfer as heat. Look for learners who: correctly say R increases with length and decreases as the area increases; state that material matters (copper versus nichrome); note that metal resistance increases with temperature. Misconception check: any learner who says 'resistance depends on the current' should be asked to revisit their Station A data.

	warms. Groups answer application questions: (a) Why do connecting wires in a Nairobi classroom use thick copper, while the heater element in a jiko electric cooker uses nichrome? (b) Why should Wanjiru not use a very long thin cable to run a kettle to a dormitory in Kericho? (c) How does the filament's $30\ \Omega$ resistance compare with copper wire, and why does the filament glow while the copper wire does not? Learners then gallery-walk the CER posters.	and that the same current causes more heating in a higher-resistance cable (state qualitatively; the quantitative $P = I^2R$ is not needed). Link to the Christmas-light string and Nairobi classroom: 'Resistance belongs to each component. Next lesson we find out what happens when several resistances are placed in one loop.' Do NOT teach the series addition rule here.		
Driving Question Board (DQB) Creation  VIDEO: Resistivity and conductivity Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Internal Resistance (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners return to the Driving Question Board with the DQ: 'Why do the other tubes stay lit when one blows out, and why do bulbs connected in series glow dimmer than identical bulbs connected in parallel to the same battery?' In groups they write on sticky notes: (1) what they now know about resistance that helps with the DQ (e.g. 'each bulb filament has its own resistance, $30\ \Omega$'; 'long thin wires have high resistance and get hot'); (2) new questions that this lesson raised (e.g. 'What happens to resistance when we put several bulbs in one loop?', 'Why do extension cables overheat when too many appliances are plugged in?', 'Can the heating in a wire cause a fire?'). They place notes beside earlier questions and, with the teacher, link them by lesson: Lesson 5 ($R = V/I$), Lesson 6 (factors of R). Questions that can be answered are marked with the lesson number where they will be tackled (e.g. Lesson 7: series resistance; Lesson 11: safety and fuses).	Say: 'Look at the board. Which of our questions have we moved closer to answering today?' Ask groups to attach one 'we now know' note and one 'new question' note. Cold-call 3 groups to read their new question and say which earlier question it connects to. WAIT TIME 10-15 seconds before each response. Ask: 'Achieng, her family uses a 12 V solar home system in Kisumu. Which of today's ideas would help her choose cables?' Take 2 responses. Cluster the new questions into themes (resistance in a loop, heating and safety, brightness) and tell learners which later lessons will address them. Do not answer the new questions yet.	Driving Question Board with Question Clustering. Revisiting the DQB makes progress visible and keeps learning tied to the anchoring phenomenon. Linking each new note to a prior lesson shows how evidence builds across the unit, and the new questions create a purpose for Lessons 7 to 11.	Check that notes are specific and linked to evidence (e.g. 'longer wire gave higher R, so a long cable heats'). Look for at least one new question per group that refers to series or parallel connections, and for learners connecting resistance to the dim series bulbs. A learner who writes only 'resistance is bad' needs a prompt: 'Where did we find resistance useful, such as the filament?'
Model	Learners revisit their evolving	Say: 'Take out your model from	Cumulative Model Revision with	Look for models that: (1) show

Building Phase  VIDEO: Introduction to circuits and Ohm's law <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Internal Resistance Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	models of the phenomenon (the Nairobi tubes, the series and parallel bulbs, and the Christmas-light string). Individually they add or revise parts of the model: (1) label each bulb filament as a resistor with its own resistance (30 Ω for the demo bulbs); (2) add a note that a component's resistance depends on its length, cross-sectional area, material and temperature; (3) add a small sketch of a long thin extension cable heating up and a thick short copper wire staying cool; (4) add an arrow showing that a larger resistance means a smaller current for the same p.d. (from $R = V/I$). Then they write a short revised explanation: 'Why do bulbs differ from copper connecting wires?' and note what is still unexplained, namely how the resistances of several bulbs combine in series and in parallel. They share their models in pairs, giving one 'glow' (a strength) and one 'grow' (a suggestion). Finally they complete a summary table.	last lesson. What new idea from today can you add? Use a different colour so we can see the revision.' Provide the guiding prompts: 'What does each bulb in our demo have that a copper wire hardly has? How does that help explain why the current in the series circuit is small?' WAIT TIME 10-15 seconds. Circulate and ask 3 learners: 'Where in your model do you show that resistance depends on the wire?' Cold-call 3 learners to share their changes. Be careful not to let learners claim the resistances add in series, which belongs to Lesson 7; if they raise it, say, 'Good thinking, hold that idea. We will test it next lesson.' Collect the models and the exit ticket: 'In one sentence, explain why a long thin extension cable heats up more than a short thick one.'	Peer Glow-and-Grow Feedback (SEP 2: Developing and Using Models). Adding colour-coded revisions shows how understanding changes over time. Labelling each bulb as a resistor with its own resistance links the microscopic idea of resistance to the bench demo. Listing the unanswered questions gives learners a reason to move to Lesson 7.	each bulb as a resistance (30 Ω); (2) state that R depends on length, area, material and temperature; (3) show a long thin cable as high resistance and heating; (4) link a larger R to a smaller current using $R = V/I$. Exit ticket success criterion: the answer mentions both high resistance and energy transferred as heat. Common errors to note: confusing thickness with length, or claiming that resistance depends on the current. Re-teach in the next lesson's starter if more than a quarter of the class make such errors.
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D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners control variables when testing length, thickness, material and temperature, or did they change more than one factor at once? (2) Which factor was hardest to understand, and what evidence changed learners' minds? (3) Did learners hold the misconception that thicker wires resist more or that resistance depends on the current, and did their own data correct it? (4) How well did the CER explanations connect the long thin extension cable to resistance and heating? (5) Did the model revisions make the link between each bulb's 30 Ω resistance and the dim series bulbs? (6) Which DQB questions are ready for Lesson 7 (series resistances) and Lesson 11 (fuses and cable safety)? (7) Was the equipment (nichrome, copper and constantan wires, ammeters, voltmeters) enough for all groups, and were the low-voltage and short-circuit-time rules followed? Record learners who need extra support with calculating $R = V/I$.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We measured V and I for wires of different length, thickness and material, and for a wire that warmed up. We calculated $R = V/I$. We saw that longer wires and thinner wires gave a smaller current and so a higher R. Copper gave a low R, nichrome a higher R, and the bulb filament (30 Ω in our demo) had a much higher R than copper connecting wire. A warm metal wire let less current
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	flow.
What did I learn?	Resistance increases with the length of a conductor, decreases when the cross-sectional area is larger (thicker wire), depends on the material, and for a metal increases as the temperature rises. $R = V/I$ (Lesson 5) lets us compare resistances. A higher resistance means less current for the same p.d. and more energy transferred as heat in the conductor.
How does this explain the phenomenon?	A long thin extension cable has a high resistance, so when current flows more energy is transferred as heat and the cable warms up, unlike a short thick copper wire. Each bulb filament is a resistor with its own resistance (30 Ω in our demo), unlike copper connecting wires. This helps explain why the way bulbs are connected changes the current, but we still need to find out how resistances combine in series and in parallel (Lessons 7 and 8).

LESSON 7 (40 minutes (one double period may be used if the bench measurements run long)): Resistors in Series: Why Three Bulbs Share the Battery and Glow Dim

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To help learners discover, through measurement, that resistors connected in series add up to a larger total resistance, that the same current flows through each one, and that the battery's p.d. is shared between them, which explains why the three series bulbs glow dim and why one failed bulb switches everything off.
Knowledge	$R(\text{total}) = R_1 + R_2 + R_3$ for resistors in series. The current is the same at every point in the single loop. The p.d. across the components adds up to the source p.d. ($3.0\text{ V} + 3.0\text{ V} + 3.0\text{ V} = 9.0\text{ V}$). Using $V = IR$ with the demo data: $90\ \Omega$ gives $9/90 = 0.10\text{ A}$, and each bulb has 3.0 V and 0.30 W . If one bulb is removed or blown, the loop is open and $I = 0$.
Skills	Connecting an ammeter in series and a voltmeter in parallel; measuring current and p.d. in a series circuit; applying $V = IR$; calculating total resistance; analysing and interpreting data in a table; constructing an explanation using evidence and reasoning (NGSS SEP: Analysing and Interpreting Data; Using Mathematics; Constructing Explanations).
Attitudes	Curiosity about everyday electrical devices; accuracy and honesty when recording readings; teamwork and responsibility when handling the battery pack and meters; respect for others' ideas during discussion; safe practice with circuits.
Key Inquiry Question	1. How do we determine the relationship between current, potential difference and resistance in an electric circuit? 2. How is current electricity applied in day-to-day life?
Purpose in Storyline	Lessons 3, 4 and 5 gave us the tools: current (same everywhere in a series loop), p.d. (measured in parallel across a component) and $V = IR$. Lesson 6 showed that resistance depends on length, area, material and temperature. Today we use these tools to explain the first half of the puzzle: why three identical bulbs in series on a 9 V battery are dim. Lesson 8 will examine parallel circuits, which explains why the other tubes stay lit.
Safety Notes	Use only the 9 V battery pack (six 1.5 V dry cells); never connect to mains in this lesson. Do not short-circuit the battery with a bare wire, since the wire and cells will heat up. Connect the ammeter in series only, never directly across the battery. Switch off or disconnect the circuit between readings. Bulbs may become warm, so avoid touching them after long use. Keep water away from the benches.

B. LESSON OVERVIEW

Mr. Otieno opens by recalling the Nairobi evening prep class where one tube blew while the others stayed lit, and Wanjiru's question: why are the three series bulbs so dim? Learners first predict what will happen to the brightness and the current as bulbs are added one at a time to a series loop. They then build the series circuit (9 V pack, three identical 9 V , 2.7 W , $30\ \Omega$ bulbs), measure current with an ammeter and p.d. across each bulb with a voltmeter, and then remove one bulb. In the Explain phase they use $V = IR$ to derive $R(\text{total}) = R_1 + R_2 + R_3 = 90\ \Omega$, calculate $I = 0.10\text{ A}$, $V = 3.0\text{ V}$ per bulb and $P = 0.30\text{ W}$ per bulb, and connect this to the old string of Christmas lights wired in series. They update the Driving Question Board and add the series half of the explanation to their models.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Resistors in series Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Series and parallel resistors Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners look at a diagram of the 9 V pack with one 30 Ω bulb in a loop, then at a diagram with a second and a third identical bulb added in the same loop (series). Individually, in their notebooks, they predict and sketch: (a) will the bulbs be brighter, dimmer or the same as with one bulb? (b) will the current at the battery increase, decrease or stay the same? (c) what will happen to the other bulbs if one is unscrewed? Learners write a one-sentence reason for each prediction using words such as current, resistance and p.d. Kipchirchir and Wanjiru's earlier ideas are displayed for reference: Wanjiru asked why series bulbs are dim, and Kipchirchir thought the battery gets used up faster in parallel.	Say: "Remember our Nairobi classroom, where one tube blew and three stayed lit. Today we look at the opposite arrangement, the one that behaves like the old Christmas-light string at home. Before we measure anything, I want your best prediction." Draw the one-, two- and three-bulb loops on the board. Give 3 minutes of silent writing, then WAIT TIME of 10-15 seconds before calling on anyone. Cold-call 4 learners (including at least two who did not speak in the previous lesson) with: "What did you predict for the current when we add the third bulb, and why?" Do not say who is right. Record predictions in three columns on the board: brighter / same / dimmer. Ask: "Does adding more bulbs in one loop make it easier or harder for current to pass? What did we learn in Lesson 6 about what makes it hard?"	Predict-Observe-Explain (POE) with a confidence vote. Making a written commitment before data are shown exposes learners' preconceptions (for example, that current gets 'used up' by the first bulb, or that each added bulb adds more light). Learners later compare the evidence against these ideas, which makes the revision of thinking visible and memorable.	Look for: (1) learners who write 'current is used up as it passes each bulb' (misconception: current is consumed); (2) learners who link brightness to current or to the energy per second in a bulb; (3) learners who predict that removing one bulb leaves the others lit (they may be mixing up series with the classroom parallel tubes). Tally predictions on the board and note the number of learners in each group. Use a thumbs-up/middle/down confidence check: how sure are you of your prediction?
Observe Phase  VIDEO: Resistors in series - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Resistors in Series (Flexbook) Source: CK-12  Search ARES for similar readings	In groups of 4 with defined roles (circuit builder, ammeter reader, voltmeter reader, recorder), learners build the series circuit: 9 V pack, three identical 9 V, 2.7 V bulbs in one loop, with the ammeter in series. Step 1: Observe the brightness (dim). Measure the current at three different points: before bulb 1, between bulbs 1 and 2, and after bulb 3. Expected: 0.10 A at each point. Step 2: Use the voltmeter in parallel to measure the p.d. across bulb 1, bulb 2, bulb 3 and across the whole battery pack. Expected: 3.0 V, 3.0 V, 3.0 V and 9.0 V. Step 3: Add up the three p.d.s and compare with the battery p.d. Step 4: Remove one bulb (unscrew bulb 2) and observe all bulbs go out; ammeter reads 0 A. Record all	Circulate with the checklist. Before they connect the battery say: "Ammeter in series, voltmeter in parallel. Show me with your finger where the current has to go through the meter." Check each circuit before the battery is connected. Prompt: "Does the reading change when you move the ammeter to a different place in the loop? Why or why not?" and "What do your three bulb p.d.s add up to?" When bulbs are removed ask: "What has happened to the loop? What is the ammeter telling you?" Allow WAIT TIME of 10-15 seconds after each question before giving hints. Stop the groups after 20 minutes and cold-call 3 groups to read out their ammeter readings and p.d.s; write them in a class table and ask:	Structured data collection and pattern-finding using a class data table (NGSS SEP: Planning and Carrying Out Investigations; Analysing and Interpreting Data). Comparing readings from different locations helps learners see for themselves that current is conserved in a series loop and that the p.d. is divided, which builds the idea of sharing energy from the source.	Observable indicators: ammeter correctly placed in series and voltmeter correctly across each bulb; readings recorded with units (A, V); learners notice that the current is the same at all three points; learners notice that three 3.0 V readings add to 9.0 V; learners link bulb removal to an open loop and $I = 0$. Use a checklist to mark each group on correct connections, accurate recording and whether they can state one pattern in their own words.

	readings in a table: Measurement Reading. Learners also compare the series brightness with Mr. Otieno's parallel set-up for reference (brighter bulbs at 9.0 V each).	"Which numbers are the same? Which are different? Do the 3.0 V values add to 9.0 V?" Remind learners that meters are not perfect, so small differences like 0.09 A or 0.11 A are acceptable; discuss why.		
Explain Phase  VIDEO: Series resistors Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Resistors in Series Practice Source: CK-12  Search ARES for similar readings	Learners use their data to explain why the bulbs are dim. Step 1: Using $V = IR$ for the whole circuit, with $V = 9\text{ V}$ and $I = 0.10\text{ A}$, they calculate $R(\text{total}) = 9/0.10 = 90\ \Omega$. Step 2: Using $V = IR$ for one bulb (3.0 V , 0.10 A) they obtain $R = 3.0/0.10 = 30\ \Omega$, which matches the bulb rating ($R = V^2/P = 81/2.7 = 30\ \Omega$). Step 3: In pairs they show that $30 + 30 + 30 = 90\ \Omega$, and so for resistors in series $R(\text{total}) = R_1 + R_2 + R_3$. Step 4: Calculate power in each bulb, $P = IV = 0.10 \times 3.0 = 0.30\text{ W}$, and compare with the rating of 2.7 W at full 9 V , which gives dim glow. Step 5: Each learner writes a claim-evidence-reasoning explanation: Why are the three series bulbs dim? Why does one blown bulb put all of them out? They then apply this to a string of old Christmas lights wired in series, where one dead bulb breaks the loop so the whole string goes dark. Extension: If a fourth identical bulb were added, what would $R(\text{total})$, I and each bulb's p.d. become? ($120\ \Omega$, 0.075 A , 2.25 V .)	Say: "Wanjiru asked why the series bulbs are dim. Use only your measurements to answer her." Pose the question: "If the same current flows through all three bulbs and the 9 V is shared between them, how much does each bulb get?" WAIT TIME of 10-15 seconds, then cold-call 3 learners. Model the first calculation on the board ($R = V/I = 9/0.10 = 90\ \Omega$), then let pairs do the rest. Ask: "Where did the extra resistance come from, and how is it related to Lesson 6, where longer paths gave more resistance?" Ask: "Compare 0.30 W with 2.7 W . How many times smaller is it?" (9 times.) Cold-call 2 pairs to read their CER statements aloud and have the class judge them against a rubric: claim, evidence with numbers, reasoning with $V = IR$. Address misconceptions: if a learner says 'the first bulb uses up the current', point to the ammeter readings, which are equal at all three points. Link to home: "Why would an old string of Christmas lights go dark when one bulb fails?" Note: do not yet explain why the classroom tubes stay lit; say that Lesson 8 will deal with it.	Claim-Evidence-Reasoning (CER) with mathematical modelling (NGSS SEP: Using Mathematics and Computational Thinking; Constructing Explanations). Moving from the measured numbers to the general rule $R(\text{total}) = R_1 + R_2 + R_3$ lets learners generate the formula from evidence rather than receiving it, and applying it to the Christmas-light string transfers the idea to a familiar Kenyan context.	Check that learners: (1) calculate $R(\text{total}) = 90\ \Omega$ correctly with units; (2) state that the current is the same and the p.d. is shared (3.0 V each); (3) calculate 0.30 W per bulb and relate it to dim glow; (4) explain that a broken bulb opens the single loop so $I = 0$; (5) use a CER structure with numbers as evidence. Use a 3-level rubric: 1 = states the observation only; 2 = adds a number or a formula; 3 = links the numbers to the dim glow and to the open loop. Exit-ticket question: 'Two $30\ \Omega$ resistors are in series on a 9 V battery. Find $R(\text{total})$, I and the p.d. across each.' ($60\ \Omega$, 0.15 A , 4.5 V .)
Driving Question Board (DQB) Creation  VIDEO: 01 Resistors In	Learners return to the Driving Question Board: 'Why do the other tubes stay lit when one blows out, and why do bulbs connected in series glow dimmer than identical bulbs connected in parallel to the	Say: "Look at our Driving Question Board. Which of our questions can we now answer with evidence we collected today, and which are still waiting?" Give 3 minutes for writing, then WAIT TIME of 10-15	Question tracking with a public Driving Question Board. Learners sort questions into answered and unanswered categories, which makes progress visible, links each lesson to the central phenomenon	Look at the sticky notes: do answered questions cite specific evidence (0.10 A , 3.0 V , $90\ \Omega$, 0.30 W)? Do new questions focus on parallel connection, current in branches or battery life? Note

Series Source: Seavuria Search ARES for similar videos HTML: Series Circuits: Voltage Drops (PLIX) Source: CK-12 Search ARES for similar readings	same battery?' In groups they write on sticky notes (a) one question that today's evidence has answered (for example, 'Why are the series bulbs dim?'), with the evidence summarised in a few words (90 Ω, 0.10 A, 3.0 V each, 0.30 W), and (b) one new question that has arisen (for example, 'What happens to the current when the bulbs are connected side by side?' or 'Why does the parallel set give 0.90 A from the battery?'). They stick the notes on the board, moving answered questions to an 'Answered/Partly answered' column. Kipchirchir's idea that the battery runs out faster in parallel is added as a question to test in Lessons 8 and 10.	seconds. Cold-call 3 groups to read one note aloud and say where it should go on the board. Ask: "Does this fully answer the question about the tubes in the Nairobi classroom? What is missing?" Guide learners to see that the series half is answered while the parallel half (why the other tubes stay lit) still needs evidence. Highlight any question about whether bulbs in parallel use more current or drain the battery faster, and tell the class: "Kipchirchir, your idea is partly right and we will test it in Lesson 8."	and creates a purpose for the next investigation (NGSS SEP: Asking Questions and Defining Problems).	groups that only repeat earlier questions and prompt them to refine. Record the number of learners who contribute at least one note.
Model Building Phase VIDEO: Resistors in series Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Series and parallel resistors Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners revise their individual models of the phenomenon (the original diagrams from Lesson 1, updated in later lessons). For the series half they add: (1) a circuit diagram of the 9 V battery with three 30 Ω bulbs in one loop, labelled with $I = 0.10$ A at every point; (2) p.d. labels of 3.0 V across each bulb that add up to 9.0 V; (3) $R(\text{total}) = 30 + 30 + 30 = 90 \Omega$; (4) power of 0.30 W per bulb, so dim glow; (5) a note showing that if one bulb is removed, the loop is open and $I = 0$. They also draw the Christmas-light string as a series loop with the failed bulb marked. In a different colour they leave a space labelled 'Parallel: what happens here? (Lesson 8)'. Finally, they complete the Summary Table row for this lesson.	Say: "Open your model from last lesson. In a new colour, add what today's evidence tells you about the series bulbs. Every arrow and number must be backed by a measurement." Circulate, and give 5 minutes for drawing. Ask: "Where in your model can I see that the current is the same everywhere? Where can I see the p.d. being shared?" WAIT TIME of 10-15 seconds, then cold-call 3 learners to explain one part of their models. Have two learners swap models and give one star (what is correct) and one wish (what is missing). Close with: "We can now explain why the series bulbs are dim. In Lesson 8 we will find out why the parallel bulbs are bright and why the other tubes stay lit."	Cumulative model revision with peer feedback (NGSS SEP: Developing and Using Models). Adding evidence-based annotations to a model that has been developing since Lesson 1 shows learners how explanations grow, and the deliberate gap for the parallel circuit highlights what remains to be explained.	Check that revised models include: a single-loop circuit with correct symbols; $I = 0.10$ A at every point; 3.0 V across each bulb and 9.0 V total; $R(\text{total}) = 90 \Omega$; 0.30 W per bulb; open loop with $I = 0$ when a bulb is removed. Mark models as Complete / Partial / Missing for each element. Collect the Summary Table entries and note learners who still say that current is used up or that the first bulb gets more of the voltage.

D. TEACHER REFLECTION

1. Did learners measure the same current at every point in the series loop, and did they accept it, or did some still believe the current is used up? 2. Which groups found it hard to place the ammeter in series and the voltmeter in parallel, and what support helped? 3. How many learners could derive $R(\text{total}) = R_1 + R_2 + R_3$ from the data rather than just recite it? 4. Were the CER explanations linked to numbers (90 Ω , 0.10 A, 3.0 V, 0.30 W)? 5. Which learners used the Christmas-light string to explain an open loop? 6. What new questions on the DQB point towards parallel circuits for Lesson 8, and how will I use Kipchirchir's idea about the battery to start it?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	In the series circuit of three identical 30 Ω bulbs on the 9 V pack, the bulbs glow dim, the ammeter reads 0.10 A at every point, the voltmeter reads 3.0 V across each bulb and 9.0 V across the battery, and when one bulb is removed all bulbs go out and the current is 0 A.
What did I learn?	For resistors in series, $R(\text{total}) = R_1 + R_2 + R_3$ (30 + 30 + 30 = 90 Ω). The current is the same everywhere in the loop ($I = 9/90 = 0.10$ A), and the battery p.d. is shared between the bulbs (3.0 V each, adding to 9.0 V). Each bulb transfers $P = IV = 0.30$ W.
How does this explain the phenomenon?	The three series bulbs are dim because their resistances add up to 90 Ω , which limits the current to 0.10 A and gives each bulb only 3.0 V and 0.30 W instead of the 2.7 W it gets at full 9 V. In a series loop, a failed or removed bulb opens the only path so $I = 0$ and everything goes out, as in an old string of Christmas lights. Why the classroom tubes and parallel bulbs behave differently is still to be investigated in Lesson 8.

LESSON 8 (80 minutes (2 x 40-minute periods)): Resistors in Parallel: Why One Failed Tube Does Not Darken the Others

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To use evidence from the 9 V bench demo and the Nairobi classroom to show that each parallel branch has the same p.d., that branch currents add up, and that the effective resistance is found from $1/R = 1/R_1 + 1/R_2 + 1/R_3$. Learners use this to explain why the other tubes stay lit when one blows and why parallel bulbs are brighter than series bulbs.
Knowledge	Learners will: (1) state that components in parallel have the same p.d. across each branch; (2) state that the total current equals the sum of the branch currents; (3) calculate effective resistance using $1/R = 1/R_1 + 1/R_2 + 1/R_3$, getting $10\ \Omega$ for three $30\ \Omega$ bulbs; (4) explain that the effective resistance of the parallel combination is less than the smallest branch resistance; (5) explain that when one branch is removed the other branches keep the same p.d. and current, and only the total current falls (0.90 A to 0.60 A in the bench demo; 0.60 A to 0.45 A in the classroom).
Skills	Planning and carrying out investigations (NGSS SEP 3): connecting ammeters in series and voltmeters in parallel. Analysing and interpreting data (SEP 4): comparing series and parallel tables. Using mathematics (SEP 5): applying $V = IR$ and the reciprocal formula. Developing and using models (SEP 2): drawing a water-pipe branching analogy and a circuit diagram. Constructing explanations (SEP 6).
Attitudes	Curiosity about why everyday lighting is wired the way it is; honesty in recording readings exactly as measured; respect for others' ideas during Talk Moves; a responsible attitude toward electrical safety at home and school.
Key Inquiry Question	1. How do we determine the relationship between current, potential difference and resistance in an electric circuit? 2. How is current electricity applied in day-to-day life?
Purpose in Storyline	Lesson 7 showed why series bulbs are dim (90 Ω , 0.10 A, 3.0 V each, 0.30 W each, and $I = 0$ if one is removed). Lesson 8 supplies the contrasting evidence for the second half of the driving question: in parallel each bulb has the full 9.0 V and 0.30 A, the effective resistance falls to $10\ \Omega$, and a failed branch does not affect the others. This gives learners the core mechanism for the Nairobi classroom tubes. Lesson 9 will then combine series and parallel rules, and Lesson 10 will use the branch currents to calculate power.
Safety Notes	Use only the 9 V battery pack (six 1.5 V cells) in class; never use mains (240 V) for the investigation. Connect the ammeter in series and the voltmeter in parallel, and never connect an ammeter directly across the battery. Disconnect the battery before changing the circuit. Bulbs and the $30\ \Omega$ resistor may become warm, so do not touch them after long use. Do not tamper with sockets or tubes in the classroom; the tube discussion is by data and diagram only.

B. LESSON OVERVIEW

Mr. Otieno reminds the class of the Nairobi evening prep scene: one of four fluorescent tubes blows and the other three stay lit. Learners first predict what happens to the brightness, the p.d. and the current when bulbs are connected in parallel to the same 9 V pack. They then measure the p.d. across, and the current through, each of three identical $30\ \Omega$ bulbs on separate branches. They find 9.0 V across each bulb, 0.30 A in each branch and 0.90 A from the battery. They compare this with the series results from Lesson 7 (0.10 A, 3.0 V each). In the Explain phase they derive the effective resistance of $10\ \Omega$, first from $R = V/I = 9/0.90$ and then from $1/R = 3/30$, and they show that removing one bulb leaves the others at 9.0 V and 0.30 A while the battery current falls to 0.60 A. They then scale this up to the classroom tubes (0.15 A each, 0.60 A, then 0.45 A at 240 V) and explain why the series Christmas-light string goes dark when one bulb fails. The DQB and the class model are updated with parallel evidence.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Parallel resistors (part 1) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Resistors in Parallel (Flexbook) Source: CK-12  Search ARES for similar readings	(10 min) Learners look at a diagram of the Nairobi classroom with 4 tubes and at the bench demo diagram with three identical 9 V, 2.7 W bulbs on separate branches of the 9 V pack. Individually, in their notebooks, they predict: (a) will the parallel bulbs be brighter, dimmer or equal to the series bulbs? (b) what p.d. will be across each bulb? (c) what will the ammeter at the battery read compared with 0.10 A in series? They draw the path of charge from the + terminal and sketch what happens when one bulb is removed. Kipchirchir's idea is shared: 'The battery will get used up faster in parallel.' Learners decide whether they agree and why. Pairs compare predictions and record one claim with a reason on a sticky note.	Display both circuit diagrams. Say: 'In Lesson 7 each series bulb had 3.0 V and 0.10 A. Before we measure anything, predict what happens when the three bulbs are on separate branches.' Give 2 minutes of silent writing. Then say: 'Turn to your partner and share your prediction and your reason.' WAIT TIME: after asking 'What do you think the voltmeter will read across one parallel bulb?' wait 10-15 seconds before taking answers. Cold-call 4 learners (including at least 2 who did not volunteer) to share predictions without evaluating them. Read Kipchirchir's prediction aloud and ask: 'What would we need to measure to test that?' Record all predictions on the board under the headings Brightness, p.d., Current.	Prediction with reasoning (Predict-Observe-Explain): committing to a written prediction before the data makes learners notice where their ideas differ from the evidence. Reasoning about the Kipchirchir idea turns a partly correct everyday idea into a testable claim about battery current.	Check notebooks while circulating. Look for: learners who predict equal 3.0 V across parallel bulbs (carry-over from series, a misconception to address); learners who write that the current is the same as in series (0.10 A); learners who say the other bulbs go out when one is removed. Tally predictions on the board (e.g. brighter / same / dimmer) and note how many learners link brightness to p.d. or current.
Observe Phase  VIDEO: Parallel resistors (part 3) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Resistors in Parallel Practice Source: CK-12  Search ARES for similar readings	(25 min) In groups of 4, learners set up the parallel circuit: the 9 V pack (six 1.5 V cells) with three identical bulbs (9 V, 2.7 W, 30 Ω each) on separate branches. Roles: builder, ammeter reader, voltmeter reader, recorder. Step 1: they note the brightness against the series circuit from Lesson 7. Step 2: they use the voltmeter (in parallel) across each bulb and across the battery and record the p.d. (expect 9.0 V each). Step 3: they place the ammeter (in series) in each branch (expect 0.30 A each) and then at the battery (expect 0.90 A). Step 4: they remove one bulb and re-read: the other two stay bright at 9.0 V and 0.30 A each, and the battery	Circulate with a checklist. Before the groups switch on, ask each group: 'Show me where the ammeter is. Is it in series? Where is the voltmeter?' Prompt: 'Is it the same p.d. across each bulb, or does it share out like in Lesson 7?' If a group reads about 0.90 A in one branch, say: 'Where did you place the ammeter? What does it measure there?' Run a mid-point stop at about 12 minutes: 'Freeze. Everyone look at the readings for one branch. Hold up your recorder's table.' Before removing a bulb, ask: 'Write what you expect the other two bulbs and the battery ammeter to do.' WAIT TIME: after the bulb is removed, wait 10-15 seconds in silence while learners	Structured data collection with a comparison table (Analysing and interpreting data): placing series and parallel results side by side makes the pattern visible (same p.d. in parallel vs shared p.d. in series; currents add in parallel vs same current in series). Using the same bulbs and the same battery isolates the connection pattern as the only variable.	Observable indicators: the ammeter is in series and the voltmeter in parallel in every group; recorded values are close to 9.0 V per bulb, 0.30 A per branch, 0.90 A total, and 0.60 A with one bulb removed; the table is completed in both columns; groups can state without prompting that the other bulbs did not change when one was removed. Note groups that report current as 'used up' in a bulb and revisit this in the Explain phase.

	current falls to 0.60 A. They fill in a table comparing series and parallel: total R, battery current, p.d. per bulb, current per bulb, brightness. They also record what happens to the series circuit when one bulb is removed (all out, $I = 0$, from Lesson 7). They then examine the Christmas-light string drawn in series and the classroom tubes drawn in parallel.	observe and record. Cold-call 3 groups to report their battery current with one bulb removed (expect 0.60 A). Show the classroom data on the board: 4 tubes, 36 W each, 240 V, 0.15 A each, 0.60 A total, then 0.45 A.		
Explain Phase  VIDEO: Resistors in parallel Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Parallel Circuits (Flexbook) Source: CK-12  Search ARES for similar readings	(25 min) Learners use their data to build the explanation in four steps. (1) Same p.d.: each branch connects directly across the battery terminals, so each bulb has 9.0 V. (2) Currents add: $I(\text{total}) = I_1 + I_2 + I_3 = 0.30 + 0.30 + 0.30 = 0.90$ A, as measured. (3) Effective resistance: $R = V/I = 9.0/0.90 = 10$ Ω, and they check by calculating $1/R = 1/30 + 1/30 + 1/30 = 3/30$, so $R = 10$ Ω. They discuss why 10 Ω is smaller than any one 30 Ω bulb: extra branches give the current more paths. (4) Power link: each parallel bulb has $P = 9.0 \times 0.30 = 2.7$ W (full brightness), compared with 0.30 W in series, which is 9 times greater. They then work on the classroom: each tube draws $36/240 = 0.15$ A, four tubes give 0.60 A, and with one blown the other three each still have 240 V and 0.15 A, so the total falls to 0.45 A. They explain, in a written paragraph with a labelled diagram, why the Christmas-light string (series) goes dark when one bulb fails but the classroom lights stay on. Finally they respond to Kipchirchir's idea: the parallel circuit draws 0.90 A from the battery, compared with 0.10 A in series, so he was partly right.	Lead a Talk Moves discussion. Ask: 'Why does each bulb get the full 9.0 V in parallel but only 3.0 V in series?' WAIT TIME: 10-15 seconds, then ask learners to turn and talk for 1 minute. Cold-call 3 learners with the follow-up: 'Say more about what the branch is connected across.' Model the calculation of $1/R$ on the board, then have learners work the classroom tubes in pairs. Ask: 'If $1/R = 3/30$, what is R? Does it make sense that R is less than 30 Ω?' Cold-call 2 pairs to explain the logic. Use a water-pipe analogy: three pipes from one tank, each pipe gets the same pressure, and the total flow is the sum. Name its limits: it does not show that the charges are already in the wire. Ask: 'Wanjiru asked why the series bulbs were dim. Use 0.30 W and 2.7 W to answer her.' Give the sentence frame: 'The tubes stay lit because ___, so each tube still has ___ V and ___ A, and only the ___ current changes.' Collect two paragraphs for feedback. Address Kipchirchir: 'Which circuit draws more current from the battery? What does that mean for battery life?' Note that real filaments get hotter and their resistance rises, so the 30 Ω value is a simplification.	Claim-Evidence-Reasoning (CER) with mathematical modelling (Using mathematics, SEP 5): learners state a claim, support it with measured values (9.0 V, 0.30 A, 0.90 A, 10 Ω) and reason with $V = IR$ and the reciprocal formula. A water-pipe analogy supports the ideas of same pressure in branches and flow adding up, and its limits are discussed.	Collect the CER paragraphs and check for: (1) a claim that parallel branches share the same p.d.; (2) correct use of 0.30 A per branch, 0.90 A total and 10 Ω; (3) a correct explanation that one failed tube leaves the others at 240 V and 0.15 A and the total at 0.45 A; (4) a contrast with the single series loop that opens. Exit ticket question: 'Three 30 Ω bulbs are in parallel across 9 V. What is the current in each, the total current and R? What is the total current if one is removed?' (Answers: 0.30 A, 0.90 A, 10 Ω, 0.60 A.) Sort responses into secure, partly secure and not yet secure, and plan a short reteach for the last group.

Driving Question Board (DQB) Creation  VIDEO: Resistors in parallel - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Parallel Circuits Practice Source: CK-12  Search ARES for similar readings	(10 min) Learners return to the class Driving Question Board. In groups they review the questions posted in earlier lessons and decide which ones can now be answered with evidence from Lesson 8 (for example, 'Why do the other tubes stay lit?', 'Why are parallel bulbs brighter?'). They move answered questions to an 'Answered by Evidence' area and write the evidence on a card (e.g. 'Each branch has 9.0 V and 0.30 A, so one failed branch leaves the others unchanged'). They add new questions, for example: 'What if we mix series and parallel in one circuit?' (Lesson 9), 'How much does it cost to run four tubes?' (Lesson 10), 'What stops a wire overheating if too many appliances are on one socket?' (Lesson 11), and 'Why does the battery run down faster in parallel?'	Say: 'Look at the DQB. Which questions do we have strong evidence for now? Which new questions did today's data raise?' Give each group 3 minutes to discuss and attach cards. WAIT TIME: after asking 'Which question is still hardest to answer?' wait 10-15 seconds. Cold-call 3 groups to read out a card and the evidence behind it. Group questions by theme (brightness, failure, energy, safety) and say which upcoming lessons will address them. Make sure Kipchirchir's battery question is recorded and linked to the 0.90 A vs 0.10 A readings.	Driving Question Board as a coherence tool: linking each question to specific evidence shows learners how their understanding has grown and keeps lessons connected to the anchoring phenomenon. New questions point to the next investigations.	Observe whether the evidence on the cards cites actual measurements (9.0 V, 0.30 A, 0.90 A, 10 Ω, 0.60 A, 0.45 A) rather than general statements. Check whether new questions are testable and connected to the phenomenon (energy, cost, safety). Note groups that move a question without evidence and ask them to justify it.
Model Building Phase  VIDEO: Parallel resistors (part 2) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Resistors in Parallel (Flexbook) Source: CK-12  Search ARES for similar readings	(10 min) Learners revise their individual models of the phenomenon. They draw (1) the series circuit with three bulbs: 90 Ω, 0.10 A, 3.0 V each; (2) the parallel circuit with three branches: 9.0 V across each, 0.30 A in each branch, 0.90 A in the battery, 10 Ω effective; and (3) the Nairobi classroom with four tubes on parallel branches across 240 V, with 0.15 A each and 0.60 A total, then 0.45 A after one blows. They label p.d. and current on each branch, add a short caption ('Each branch has its own path and the same p.d.'), and note what remains unexplained, such as the cost of energy, the combination of series and parallel in one circuit, and safety of the whole wiring system. They add a note about why the Christmas-light string in	Say: 'Add what you learned today to your model. Show where the current splits and where it joins again, and show the p.d. across each branch.' Circulate and ask: 'Where on your diagram does the failed tube sit? Show me how the current reaches the other tubes.' WAIT TIME: after asking 'What would your model predict if we removed two bulbs?' wait 10-15 seconds. Cold-call 3 learners to explain their model to the class. Prompt: 'What does the model still not explain?' Close by asking each learner to add one sentence to their model that connects 'same p.d. in each branch' to 'the other tubes stay lit'. Preview Lesson 9: 'Next time we will combine series and parallel in one circuit.'	Cumulative model revision (Developing and using models, SEP 2): learners update the same model each lesson, showing quantities on every branch. Seeing the series and parallel models side by side allows them to use their model to predict what happens when a component fails.	Check each model for: separate branches directly across the source; 9.0 V on every parallel bulb and 240 V on every tube; currents labelled and adding at the junction ($0.30 + 0.30 + 0.30 = 0.90$ A; $0.15 \times 4 = 0.60$ A); the correct change after a failure (0.60 A and 0.45 A); and a contrast with the series loop. Use a quick rubric: complete, mostly complete, or needs support. Offer a worked example to learners who still draw the failed bulb breaking the whole circuit.

	series goes dark.			
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D. TEACHER REFLECTION

After the lesson, review the exit tickets and models. How many learners can state that each parallel branch has the same p.d., and how many still think the p.d. shares out as in series? Did groups connect the ammeter and voltmeter correctly? Which learners can explain why the effective resistance ($10\ \Omega$) is smaller than $30\ \Omega$ without just quoting the formula? Did anyone confuse the total current falling to $0.60\ \text{A}$ with the branch currents changing? Was there enough time for the Explain phase and the classroom tubes ($0.15\ \text{A}$ each, $0.60\ \text{A}$, then $0.45\ \text{A}$)? Reflect on whether the water-pipe analogy helped or caused confusion about current being 'used up'. Plan targeted support before Lesson 9 for learners who need more practice with the reciprocal formula, and note which DQB questions to prioritise next.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We connected three identical $9\ \text{V}$, $2.7\ \text{W}$ ($30\ \Omega$) bulbs on separate branches of the $9\ \text{V}$ pack. Each had $9.0\ \text{V}$ across it and $0.30\ \text{A}$ through it, the battery current was $0.90\ \text{A}$ and all bulbs glowed brightly ($2.7\ \text{W}$ each). When one bulb was removed, the other two stayed bright at $9.0\ \text{V}$ and $0.30\ \text{A}$ and the battery current fell to $0.60\ \text{A}$. In the Nairobi classroom, each tube draws $0.15\ \text{A}$, the four draw $0.60\ \text{A}$, and with one blown the other three draw $0.45\ \text{A}$.
What did I learn?	In a parallel circuit each branch has the same p.d. as the source, the branch currents add up to the total current, and the effective resistance is found from $1/R = 1/R_1 + 1/R_2 + 1/R_3$, which gives $10\ \Omega$ for three $30\ \Omega$ bulbs. The effective resistance is less than the smallest branch resistance. A break in one branch does not change the p.d. or current in the other branches.
How does this explain the phenomenon?	The other tubes stay lit because each tube is on its own branch directly across the $240\ \text{V}$ supply, so when one branch is broken the others still have the same voltage and current ($0.15\ \text{A}$ each). Parallel bulbs glow brighter than series bulbs because each gets the full $9.0\ \text{V}$ and $0.30\ \text{A}$ ($2.7\ \text{W}$), compared with $3.0\ \text{V}$, $0.10\ \text{A}$ and $0.30\ \text{W}$ in series. In a series string such as old Christmas lights, one failed bulb opens the only loop, so the whole string goes dark.

**LESSON 9 (40 minutes (single lesson; extend to a double lesson of 80 minutes if students build the mixed circuit on the bench)):
Combined Circuits and Cells: Reducing a Mixed Circuit and Powering Homes from Kisumu to Kericho**

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students reduce a mixed circuit step by step using the series rule (L7) and the parallel rule (L8), and compare cells connected in series (e.m.f. adds) with cells connected in parallel (same e.m.f. for longer). They use this to move closer to explaining why real lighting systems, such as the Nairobi classroom tubes or Achieng's 12 V solar home system, are wired the way they are.
Knowledge	Learners will know that (1) a mixed circuit is solved by first reducing each parallel group to one equivalent resistance, then adding the series parts; (2) for one 30 Ω bulb in series with two 30 Ω bulbs in parallel on a 9 V battery, the parallel pair is 15 Ω , total $R = 45 \Omega$ and $I = 9/45 = 0.20 \text{ A}$; the series bulb has 6.0 V and each parallel bulb has 3.0 V and 0.10 A ($6.0 + 3.0 = 9.0 \text{ V}$); (3) each dry cell is 1.5 V, three cells in series give 4.5 V, six in series give 9.0 V; (4) cells in parallel keep 1.5 V but last longer; (5) Achieng's family solar home system in Kisumu uses a 12 V battery.
Skills	Analysing and interpreting data, using mathematics and computational thinking (Ohm's law $V = IR$ with the series and parallel rules), developing and using models (circuit diagrams and redrawn equivalent circuits), constructing explanations, and carrying out an investigation by measuring current with an ammeter in series and p.d. with a voltmeter in parallel.
Attitudes	Learners appreciate how step-by-step reasoning helps solve real wiring problems at home and at school, show curiosity about solar home systems in Kenyan homes, work with responsibility and respect when sharing roles and ideas in groups, and demonstrate careful, safe handling of batteries and meters.
Key Inquiry Question	How do we determine the relationship between current, potential difference and resistance in an electric circuit? How is current electricity applied in day-to-day life?
Purpose in Storyline	Lessons 7 and 8 gave the series rule ($R = R_1 + R_2 + R_3$, same current, p.d. shared) and the parallel rule (same p.d. on each branch, currents add, $1/R = 1/R_1 + 1/R_2 + 1/R_3$). Lesson 9 combines them in a mixed circuit and adds the idea that the source itself can be arranged (cells in series or parallel). This prepares Lesson 10 (energy and power) and shows students that the dim series bulbs, the bright parallel bulbs and a mixed circuit are all explained by the same few rules.
Safety Notes	Use only the 9 V battery pack (six 1.5 V dry cells) and 9 V, 2.7 W bulbs; never use mains 240 V sockets for the bench work. Connect the ammeter in SERIES and the voltmeter in PARALLEL; never connect an ammeter directly across a battery (short circuit, hot wires, flat cells). Disconnect the battery between readings because bulbs and wires can become warm. Do not touch bare wire ends together across the pack. Dispose of old dry cells responsibly and do not mix old and new cells. Mr. Otieno checks every circuit before students close the switch.

B. LESSON OVERVIEW

Mr. Otieno returns to the Nairobi evening prep class where one of four fluorescent tubes blew and the other three stayed lit, and to the 9 V bench demo where three series bulbs were dim (90 Ω , 0.10 A, 3.0 V each, 0.30 W each) and three parallel bulbs were bright (10 Ω , 0.30 A each, 2.7 W each). Today he asks: what if a circuit has BOTH series and parallel parts, and what if we change the source instead of the bulbs? Students predict the brightness of a mixed circuit (one 30 Ω bulb in series with two 30 Ω bulbs in parallel), then measure it, reduce it step by step (15 Ω , then 45 Ω , then 0.20 A, then 6.0 V and 3.0 V) and explain it. They then compare cells in series (1.5 V, 4.5 V, 9 V) and cells in parallel (1.5 V that lasts longer), building on Kipchirchir's prediction that parallel circuits use up the battery faster, and link the ideas to Achieng's 12 V solar home system in Kisumu. Finally they add to the DQB and revise their models.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Simplifying resistor networks Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Combined Series-Parallel Circuits (Flexbook) Source: CK-12  Search ARES for similar readings	Students look at a diagram on the board of a MIXED circuit: one 30 Ω bulb (A) in series with a pair of 30 Ω bulbs (B and C) in parallel, all on the 9 V pack. Individually, they write a prediction in their notebooks: (a) Which bulb will be brightest? (b) Will bulb B be as bright as in the full parallel demo, where each bulb got 9.0 V? (c) What will happen to the others if bulb B is removed? They then sketch what they think the p.d. across each bulb is. Next, they predict what happens to the brightness if the 9 V pack is replaced by only three 1.5 V cells (4.5 V). They connect this to the phenomenon: the series bulbs were dim and the parallel bulbs were bright, so where does a mixed circuit fit?	Display the mixed circuit diagram and say: "Last week we met two patterns. Today there is a third pattern that has both. Before I show you any numbers, I want your best prediction." Ask: "Which bulb do you think is brightest, A, B or C, and why?" Give WAIT TIME of 10-15 seconds in silence, then have students write for 2 minutes. Cold-call 4 students (include Wanjiru's group, who asked why series bulbs are dim) to share predictions without saying right or wrong; record each on the board using a tally. Say: "Wanjiru asked why series bulbs were dim. Use what you know about sharing p.d. to give a reason for your prediction." Ask: "If I remove bulb B, what do you think happens to A and C?" WAIT TIME of 10 seconds, cold-call 2 more students.	Prediction with reasoning (Predict-Observe-Explain): by committing to a prediction and a reason in writing before seeing data, students surface their ideas from L7 and L8 (series shares p.d.; parallel gives each branch the full p.d.), which they can later confirm, refine or revise. The tally on the board makes the class's range of ideas visible.	Look for: (1) whether students write a reason using p.d. or current rather than only 'it is parallel so it is bright'; (2) whether students recognise that A is in the main loop and carries the current of both B and C; (3) common misconceptions to note: 'the first bulb uses up the current', 'all bulbs in the circuit get 9 V', 'removing B puts out everything'. Tally the number of students who predict A brightest (about 30-50% expected) and note the reasons given.
Observe Phase  VIDEO: Simplifying resistor networks Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Combined Series-Parallel Circuits Practice Source: CK-12  Search ARES for similar readings	In groups of four with roles (builder, ammeter reader, voltmeter reader, recorder), students build the mixed circuit from the 9 V pack (six 1.5 V dry cells in a holder) and three identical 9 V, 2.7 W bulbs. They observe the brightness of A, B and C. They measure: (1) the current from the battery with the ammeter in SERIES (expected 0.20 A); (2) the p.d. across A (expected 6.0 V) and across B and across C with the voltmeter in PARALLEL (expected 3.0 V each); and (3) the current in B or C (expected 0.10 A each). They record results in a	Circulate and say: "Ammeter in series, voltmeter in parallel. Show me where the meter sits on your circuit diagram before you close the switch." Check every circuit before the switch is closed. Ask each group: "Which bulb is brightest? Point to it. Now use the voltmeter to tell me why." Pause: "Predict first: if bulb B is removed, what will the ammeter show?" WAIT TIME of 10 seconds, then have them test. Ask 3 cold-call groups to read their values aloud and write them on the board in one class table (battery current 0.20 A; V across A 6.0 V; V across B and	Data collection with structured tables and a class data pool: groups record in the same table so patterns (A has more p.d. than B or C; the currents in B and C add up to the current in A) are visible across groups. Pooling results reduces the effect of measurement error and shows the pattern is general.	Observe: (1) meters connected correctly (ammeter in series, voltmeter across the component) in at least 80% of groups; (2) groups record values near 0.20 A, 6.0 V, 3.0 V, 0.10 A (allow ± 0.02 A, ± 0.3 V); (3) students say that C stays lit when B is removed while the brightness of A changes; (4) students state that three cells in series gave 4.5 V. Prompt groups that read a voltmeter in series or that confuse e.m.f. with p.d.

	table. They then unscrew bulb B and observe that C stays lit but A changes (the current falls). Next, they compare cells: they measure the e.m.f. of one 1.5 V cell, then three cells in series (4.5 V), then all six in series (9 V), then two cells in parallel (1.5 V). They watch a bulb connected to 4.5 V versus 9 V. Mr. Otieno also shows Achieng's photograph/description of the 12 V battery used in her Kisumu solar home system, and a card describing how long cells in parallel last.	C 3.0 V each; current in B and C 0.10 A each). For cells say: "Kipchirchir predicted that the battery gets used up faster in parallel. Hold that idea. We will test it with numbers." Remind groups to disconnect between readings.		
Explain Phase  VIDEO: Parallel resistors (part 3) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Combined Series-Parallel Circuits (Flexbook) Source: CK-12  Search ARES for similar readings	Students explain the data by reducing the mixed circuit step by step on mini whiteboards, using the same sequence each time: STEP 1: reduce the parallel pair B and C: $1/R = 1/30 + 1/30 = 2/30$, so $R = 15\ \Omega$. STEP 2: redraw the circuit as A (30 Ω) in series with one 15 Ω resistor and add: $R\text{ total} = 30 + 15 = 45\ \Omega$. STEP 3: find the main current $I = V/R = 9/45 = 0.20\text{ A}$. STEP 4: find the p.d. across A: $V = IR = 0.20 \times 30 = 6.0\text{ V}$. STEP 5: the rest of the battery p.d. is across the parallel pair: $9.0 - 6.0 = 3.0\text{ V}$ (check: $0.20 \times 15 = 3.0\text{ V}$). STEP 6: split the current: each identical branch carries $3.0/30 = 0.10\text{ A}$ ($0.10 + 0.10 = 0.20\text{ A}$). They compare to measured values. They then explain why bulb A is brightest (power $6.0 \times 0.20 = 1.2\text{ W}$ against $3.0 \times 0.10 = 0.30\text{ W}$ for B or C; teacher notes power fully in L10, so only the comparison 'more p.d. and more current' is used here). For cells, they explain that cells in series add e.m.f. ($3 \times 1.5 = 4.5\text{ V}$; $6 \times 1.5 = 9\text{ V}$), that cells in parallel keep 1.5 V, and that each cell supplies only a share of the current so the set	Model Step 1 only on the board: "Watch me reduce the parallel pair. I never add 30 and 30 here. I use 1 over R." Then say: "You do Step 2 on your whiteboards. Show me in 60 seconds." Use WAIT TIME of 10-15 seconds before calling on students for each step. Cold-call 3 students to present Steps 2, 3 and 4 and ask the class: "Thumbs up if you agree, sideways if unsure." Ask: "Why is the p.d. across A bigger than across B?" WAIT TIME 15 seconds, then cold-call 2 students. Say: "Compare 0.20 A on the ammeter to our calculated 0.20 A. What does that tell you about our model?" For cells ask: "Kipchirchir said the battery gets used up faster in parallel. What do the currents 0.90 A and 0.10 A tell us?" Cold-call 2 students. Ask: "Achieng's family uses a 12 V battery. How many 1.5 V cells in series would give 12 V?" (8) and "Why would they wire the home lights in parallel?" Cold-call 2 students. Do not introduce power calculations beyond the comparison (L10 teaches $P = IV$).	Worked-example fading with a reduction-and-redraw model: the teacher models one step, students complete the rest, so students learn a transferable procedure (reduce parallel groups, add series parts, then work back for current and p.d.) rather than a single answer. Claim-Evidence-Reasoning is used for the cells question: claim (cells in series add e.m.f.), evidence (1.5 V, 4.5 V, 9 V readings), reasoning (each cell adds energy per unit charge, $1\text{ V} = 1\text{ J/C}$).	Collect whiteboards. Look for: (1) $1/R$ used correctly for the parallel pair giving 15 Ω, not 60 Ω or 30 Ω; (2) total 45 Ω, not 75 Ω; (3) $I = 0.20\text{ A}$ and V across A = 6.0 V; (4) the p.d. across B and C = 3.0 V found by subtraction or by IR; (5) the current split $0.10\text{ A} + 0.10\text{ A} = 0.20\text{ A}$; (6) a correct statement that cells in series add e.m.f. while cells in parallel keep 1.5 V. Tally correct whiteboards: aim for 80% by Step 4. Reteach in a small group those who add $30 + 30$ for the parallel pair.

	lasts longer. They answer Kipchirchir's prediction: the parallel bulbs drew 0.90 A against 0.10 A in series, so the battery does supply more current in parallel; he is partly right. They finally explain Achieng's 12 V solar home system: the lights in the home are wired in parallel so each gets the full 12 V and one failing does not put the rest out.			
Driving Question Board (DQB) Creation  VIDEO: Series resistors <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Combined Series-Parallel Circuits Practice Source: CK-12  Search ARES for similar readings	Students return to the Driving Question Board (DQB) with sticky notes. They (1) mark which existing questions are now partly answered by today's evidence (for example 'What if the circuit has both series and parallel parts?'; 'Why does the battery run down faster with more bulbs in parallel?'), (2) write one new question on a sticky note, for instance 'Why do the parallel bulbs use more energy from the battery even though they are brighter?' (leading to L10) or 'What happens in the solar home system if the battery is flat?', and (3) add a note to the 'What we have learned' column such as 'Mixed circuit: $R = 45\ \Omega$, $I = 0.20\ \text{A}$, $6.0\ \text{V}$ and $3.0\ \text{V}$ '. Students link at least one note to the phenomenon of the blown tube in Nairobi or the dim series bulbs.	Say: "Go back to our driving question about the tubes and the dim bulbs. Which questions on the board did today's evidence help with?" Give WAIT TIME of 10-15 seconds, then cold-call 3 students to move or annotate a question. Ask: "What new question do you have after today? Write it on a sticky note." Give 2 minutes. Ask 3 students to read their notes and place them on the board. Highlight Kipchirchir's idea: "We said it is partly right. Which question captures what we still need to know about energy and cost?" Direct students to a question that leads to Lesson 10 (energy and power) and a question that leads to Lesson 11 (fuses and safety).	Driving Question Board as a public record of shared sensemaking: moving, annotating and adding questions shows students that the class's knowledge is growing and that new evidence creates new, more precise questions, motivating the next lessons.	Check that (1) at least half of students can point to a DQB question and say which evidence from today answers it (look for the numbers $45\ \Omega$, $0.20\ \text{A}$, $6.0\ \text{V}$, $3.0\ \text{V}$, or $4.5\ \text{V}$ and $9\ \text{V}$); (2) new questions are testable and linked to the phenomenon (energy, cost, fuse, battery lasting); (3) students use correct terms (p.d., e.m.f., series, parallel). Note students whose questions show confusion between e.m.f. and p.d. for a follow-up.
Model Building Phase  VIDEO: Parallel resistors (part 1) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES	Students revise their model of the phenomenon in their notebooks. The model has three parts: (1) a labelled diagram of the Nairobi classroom lighting (4 tubes on separate branches across the 240 V supply), the 9 V series demo and the 9 V parallel demo, now with a third diagram of the mixed circuit; (2) a table of totals: series $90\ \Omega$ and $0.10\ \text{A}$, parallel $10\ \Omega$ and $0.90\ \text{A}$, mixed $45\ \Omega$ and $0.20\ \text{A}$; (3)	Say: "Open your model from last lesson. What do you need to add or change after today?" Give WAIT TIME of 10-15 seconds and then 5 minutes for writing. Circulate and ask: "Where in your model does the $0.20\ \text{A}$ flow? Draw it." Ask 3 students to share one change they made and why (cold-call). Prompt: "Use the words in series, in parallel, p.d., current and e.m.f. in your explanation." Remind them:	Cumulative model revision (modelling): students extend a single model across lessons, linking diagrams, numbers and words. Adding the mixed circuit and the cells shows how a model can be refined rather than replaced, and the 'still unexplained' list directs the next investigations.	Look for: (1) a mixed circuit diagram with $45\ \Omega$, $0.20\ \text{A}$, $6.0\ \text{V}$ and $3.0\ \text{V}$ correctly placed; (2) correct comparison of totals ($90\ \Omega$, $45\ \Omega$ and $10\ \Omega$ for the three arrangements); (3) correct statement that cells in series add e.m.f. ($1.5\ \text{V}$, $4.5\ \text{V}$, $9\ \text{V}$) while parallel cells keep $1.5\ \text{V}$; (4) an explanation of why Achieng's home lights are in parallel on the 12 V system; (5) an honest list of

for similar videos  HTML: Series and Parallel Circuits (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	a short explanation, with an arrow and a note for the source: 'Cells in series add e.m.f.; cells in parallel keep the same e.m.f. but last longer.' They add Achieng's 12 V solar home system to the model and state why lights in the home are wired in parallel. They then write a claim-evidence-reasoning paragraph answering: 'How does today's mixed circuit help explain why the connection pattern changes brightness?' and mark what is still unexplained (the energy used and the cost, to be completed in Lesson 10, and the protection against excess current in Lesson 11).	"If one tube in the Nairobi classroom blows, which numbers in your model stay the same?" (240 V and 0.15 A in each remaining tube) and "Which number changes?" (total current 0.60 A to 0.45 A). Ask for the Christmas-lights contrast: "Why does the old string of lights go dark when one bulb fails?" (single series loop broken).	what remains unexplained. Use a 3-2-1 exit check: 3 values from the mixed circuit, 2 differences between cells in series and parallel, 1 question remaining.
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D. TEACHER REFLECTION

After the lesson, consider: Did students use the reduce-and-redraw sequence on their own, or did they add 30 + 30 for the parallel pair? Which groups connected the meters correctly without prompting? Did students link the 0.90 A and 0.10 A currents to Kipchirchir's prediction about the battery running down faster in parallel? Did the mention of Achieng's 12 V solar home system make the cells idea meaningful for students from rural homes in the Rift Valley and around Kisumu? Which DQB questions still need evidence (energy, cost, fuses)? Note students who confuse e.m.f. (source) with p.d. (across a component) and plan a short reteach in Lesson 10. If measured values differed from the calculated ones, discuss real cells (small internal resistance) and real filaments (resistance rises as they get hotter) as reasons, while keeping the model of constant 30 Ω .

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We built a mixed circuit with one 30 Ω bulb in series with two 30 Ω bulbs in parallel on the 9 V pack. The bulb in series was brightest; the battery current was 0.20 A, the p.d. across the series bulb was 6.0 V, and each parallel bulb had 3.0 V and 0.10 A. When one parallel bulb was removed the other stayed lit. One cell gave 1.5 V, three cells in series gave 4.5 V and six cells in series gave 9 V; two cells in parallel still gave 1.5 V.
What did I learn?	To solve a mixed circuit, reduce the parallel group first ($1/R = 1/30 + 1/30$, so 15 Ω), then add the series resistance ($30 + 15 = 45 \Omega$), then use $I = V/R = 9/45 = 0.20 \text{ A}$ and work back to find the p.d. and current in each part. Cells in series add their e.m.f.; cells in parallel give the same e.m.f. but last longer. Achieng's 12 V solar home system in Kisumu uses a higher e.m.f. source with lights wired in parallel.
How does this explain the phenomenon?	The blown tube in Nairobi does not affect the other tubes because each tube is on its own parallel branch across the 240 V supply, while three series bulbs share the 9 V (3.0 V each) and carry only 0.10 A so they glow dim. A mixed circuit is a combination of both: the series part takes a bigger share of the p.d. and carries the total current, while the parallel part splits the current. Choosing how cells and bulbs are connected decides the p.d. and current each bulb receives, which explains the differences in brightness.

LESSON 10 (40 minutes (one double lesson of 80 minutes may be used for the full cost task)): Electrical Energy and Power: Why Series Bulbs Glow Dim and What the Tubes Cost

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To use $P = IV$ and $E = IVt = Pt$ to explain brightness quantitatively (0.30 W per series bulb against 2.7 W per parallel bulb) and to calculate the energy used and the cost in kWh for the Nairobi classroom tubes and for 60 W and 9 W bulbs.
Knowledge	Power P is the rate at which a component transfers electrical energy: $P = IV$, in watts (1 W = 1 J/s). Energy $E = IVt = Pt$ in joules; 1 kWh = 1000 W for 1 hour = 3.6 million J. Cost = kWh x tariff (KSh 25 per kWh). A bulb's brightness depends on its power, which depends on both the p.d. across it and the current through it.
Skills	Developing and using models, Analysing and interpreting data, Using mathematics and computational thinking (substituting into $P = IV$ and $E = Pt$, converting W to kW and calculating cost), Constructing explanations from evidence, Engaging in argument from evidence.
Attitudes	Appreciate energy saving at school and at home; value careful, honest use of data; show curiosity and respect for others' ideas when comparing predictions.
Key Inquiry Question	How is current electricity applied in day-to-day life? (and, linked to Key Inquiry 1: how do current, potential difference and resistance determine how much energy a circuit transfers?)
Purpose in Storyline	Lessons 7 and 8 gave us the currents and p.d.s that explain why series bulbs are dim and parallel bulbs are bright. Lesson 10 turns those numbers into power and energy (builds on L4: p.d. is energy per unit charge, and L8: the current in each bulb). It lets learners say how much brighter (9 times) and what it costs, and prepares for Lesson 11 (fuses and overload, where $P = IV$ gives the current of appliances) and Lesson 12 (the final model and a classroom lighting design).
Safety Notes	Use only the 9 V battery pack and 9 V, 2.7 W bulbs for the demo; do not connect learners to mains. Do not touch bulbs after a few minutes because they become warm. Kilowatt-hour calculations are paper-based, and no mains appliance is tested in class. Keep the battery leads short and switch off between observations.

B. LESSON OVERVIEW

Mr. Otieno reconnects the 9 V pack with three identical 9 V, 2.7 W (30 Ω) bulbs in series and then in parallel. Learners already know the series current is 0.10 A with 3.0 V across each bulb, and that in parallel each bulb has 9.0 V and 0.30 A. In this lesson they predict how much brighter parallel is, then calculate $P = IV$: $3.0 \times 0.10 = 0.30$ W per series bulb against $9 \times 0.30 = 2.7$ W per parallel bulb, which is 9 times the power. They then scale up to the Nairobi classroom: four 36 W tubes ($144 \text{ W} = 0.144 \text{ kW}$) for 5 h use 0.72 kWh, which costs KSh 18 at KSh 25/kWh. Finally they compare a 60 W filament bulb with a 9 W LED over 5 h (0.300 kWh, KSh 7.50 against 0.045 kWh, KSh 1.13). They add power and cost to the DQB and to their models, linking dim, bright and cost to the driving question.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners look at the series set	Reconnect both circuits. Say:	Predict-Observe-Explain with	Check that at least 80% of

 VIDEO: Electrical Energy - Example 1 Source: CK-12  Search ARES for similar videos  HTML: Electric Power and Electrical Energy Use (Flexbook) Source: CK-12  Search ARES for similar readings	(three bulbs, dim) and the parallel set (three bulbs, bright) from the 9 V pack. In pairs they write a prediction in their notebooks: 'How many times brighter is one parallel bulb than one series bulb? What quantity could we calculate to compare them? What do you think running the four Nairobi classroom tubes for a whole evening costs?' Wanjiru asks why the series bulbs are dim, and Kipchirchir suggests the parallel battery is used up faster. Learners record a numerical guess and a reason.	'Remember the tubes that stay lit when one blows. Today we ask how much energy they use and what that costs.' Ask: 'Brightness is about how fast a bulb transfers energy. Which two quantities from Lessons 4 and 8 might decide that?' Give 15 seconds WAIT TIME, then have learners write silently for 2 minutes. Cold-call 4 pairs for their numerical guess, and ask each 'What is your reason?' without saying who is right. Record guesses on the board in a column headed 'Our predictions'. Ask Kipchirchir to restate his idea about the battery being used up faster and say: 'Keep that idea, we will test it with numbers today.'	numerical estimation: committing to a number before calculating makes learners' ideas visible, creates a reason to calculate, and lets them later compare their intuition with evidence about the brightness gap.	notebooks contain a numerical prediction with a reason. Look for responses that use current or p.d. as reasons, for example 'parallel has more current' or 'series bulbs share the voltage'. Note learners who think only the battery decides brightness, or who say 'more current always means brighter regardless of p.d.', and revisit them in the Explain Phase.
Observe Phase  VIDEO: Electric power Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Electric Power and Electrical Energy Use (Flexbook) Source: CK-12  Search ARES for similar readings	Groups of 4 re-measure and record the demo data in a table. For series: ammeter reading 0.10 A and voltmeter reading across one bulb 3.0 V. For parallel: 0.30 A in one branch, 9.0 V across one bulb and 0.90 A total from the battery. They then place a thin strip of paper behind each bulb to compare brightness and rank them. Using their data, each group calculates $P = IV$ for one series bulb and one parallel bulb, and for the whole battery in each circuit. They then read a short data card with the tubes in the Nairobi classroom (4 tubes, each 36 W, 240 V mains) and the energy-use card for the 60 W and 9 W bulbs.	Circulate with the question: 'Is your voltmeter across the bulb and your ammeter in the loop?' Remind learners that $P = IV$: 'power is current times p.d., and 1 W means 1 joule every second'. Link to L4: 'Remember $1 \text{ V} = 1 \text{ J/C}$, so volts times amperes gives joules per coulomb times coulombs per second, which is joules per second.' Ask each group to check $3.0 \text{ V} \times 0.10 \text{ A} = 0.30 \text{ W}$ and $9 \text{ V} \times 0.30 \text{ A} = 2.7 \text{ W}$. WAIT TIME 10 seconds after asking 'What do you notice about the two powers?'. Cold-call 3 groups to report their power values and the ratio ($2.7 / 0.30 = 9$). Ask: 'Which circuit takes more power from the battery overall?' (series: $9 \times 0.10 = 0.90 \text{ W}$; parallel: $9 \times 0.90 = 8.1 \text{ W}$). Ask Kipchirchir's group to compare these two values.	Data table and pattern-finding (analysing and interpreting data): learners organise measured V and I values in a table and then compute P so that the pattern 'bright bulb = high P' emerges from numbers they collected themselves, not from teacher statement.	Observe that learners place the ammeter in series and the voltmeter in parallel. Check that tables show 0.30 W (series bulb), 2.7 W (parallel bulb), 0.90 W (battery, series) and 8.1 W (battery, parallel) with units. Use a quick tally: how many groups obtained a ratio of 9? Support any group that multiplies the total battery voltage by a single bulb's current.
Explain Phase  VIDEO: Electrical Energy	Learners write an explanation in the claim-evidence-reasoning format: 'Claim: ... Evidence: ...	Model one calculation on the board and then let learners attempt the next one. Say: 'Series	Claim-Evidence-Reasoning with worked-example fading: the teacher models one step of the	Listen for correct units (W, kW, kWh, KSh) and correct answers: 0.30 W, 2.7 W, 0.72 kWh, KSh 18,

- Example 1 Source: CK-12 Search ARES for similar videos  HTML: Work Energy And Power <i>Related topic: not an exact match for this lesson.</i> Source: PhET Search ARES for similar readings	Reasoning: ...' for why the series bulbs are dim and the parallel bulbs are bright, using $P = IV$. They then work through the cost problems: (a) $E = Pt$ for the four 36 W tubes: $144 \text{ W} = 0.144 \text{ kW}$; $0.144 \text{ kW} \times 5 \text{ h} = 0.72 \text{ kWh}$; cost $0.72 \times 25 = \text{KSh } 18$. (b) 60 W bulb for 5 h: $0.060 \times 5 = 0.300 \text{ kWh} = \text{KSh } 7.50$; 9 W LED: $0.009 \times 5 = 0.045 \text{ kWh} = \text{KSh } 1.125$, about $\text{KSh } 1.13$. (c) Energy in joules for one parallel bulb in 60 s: $E = IVt = 0.30 \times 9 \times 60 = 162 \text{ J}$, and compare it to $1 \text{ kWh} = 3.6 \text{ million J}$. Learners also say what happens to the cost if one of the four tubes blows (three tubes: 108 W, 0.54 kWh, $\text{KSh } 13.50$).	bulbs are dim not because the current is used up, but because each bulb gets only 3.0 V and 0.10 A, so $P = 0.30 \text{ W}$.' Ask: 'Why do we divide watts by 1000 for kWh?' with 10 seconds WAIT TIME. Use cold calls: 3 learners for the unit conversion, 3 for the cost calculation, 2 for the LED comparison. Ask: 'If the LED costs $\text{KSh } 1.13$ and the filament bulb $\text{KSh } 7.50$ for the same 5 hours, what does that tell a school bursar in Kericho?' Confront the misconception 'current is used up by the first bulb': current is 0.10 A everywhere in the series loop (L3); what changes is the energy transferred per coulomb at each bulb. Ask learners to explain Kipchirchir's idea: parallel draws 0.90 A and series 0.10 A, so the battery supplies energy faster in parallel (8.1 W against 0.90 W), meaning his idea is partly right.	cost calculation and learners complete the others, so they link the equation $P = IV$ to measured data and build a justified explanation that connects power, energy, brightness and cost.	$\text{KSh } 7.50$, $\text{KSh } 1.13$. A strong response states that power depends on both p.d. and current, and that 9 times the power means 9 times the energy per second. Collect 5 exit explanations and sort them into three levels: uses $P = IV$ with numbers, only states 'more current', or confuses power with energy. Use the sort to plan small-group support.
Driving Question Board (DQB) Creation  VIDEO: Electric power Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Powers of complex numbers <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners return to the class DQB and add evidence cards in the columns for the driving question. They write on sticky notes: 'Series bulb: $3.0 \text{ V} \times 0.10 \text{ A} = 0.30 \text{ W}$ (dim)', 'Parallel bulb: $9 \text{ V} \times 0.30 \text{ A} = 2.7 \text{ W}$ (9 times brighter)', 'Four tubes, 5 h = $0.72 \text{ kWh} = \text{KSh } 18$', and 'LED 9 W costs $\text{KSh } 1.13$ against $\text{KSh } 7.50$ for a 60 W bulb'. They also add new questions, for example: 'Why does an extension cable get hot if power is $P = IV$?', 'How big a current does a kettle draw?' and 'Why do we use a fuse?'. Learners mark which earlier questions are now answered.	Say: 'Look at our board. Which of our questions can we now answer with numbers?' Give 10 seconds WAIT TIME, then ask 4 learners to place a card and justify it. Prompt: 'Which question do we still not understand?' Point to the questions about heating and fuses and say: 'Lesson 11 will use $P = IV$ for kettles and heaters.' Ask Achieng: 'If your 12 V solar system powers a 36 W light, what current does it draw?' and cold-call 2 others to calculate ($36 / 12 = 3 \text{ A}$) as a quick transfer of $P = IV$. Keep the board visible for the next lesson.	Driving Question Board as a public record of knowledge growth: placing evidence beside the original questions shows learners how their understanding has progressed and which questions to investigate next, which keeps the lesson tied to the phenomenon.	Count the number of new evidence cards that include a calculation and units. Note whether learners place cards under the correct question. Look for new questions that show understanding (for example about the 13 A fuse and thick or thin cables). Use stickies to identify learners who ask about 'how much' (quantitative) rather than only 'why'.

Model Building Phase  VIDEO: Voltage <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Function as a geometric series <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners revise their individual models (a labelled diagram with a short written explanation) of the phenomenon. They add: the power in each series bulb (0.30 W) and each parallel bulb (2.7 W) beside the diagrams; the Nairobi classroom tubes in parallel, each at 240 V and 0.15 A with 36 W, and an energy and cost note (0.72 kWh = KSh 18 for 5 h); and one sentence on how the energy is transferred (from the battery or mains, through the lamp, as light and heat). In pairs, they check each other's model against the data and write one 'energy-saving' recommendation for the school, such as switching from 60 W bulbs to 9 W LEDs.	Say: 'Use a different colour pen to show what is new from today.' Give 15 seconds WAIT TIME after asking: 'What does your model say about why the other tubes stay lit, and what does it say about energy?' Circulate and ask: 'Where does the energy go in the bulb?' and 'How does your model show that the series current is 0.10 A everywhere?' Cold-call 3 learners to describe one change they made. Reinforce the vocabulary: p.d. across the component, e.m.f. of the source. Ask them to include units on every number. Close with: 'In Lesson 11 we will see what happens if too many appliances, such as a kettle and a heater, are on one socket.'	Cumulative model revision with peer review: revising the same model each lesson in a different colour shows learners how new evidence adds to, or changes, earlier ideas, and peer checking against the data promotes argument from evidence.	Check models for: (1) power values correctly labelled on series and parallel bulbs; (2) a statement that $P = IV$ with the p.d. across and the current through each bulb; (3) the cost example with the right units; (4) an energy-saving recommendation backed by numbers. Collect 6 models for a quick rubric (complete, partial, missing). Use an exit ticket: 'A 36 W tube runs for 5 h at KSh 25/kWh. What does it cost?' ($0.036 \times 5 = 0.18 \text{ kWh} = \text{KSh } 4.50$).
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D. TEACHER REFLECTION

Which learners confused power with energy or current, and which steps of the kWh conversion (W to kW, then multiplying by hours, then by tariff) caused most errors? Did learners link the 9-times difference in power (0.30 W against 2.7 W) to the original puzzle of dim and bright bulbs? Was Kipchirchir's partly correct idea about the parallel circuit drawing more current (0.90 A against 0.10 A) used well to build understanding? What evidence from the DQB shows that learners are ready for the safety questions in Lesson 11?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the three series bulbs (3.0 V and 0.10 A each) glowed dim, and the three parallel bulbs (9.0 V and 0.30 A each) glowed bright. We saw that the Nairobi classroom tubes are all on 240 V and stay lit when one blows. Our data card showed that a 60 W bulb uses more energy than a 9 W LED in the same time.
What did I learn?	We learned that power $P = IV$ (in watts, $1 \text{ W} = 1 \text{ J/s}$) and energy $E = IVt = Pt$. We learned that $1 \text{ kWh} = 1000 \text{ W for } 1 \text{ hour} = 3.6 \text{ million J}$, and that $\text{cost} = \text{kWh} \times \text{KSh } 25$. We calculated 0.30 W for a series bulb, 2.7 W for a parallel bulb, $0.72 \text{ kWh} = \text{KSh } 18$ for four tubes in 5 h, and KSh 7.50 for a 60 W bulb against KSh 1.13 for a 9 W LED.
How does this explain the phenomenon?	Series bulbs are dim because each gets only 3.0 V and 0.10 A, which gives 0.30 W, while parallel bulbs each get the full 9 V and 0.30 A, which gives 2.7 W (9 times more). The parallel circuit takes more total power from the battery (8.1 W against 0.90 W), so its energy is used faster, which is why Kipchirchir was partly right. The other tubes stay lit because each tube is on its own branch at

	240 V, and the school pays for the energy by the kWh, which can be reduced by using LEDs.
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LESSON 11 (40 minutes (one double lesson of 2 x 40 minutes recommended if the fuse demonstration is performed)): Domestic Wiring and Safety: Why the Fuse Blows When a Socket Is Overloaded

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students use the parallel-wiring idea behind the Nairobi classroom tubes, together with $P = IV$, to explain why a 13 A fuse blows when a kettle and a heater share one socket, and how fuses, circuit breakers and earthing protect people and property.
Knowledge	Appliances in a home are wired in parallel across the 240 V mains, so their currents add up. Current in an appliance is found from $I = P/V$. Too much current (overload or short circuit) heats the wires, and a fuse or circuit breaker on the live wire breaks the circuit. The earth wire gives shock protection by carrying fault current safely to the ground.
Skills	Science and Engineering Practices: Using mathematics (calculate $I = P/V$ for each appliance and add currents); Constructing explanations (explain why a fuse blows); Analysing and interpreting data (compare total current with fuse rating); Developing and using models (revise the circuit model to include protection devices).
Attitudes	Responsibility and safety consciousness when using electricity at home and school; respect for evidence and for others' ideas; curiosity about how everyday devices protect us.
Key Inquiry Question	KICD Key Inquiry Questions: (1) How do we determine the relationship between current, potential difference and resistance in an electric circuit? (2) How is current electricity applied in day-to-day life? Lesson question: Why does a fuse blow when a socket is overloaded, and how do fuses, circuit breakers and earthing keep a parallel wiring system safe?
Purpose in Storyline	Lessons 6, 8 and 10 gave us three ideas: long thin cables heat up (L6), parallel appliances each take their own current from the same voltage and these currents add up (L8), and $P = IV$ (L10). In this lesson students put the three ideas together to explain why the classroom parallel lighting circuit and the home sockets need a fuse or breaker on the live wire and earthing. This prepares for Lesson 12, where students design the safe parallel classroom circuit. No new circuit-reduction rules are taught here.
Safety Notes	NEVER connect any student-built circuit to 240 V mains. Use only the 9 V battery pack, low-voltage supply, or a teacher-controlled low-current fuse-wire demonstration (thin fuse wire or a thin strand of steel wool across a 6-9 V supply, behind a safety screen, with students at least 2 m away, goggles on). Use pictures, a dismantled plug and a non-working extension cable only for inspection. Do not touch hot wire; keep water away. Have a fire blanket and a bucket of sand ready. No student should open a mains socket at home; tell parents or a qualified electrician.

B. LESSON OVERVIEW

Mr. Otieno returns to the Nairobi evening prep class. After the blown tube, the other three stayed lit because each tube is on its own parallel branch (0.15 A each, 0.60 A for four, 0.45 A for three). Today the question is: what if too many appliances are put on one branch? Students predict what happens when a 3000 W kettle and a 2000 W heater share one socket with a 13 A fuse. They calculate $I = P/V$ for each (12.5 A and 8.33 A), add them (20.8 A) and compare with the 13 A fuse. They observe a safe low-voltage fuse-wire demonstration, examine a plug, a miniature circuit breaker picture and an extension cable, and explain earthing. They add the DQB entries and add protection devices to their class model of the parallel classroom circuit.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: current electricity fuse circuit  HTML: Circuit Construction Kit: DC <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English) Search ARES for similar readings	Wanjiru's family cooks ugali on an electric stove in Kericho and a student plugs a 3000 W kettle and a 2000 W heater into one extension socket on a cold morning. Students are shown a picture of the plug with a 13 A fuse and asked individually in their notebooks: 'What will happen when both are switched on at the same time, and why?' They draw a quick sketch of the socket, the two appliances (as parallel branches) and the fuse, and mark where they think the current is biggest. They also predict: 'If the fuse were replaced with a thick nail, what might happen to the extension cable from Lesson 6?'	Display the picture and say: 'Think back to the Nairobi classroom. One tube blew and the others stayed lit. Now we ask a different question: what if we put too much on one socket?' Give 2 minutes of silent individual writing, then WAIT TIME of 10-15 seconds before any talk. Cold-call 4 students (not volunteers) to share a prediction, and say: 'I am not looking for the right answer, I am looking for your reasoning.' Record predictions on the board in three columns: 'Nothing happens', 'Fuse blows', 'Cable heats/fire'. Ask: 'What would you need to know to decide between these?' Record the answers (appliance power, mains voltage, fuse rating) for use in the next phase.	Predict-Observe-Explain (POE) with individual commitment: writing a prediction before instruction exposes prior ideas, such as 'a fuse just stops electricity from being used', so they can be tested against evidence. Sorting predictions into columns shows students the range of ideas the class needs to test.	Look for: (1) whether students draw the two appliances as separate parallel branches from the same 240 V supply; (2) whether any mention 'too much current' or 'heating' versus 'too much voltage'; (3) whether they link the thick nail idea to the heating cable from Lesson 6. Tally on a clipboard how many students predicted each column. Misconceptions to note: 'the fuse uses up the electricity', 'higher voltage in the socket when two appliances are plugged in'.
Observe Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: current electricity fuse circuit  HTML: Circuit Construction Kit: DC <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)	Part A (calculation from data): In groups of 4, students use the table of appliance data from Lesson 10 style values: kettle 3000 W, heater 2000 W, mains 240 V. They calculate the current in each using $I = P/V$: kettle $3000/240 = 12.5$ A, heater $2000/240 = 8.33$ A. They add the currents because the appliances are in parallel: $12.5 + 8.33 = 20.8$ A, then compare with the 13 A fuse rating. They also compute the total power 5000 W and check $I = 5000/240 = 20.8$ A. Part B (teacher demonstration at low voltage): Mr. Otieno connects a short piece of thin fuse wire in series with a 9 V supply and a bulb bank, then increases the number of parallel bulbs (or reduces the series resistance) until the thin wire glows and melts. Students	Part A: circulate and ask each group: 'Why do we ADD the currents and not the resistances here?' and 'What was the current in each Nairobi tube branch, and how is this socket similar?' Cold-call 3 groups to report their totals. Part B: Say: 'Watch the ammeter and the thin wire, not the bulbs. What do you notice when the current rises?' After the wire melts, WAIT TIME of 10-15 seconds, then ask: 'Where exactly in the circuit did the break happen, and why there and not in the thick wires?' Part C: Ask at the plug: 'Which wire does the fuse sit on, and why must it be the live one?' Cold-call 3 students. Emphasise: fuse and switch go on the live wire so that when they break the circuit the appliance is not left 'live' at 240	Data-based reasoning with a claim-evidence-reasoning (CER) chart: students compare calculated current (20.8 A) with the fuse rating (13 A) and with the observed melting of thin wire. Using different forms of evidence (calculation, demonstration, physical objects) lets students triangulate the same idea: too much current heats the thin part, which melts and opens the circuit.	Check that: (1) each group's currents are 12.5 A, 8.33 A and 20.8 A with correct units; (2) students state clearly that currents add in parallel (L8) and not that voltage adds; (3) during the demonstration, students record that the wire melted when current exceeded a certain value; (4) students correctly identify live, neutral and earth wires and the fuse position in the plug. Use a thumbs up/sideways/down check: 'Can you explain why 20.8 A exceeds 13 A?' Note students who still say 'the fuse has 13 volts'.

Search ARES for similar readings	record the number of bulbs and the current from an ammeter when the wire melts, observe that the bulbs go out, and note that the rest of the circuit stays cool. Part C (inspection stations, 3 minutes each): (1) a dismantled 3-pin plug with fuse, live (brown), neutral (blue), earth (green and yellow) wires; (2) a picture/model of a miniature circuit breaker (MCB) showing its switch and tripping lever; (3) a long thin extension cable (non-working) compared with a thicker cable, linking to heating in thin wires from Lesson 6.	V. Repeat safety rules: no mains experiments.		
Explain Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: current electricity fuse circuit  HTML: Circuit Construction Kit: DC <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English) Search ARES for similar readings	Students write a short CER explanation in groups: Claim: the 13 A fuse blows when the kettle and heater share the socket. Evidence: $12.5\text{ A} + 8.33\text{ A} = 20.8\text{ A}$, which is more than 13 A; the thin fuse wire melted at a high current in the demonstration. Reasoning: current in the wire produces heat (links to L6: a thin wire has higher resistance and heats up); $P = IV$ means a greater total power at 240 V gives a greater current; the fuse is the weakest, thinnest part so it melts first and opens the circuit before the cable gets hot enough to burn. They then answer three application questions: (a) Why is a 13 A fuse better than a thick nail in the plug? (b) How does a circuit breaker differ from a fuse (it trips and can be reset, no replacement; it works with an electromagnet or bimetallic strip)? (c) Earthing: if a faulty live wire touches the metal body of the kettle, the earth wire provides a low-resistance path to the ground, a large current flows, the fuse blows, and the person is	Say: 'Use the words from the evidence, not just "it overloaded". Which numbers prove it?' Give 3 minutes of group writing, then WAIT TIME of 10-15 seconds before sharing. Cold-call 3 groups to read their reasoning aloud and ask the class to 'add one thing or challenge one thing'. Probe: 'The kettle alone is 12.5 A, which is under 13 A. Why is it still a risk to use a 13 A fuse for a kettle plus anything else?' Then ask: 'On the Nairobi classroom circuit, the four tubes took 0.60 A in total. Why is it fine for them to share one fuse or breaker?' Link back to the phenomenon: 'Parallel wiring lets tubes work independently, but because currents add, the supply needs protection.' Correct the idea that a fuse reduces the current; it only cuts the circuit if the current is too high.	Claim-Evidence-Reasoning with peer critique: the structured written argument forces students to connect numbers (20.8 A vs 13 A), a mechanism (heating in wires) and a safety device. Peer challenge and 'add one thing' build collective sensemaking and expose gaps such as confusing fuse rating with voltage.	Success indicators: a complete explanation includes (1) both currents calculated from $I = P/V$, (2) addition because the appliances are in parallel, (3) comparison with 13 A, (4) heating of wire leading to the fuse melting and breaking the circuit, (5) the fuse on the live wire. For earthing, students mention a low-resistance path to the ground and a large current that blows the fuse. Quick exit slip: 'A 2000 W heater and a 1000 W iron share a 13 A socket at 240 V. Will the fuse blow? Show the calculation.' ($8.33\text{ A} + 4.17\text{ A} = 12.5\text{ A}$, which is just below 13 A, so it should not blow, but it is close.)

	not shocked. They also note that overload (too many appliances) and short circuit (live touching neutral, very low resistance, very large current) are two ways in which the current becomes too high.			
Driving Question Board (DQB) Creation  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: current electricity fuse circuit  HTML: Circuit Construction Kit: DC <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Students revisit the class DQB. They place sticky notes with new learning under the headings 'Answered', 'Partly answered' and 'New questions'. Likely additions: Answered: 'Why do the other tubes stay lit?' (parallel branches, each at 240 V, one broken branch leaves others unaffected); 'Why is the Christmas-light string dark when one bulb fails?' (series, single loop). Partly answered: 'Why do series bulbs glow dimmer?' (more total resistance, less current: 90 Ω , 0.10 A, 0.30 W each) with a link to the safety lesson about power and current. New questions from today: 'How does a circuit breaker know when to trip?', 'What if the earth wire is missing?', 'Why does the fuse go on the live wire and not the neutral?', 'How should the classroom lighting circuit be protected?' Each student adds at least one note.	Say: 'Look at our DQB. Which questions can we now answer using evidence from the last 11 lessons, and which new ones do we have?' Give individual silent writing for 2 minutes. WAIT TIME of 10-15 seconds, then cold-call 4 students to place and justify a note. Ask: 'What evidence would you need to answer the question about the circuit breaker?' Mark which questions will be addressed in Lesson 12 (design of a safe classroom parallel circuit with fuse and earthing).	DQB sorting and gap analysis: Categorising questions into answered, partly answered and new makes students aware of their progress and what remains unexplained, and keeps the whole storyline connected to the original phenomenon.	Check that notes cite evidence (e.g., '20.8 A exceeds 13 A') rather than just stating facts. Count how many students propose a new, testable question. Listen for students linking safety devices back to parallel circuits and the blown tube. Note any students still saying that the fuse is the cause of the tubes staying lit.
Model Building Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: current electricity fuse circuit	Students revise their group model of the Nairobi classroom lighting circuit. They redraw four tubes in parallel across the 240 V mains and add: a fuse or circuit breaker on the live wire before the branches, a switch on the live wire, and an earth wire connected to the metal casings. Next to the drawing they add a 'what happens if' box: (1) one tube blows: the other three stay lit, current falls from 0.60 A to 0.45 A; (2) too many appliances on	Say: 'Your model must now explain the tubes, the dim series bulbs AND the safety devices. Show where the current flows in each case with arrows.' Give 8 minutes. Circulate and ask: 'Where is the fuse in your drawing, and what would happen to the other tubes if the fuse blew?' (all go out, because it protects the whole circuit). Cold-call 3 groups to present using the gallery walk. WAIT TIME of 10-15 seconds after	Model-based reasoning (Developing and Using Models): students revise a visual model by adding protection devices, which turns it into an explanatory tool that predicts what happens in overload, short circuit and earth-fault situations. Gallery walk with peer feedback improves precision.	Look for: (1) four tubes drawn in parallel with the fuse/breaker and switch on the live wire; (2) an earth wire connected to the metal casing; (3) correct currents of 0.60 A and 0.45 A; (4) a correct explanation of the series string as a single loop. Rubric: 3 = all components and explanations correct; 2 = one element missing (e.g., earth); 1 = fuse in the wrong place or confusion between voltage and current. Collect

 HTML: Circuit Construction Kit: DC <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English) Search ARES for similar readings	a socket: total current exceeds the fuse rating, the fuse blows; (3) a live wire touches a metal casing: earth wire carries a large current, the fuse blows and the casing is not live. They also add a small comparison: series Christmas-light string has no independent branches, so one failure darkens everything.	each question before calling on someone. Use prompts such as: 'Which part of your model shows why the current is 0.60 A?' and 'Which part shows why a person is safe if the casing becomes live?' Close by asking each student to say one safety rule they will follow at home in Kisumu, Kericho or Nairobi.	models for use in Lesson 12.
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D. TEACHER REFLECTION

Which students used $I = P/V$ correctly and added currents in parallel rather than resistances? Did any still believe that a fuse 'reduces' current or that sockets change the mains voltage? Did the low-voltage fuse demonstration make the link to heating in thin wires (Lesson 6) clear? Which students could explain earthing without prompting? What adjustments are needed before Lesson 12, where students must design a safe parallel classroom lighting circuit independently? Note any students needing support with unit conversion and division of watts by volts.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that a kettle (12.5 A) and a heater (8.33 A) on one socket draw 20.8 A, more than a 13 A fuse allows; a thin fuse wire melted when the current became too large; the plug has a fuse on the live wire and an earth wire; an extension cable that is long and thin heats up.
What did I learn?	Appliances in a home are in parallel, so their currents add. $I = P/V$. Too much current (overload or short circuit) heats the wires; a fuse or circuit breaker on the live wire cuts the current. Earthing gives fault current a safe path and protects people from shock.
How does this explain the phenomenon?	The fuse blows because the total current of 20.8 A exceeds the 13 A rating, so the thin fuse wire heats and melts and breaks the circuit before the cable overheats. The other tubes in the Nairobi classroom stay lit when one blows because they are in parallel, but the whole parallel system needs a fuse or breaker and earthing for safety.

LESSON 12 (80 minutes (2 x 40-minute periods)): Model Revision and Application: Why the Other Tubes Stay Lit and Why Series Bulbs Glow Dim

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To synthesise all the ideas from Lessons 1-11 into a Final Explanation and a revised model of the anchoring phenomenon, to resolve the Driving Question Board, and to apply the ideas by designing a safe parallel lighting circuit for a Nairobi classroom. No new physics concept is taught.
Knowledge	Learners use ideas already built in Lessons 1-11. Series: one loop, total $R = 90\ \Omega$, $I = 0.10\ \text{A}$ everywhere, $3.0\ \text{V}$ per bulb, $0.30\ \text{W}$ per bulb, so the bulbs are dim, and if one bulb fails $I = 0$ and all go out (the Christmas-light string). Parallel: each branch has the full $9.0\ \text{V}$, $0.30\ \text{A}$ per branch, $0.90\ \text{A}$ total, effective $R = 10\ \Omega$, $2.7\ \text{W}$ per bulb, and if one bulb is removed the others stay at $9.0\ \text{V}$ and $0.30\ \text{A}$ (total $0.60\ \text{A}$). The Nairobi classroom has 4 tubes of $36\ \text{W}$ on $240\ \text{V}$, $0.15\ \text{A}$ per tube, $0.60\ \text{A}$ in total, then $0.45\ \text{A}$ when one blows. $144\ \text{W}$ for $5\ \text{h} = 0.72\ \text{kWh} = \text{KSh } 18$ at $\text{KSh } 25/\text{kWh}$. Protection: a fuse ($13\ \text{A}$) or breaker on the live wire and earthing (Lesson 11).
Skills	Developing and using models (revising a circuit model); constructing explanations using evidence and quantitative reasoning ($V = IR$, $P = IV$, series and parallel rules); analysing and interpreting data; designing a solution that meets criteria and constraints (safe, low cost, reliable); engaging in argument from evidence; communicating findings in diagrams and in writing.
Attitudes	Appreciate the importance of safe electrical installations in homes and schools; show responsibility and respect when working in groups and when critiquing the designs of others; show curiosity about how Physics explains everyday observations in Kenya.
Key Inquiry Question	1. How do we determine the relationship between current, potential difference and resistance in an electric circuit? 2. How is current electricity applied in day-to-day life?
Purpose in Storyline	This is the culminating lesson. Learners compare their Lesson 1 initial model with the final model, answer the remaining DQB questions with evidence, write the Final Explanation, and then transfer the ideas to a design task: a safe classroom lighting circuit that keeps the lights on when one tube fails.
Safety Notes	Use only the $9\ \text{V}$ battery pack (six $1.5\ \text{V}$ dry cells) at the bench; never connect a bulb directly to mains. The classroom design is on paper only. Remind learners that mains at $240\ \text{V}$ can kill, that only qualified electricians may wire or repair mains circuits, and that they must never replace a fuse with a higher-rated fuse, wire or foil. Keep wires and cells away from the face and do not short-circuit the battery pack because wires and cells heat up.

B. LESSON OVERVIEW

Mr. Otieno returns to the evening prep class in Nairobi where one tube has blown and three stay lit, and to the $9\ \text{V}$ bench demo where three series bulbs glow dim while three parallel bulbs glow brightly. Learners start by predicting what their designed classroom circuit must do if a tube fails and what would go wrong if the tubes were wired in series. They then test the revised model on a quick bench check (series $0.10\ \text{A}$, $3.0\ \text{V}$ each, $0.30\ \text{W}$; parallel $0.30\ \text{A}$ per branch, $9.0\ \text{V}$, $2.7\ \text{W}$; then one bulb removed from each) and use the Lesson 1-11 evidence to write the Final Explanation. They use a PEOE-style comparison of their Lesson 1 model with the final model, complete the DQB by matching each question to evidence, and finish by designing and critiquing a safe parallel lighting circuit for the classroom, with a $13\ \text{A}$ fuse or breaker on the live wire, earthing, a current and cost calculation ($0.60\ \text{A}$, then $0.45\ \text{A}$; $144\ \text{W}$, $0.72\ \text{kWh}$, $\text{KSh } 18$). Acheng's $12\ \text{V}$ solar home system in Kisumu and the Kericho dormitory are used as transfer contexts.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Introduction to circuits and Ohm's law <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Combined Series-Parallel Circuits (Flexbook) Source: CK-12  Search ARES for similar readings	Learners see the Lesson 1 photo of the Nairobi classroom with one blown tube and the dim series bulbs. Individually they write a prediction in their notebooks: 'If you were the electrician for a new Kericho dormitory, how would you wire the tubes so that one failure does not darken the room? What would happen if you wired them in series like the old Christmas-light string?' They sketch a quick circuit and predict, with numbers, the current when four 36 W tubes are all on 240 V and when one blows. They then share with a partner.	Display the Lesson 1 initial model posters and the DQB. Say: "Today you are the electricians. Before we test anything, predict: how should the tubes be connected, and what will the current be?" Give WAIT TIME of 10-15 seconds of silent thinking before writing. Circulate and note which learners draw series or parallel. Cold-call 3 pairs, one with a series sketch, one with a parallel sketch and one with numbers, without evaluating them. Ask: "Wanjiru asked why the series bulbs are dim. What do you now expect to be the answer?" Record the predictions on the board and leave them up to revisit.	Eliciting prior knowledge through prediction (Predict-Observe-Explain). Writing a prediction with numbers before the evidence activates the ideas from Lessons 1-11 and makes later model revision visible, because learners can compare what they predicted with what they found.	Look for: (1) sketches that show separate branches for each tube directly across the supply; (2) predictions that use $I = P/V$ ($36/240 = 0.15$ A per tube) or at least say that the current is larger with more tubes; (3) any misconception, such as 'the current is used up by the first bulb' or 'the parallel circuit has less current'. Record the number of learners holding each idea to target the Explain Phase.
Observe Phase  VIDEO: Introduction to circuits and Ohm's law <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Combined Series-Parallel Circuits Practice Source: CK-12  Search ARES for similar readings	In groups of four, learners rebuild the 9 V bench demo using the battery pack and three identical 9 V, 2.7 W bulbs (30 Ω each). Roles: builder, ammeter reader, voltmeter reader, recorder. They (a) connect the bulbs in series, record the current (0.10 A) and the p.d. across each bulb (3.0 V), and compare the brightness; (b) remove one bulb and record $I = 0$ and that all go out; (c) connect the bulbs in parallel, record the battery current (0.90 A), the branch current (0.30 A) and the p.d. across each bulb (9.0 V); (d) remove one bulb and record that the other two stay bright, each at 9.0 V and 0.30 A, with a total of 0.60 A. They complete a data table, calculate $R = V/I$ (90 Ω series, 10 Ω parallel) and $P = IV$	Say: "Remember: the ammeter goes in series and the voltmeter goes across the component, in parallel. Check your partner's connections before you switch on." Remind groups that the pack is 9 V and must never be short-circuited. Give WAIT TIME of 10-15 seconds after each reading before they record, and ask them to say what they expect before the bulb is removed. Ask the voltmeter reader: "What is the sum of the three p.d.s in series? What does that tell you?" Cold-call 4 groups, one for each of the four measurements, to report values to the class and write them on a class data table. Ask: "Do these numbers match the numbers your group predicted?"	Structured data collection and pattern finding. Learners measure the same quantities in both circuits so that the contrast (0.10 A vs 0.30 A per branch, 3.0 V vs 9.0 V, 0.30 W vs 2.7 W) appears as a pattern in one table that they can use as evidence.	Check that: ammeters are in series and voltmeters in parallel; values are close to 0.10 A, 3.0 V, 0.30 A, 9.0 V and 0.90 A; the recorder notes $I = 0$ when a series bulb is removed; learners calculate 90 Ω and 10 Ω correctly. Use a quick checklist on a clipboard. Learners who read 3.0 V as the battery p.d. receive a prompt: 'Which component is the voltmeter across?'

	(0.30 W and 2.7 W per bulb). Finally, they compute the Nairobi classroom values from the data: 4 tubes at 36 W give 0.15 A each, 0.60 A in total, then 0.45 A when one blows.			
Explain Phase  VIDEO: Introduction to circuits and Ohm's law <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Series and Parallel Circuits (Flexbook) Source: CK-12  Search ARES for similar readings	Learners individually write the Final Explanation, using a claim-evidence-reasoning frame, answering: 'Why do the other tubes stay lit when one blows, and why do series bulbs glow dimmer than identical bulbs in parallel on the same battery?' They must use at least four pieces of evidence from the bench data and the classroom data: (1) in series the total resistance is 90 Ω, so $I = 9/90 = 0.10$ A, the p.d. is shared as 3.0 V per bulb and $P = 0.30$ W, so each glows dim; (2) in parallel each bulb has the full 9.0 V, 0.30 A and 2.7 W (nine times the power), effective $R = 10$ Ω and a battery current of 0.90 A; (3) a failed series component opens the single loop so $I = 0$ and the whole old Christmas-light string goes dark; (4) a failed parallel branch leaves the other branches with the same p.d. and current, so in the Nairobi classroom the current drops from 0.60 A to 0.45 A while each remaining tube is still at 240 V. They also address Kipchirchir's prediction (the parallel circuit draws more current, 0.90 A vs 0.10 A, so the battery is used up faster, which is partly right) and the simplification that real filaments change resistance as they heat up. Then they swap explanations with a partner and use a checklist to give feedback.	Say: "Your explanation must have a claim, evidence with numbers, and reasoning that links $V = IR$ and $P = IV$. Don't just say 'parallel is better'; tell me why." Give WAIT TIME of 10-15 seconds before writing starts. Circulate and ask: "Where does the 3.0 V come from?" and "Why is the current in parallel nine times... no, three times bigger in each branch but nine times more power?" Cold-call 3 learners to read one sentence of their reasoning and ask 2 others to say whether the evidence supports it. Share the model exemplar only after peer feedback. Address misconceptions directly: current is not used up, the p.d. across parallel branches is the same, and fuses go in the live wire.	Claim-Evidence-Reasoning with peer critique. Writing a full causal explanation forces learners to link observations (brightness, failure behaviour) to the mechanisms (resistance, p.d. sharing, branch currents), and peer critique exposes gaps in reasoning.	Use a 4-point rubric: (1) states that parallel gives each tube the full supply p.d. on its own branch; (2) uses 90 Ω, 0.10 A, 3.0 V and 0.30 W for series and 10 Ω, 0.30 A, 9.0 V and 2.7 W for parallel; (3) explains the open loop in series vs the unaffected branches in parallel; (4) correctly applies the classroom numbers (0.15 A per tube, 0.60 A then 0.45 A). Note learners at level 1-2 for a follow-up mini-conference.
Driving Question	Learners revisit the DQB from Lesson 1 and the questions added	Say: "Pick a question on our board. Which lesson gave us the	Driving Question Board as a shared record of learning.	Observe whether learners name the correct lesson evidence and

Board (DQB) Creation  VIDEO: Introduction to circuits and Ohm's law <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Resistance in Parallel Circuits Vs. Series Circuits (PLIX) Source: CK-12  Search ARES for similar readings	since. In groups they sort the sticky notes into 'answered', 'partly answered' and 'new questions', and for each answered question they attach an evidence label (for example 'L3 current', 'L4 p.d.', 'L7 series', 'L8 parallel', 'L10 power and cost', 'L11 fuse'). They answer questions such as 'Why are the series bulbs dim?', 'Why do the other tubes stay lit?', 'Does the battery get used up faster in parallel?' and 'Why does the extension cable get hot?'. They also add questions for further inquiry, for example 'How does a circuit breaker work inside?' or 'How do LED tubes save more money?'. The class agrees on a final DQB summary statement.	evidence to answer it? Show me the number." Give WAIT TIME of 10-15 seconds after each question. Cold-call 5 learners to move a note and justify it with evidence. Ask Achieng to link the 12 V solar home system in Kisumu to the idea that cells in series add e.m.f. Ask Kipchirchir to revise his prediction using 0.90 A against 0.10 A. Point out any unanswered note and describe how it could be investigated further, without introducing new physics.	Matching each question to specific evidence shows learners how the lessons built on one another and how an explanation is justified by data rather than authority.	numbers for each question (for example, 'L8: each branch has 9.0 V and 0.30 A'). Count how many Lesson 1 questions are classified as answered with evidence. Notice questions that remain 'partly answered' and decide whether to revisit them next term.
Model Building Phase  VIDEO: Introduction to circuits and Ohm's law <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Series Circuits: Voltage Drops (PLIX) Source: CK-12  Search ARES for similar readings	Part 1 (model comparison): Groups pull out their Lesson 1 initial model and draw a final model on a large sheet showing the battery, the series loop and the parallel branches, with labelled I, V, R and P values (90 Ω, 0.10 A, 3.0 V, 0.30 W; 10 Ω, 0.30 A per branch, 0.90 A, 9.0 V, 2.7 W) and the Nairobi classroom (240 V, 0.15 A per tube). They mark what changed: 'then' vs 'now'. Part 2 (application design): Each group designs a safe lighting circuit for a Nairobi classroom with four 36 W tubes: a circuit diagram with the tubes in parallel, a switch and a 13 A fuse or circuit breaker in the live wire, an earth wire connected to the metal fittings, and neutral. They calculate the current (0.60 A with all tubes on, 0.45 A with one blown), confirm that this is well below the 13 A rating, and calculate the cost of running the	Say: "Put your old model next to your new one. Tell me three things that you changed and which evidence made you change them." Give WAIT TIME of 10-15 seconds before groups begin sketching. Cold-call 4 groups to state one change and one remaining uncertainty (for example the real filament's resistance rising as it heats). For the design: "Your design must keep the lights on when one tube fails and protect people from shock and fire. Where does the fuse go, and why?" Check that the fuse is on the live wire and that earthing is shown. Pose the challenge: "Could we ever wire the Kericho dormitory tubes in series? What would happen on a 240 V supply?" During the gallery walk, require each learner to give one 'two stars and a wish' comment. Remind them that only qualified	Model revision by comparison and design-based transfer. Placing the Lesson 1 model beside the final model makes conceptual change explicit, and applying the model to a real design problem (with constraints of safety and cost) tests whether learners can use their understanding in a new context.	Check that the final model includes: parallel branches with 9.0 V (or 240 V) across each, labelled currents that add up ($0.30 + 0.30 + 0.30 = 0.90$ A; $4 \times 0.15 = 0.60$ A), and an explanation of the series case (shared p.d., 0.10 A, open loop). Check that the design has the fuse/breaker in the live wire, earthing, a correct current of 0.60 A then 0.45 A and a cost of KSh 18. Use the checklist from the gallery walk as an exit rubric for each group.

	tubes for 5 h ($144 \text{ W} = 0.144 \text{ kW}$, $0.72 \text{ kWh} = \text{KSh } 18$ at KSh 25/kWh). They compare a 60 W filament bulb with a 9 W LED for 5 h (KSh 7.50 vs KSh 1.13). They then do a gallery walk to critique two other designs using the criteria: lights stay on when one fails, fuse in the live wire, earthing, cost, and safe cable use. Finally, they complete the summary table.	electricians install mains wiring.		
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D. TEACHER REFLECTION

After the lesson, review the Final Explanations and the designs. How many learners could use numbers (90Ω vs 10Ω , 0.10 A vs 0.30 A , 0.30 W vs 2.7 W , 0.60 A vs 0.45 A) to justify their claims? Which misconceptions persist (for example 'current is used up' or 'a fuse can go on the neutral wire')? Did learners use evidence from earlier lessons rather than only the demo of the day? Which DQB questions remain open, and how can they be used as starting points for Section 3.3 Introduction to Electronics? Consider how the Kenyan contexts (the Nairobi classroom, the Kericho dormitory, Achieng's solar system in Kisumu, ugali cooking on an electric stove) helped or distracted from the physics, and how to support learners who need more time with the quantitative reasoning.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that three identical bulbs on a 9 V battery pack glow dim in series ($I = 0.10 \text{ A}$, 3.0 V each, 0.30 W each, total $R = 90 \Omega$) and bright in parallel (9.0 V and 0.30 A per branch, 2.7 W each, 0.90 A from the battery, effective $R = 10 \Omega$). Removing one bulb in series puts all out, but in parallel the others stay lit, with the total current falling to 0.60 A . In the Nairobi classroom, when one of 4 tubes blows, the other three stay lit.
What did I learn?	We learned that current is the flow of charge measured in series, that p.d. is energy per unit charge measured in parallel across a component, that $V = IR$, that resistances add in series ($R = R_1 + R_2 + R_3$) and combine in parallel ($1/R = 1/R_1 + 1/R_2 + 1/R_3$), that $P = IV$ and $E = Pt$ (so 144 W for $5 \text{ h} = 0.72 \text{ kWh} = \text{KSh } 18$), and that fuses or breakers on the live wire and earthing protect people and wiring.
How does this explain the phenomenon?	The other tubes stay lit because each tube is on its own branch directly across the 240 V supply, so if one branch breaks the others still have the same p.d. and current (0.15 A each; the total falls from 0.60 A to 0.45 A). Series bulbs are dim because they share the 9 V (3.0 V each) in one loop of 90Ω , so the current is only 0.10 A and the power only 0.30 W , whereas parallel bulbs each get 9 V , 0.30 A and 2.7 W . In series a failed bulb opens the one loop, like the old Christmas-light string. A classroom should therefore be wired in parallel with a fuse or breaker on the live wire and earthing.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.